

Chapter 1: Introduction to Project Management

Information Technology Project Management, Eighth Edition

Note: See the text itself for full citations.



Learning Objectives, Part 1

- ▶ Understand the growing need for better project management, especially for information technology (IT) projects
- ▶ Explain what a project is, provide examples of IT projects, list various attributes of projects, and describe the triple constraint of project management
- ▶ Describe project management and discuss key elements of the project management framework, including project stakeholders, the project management knowledge areas, common tools and techniques, and project success

Learning Objectives, Part 2

- ▶ Discuss the relationship between project, program, and portfolio management and the contributions each makes to enterprise success
- ▶ Understand the role of project managers by describing what they do, what skills they need, and career opportunities for IT project managers
- ▶ Describe the project management profession, including its history, the role of professional organizations like the Project Management Institute (PMI), the importance of certification and ethics, and the advancement of project management software

Introduction

- ▶ Many organizations today have a new or renewed interest in project management
- ▶ Worldwide IT spending was \$3.8 trillion in 2014, a 3.2 percent increase from 2013 spending
- ▶ The Project Management Institute estimates demand for 15.7 million project management jobs from 2010 to 2020, with 6.2 million of those jobs in the United States

Project Management Statistics

- ▶ In 2013 (the most recent year of PMI's salary survey), the average salary in U.S. dollars for someone in the project management profession was \$108,000 per year in the United States; \$134,658 in Australia, (the highest-paid country); and \$24,201 in Egypt (the lowest-paid country)
- ▶ The top skills employers look for in new college graduates are all related to project management: team-work, decision-making, problem-solving, and verbal communications
- ▶ Organizations waste \$109 million for every \$1 billion spent on projects, according to PMI's Pulse of the Profession® report

Motivation for Studying Information Technology (IT) Project Management

- ▶ IT Projects have a terrible track record, as described in the What Went Wrong?
- ▶ A 1995 Standish Group study (CHAOS) found that only 16.2% of IT projects were successful in meeting scope, time, and cost goals; over 31% of IT projects were canceled before completion
- ▶ A PricewaterhouseCoopers study found that overall half of all projects fail and only 2.5% of corporations consistently meet their targets for scope, time, and cost goals for all types of project.

Advantages of Using Formal Project Management

- ▶ Better control of financial, physical, and human resources
- ▶ Improved customer relations
- ▶ Shorter development times
- ▶ Lower costs
- ▶ Higher quality and increased reliability
- ▶ Higher profit margins
- ▶ Improved productivity
- ▶ Better internal coordination
- ▶ Higher worker morale

What Is a Project?

- ▶ A **project** is “a temporary endeavor undertaken to create a unique product, service, or result” (PMBOK® Guide, Fifth Edition, 2013)
- ▶ Operations is work done to sustain the business
- ▶ Projects end when their objectives have been reached or the project has been terminated
- ▶ Projects can be large or small and take a short or long time to complete

Examples of IT Projects

- ▶ A team of students creates a smartphone application and sells it online
- ▶ A company develops a driverless car
- ▶ A government group develops a system to track child immunizations
- ▶ A global bank acquires other financial institutions and needs to consolidate systems and procedures

Top Strategic Technologies for 2012 (Gartner)

- ▶ Computing everywhere
- ▶ The Internet of things
- ▶ 3D printing
- ▶ Advanced, pervasive, and invisible analytics

Media Snapshot: Unproductive Apps

- ▶ Gartner predicted that by 2014, there would be more than 70 billion mobile application downloads every year, but it was almost double
- ▶ Facebook is by far the most downloaded app, and the most popular category of all apps continues to be games
- ▶ The challenge is to develop useful apps and get workers to focus on them instead of the many distracting options available

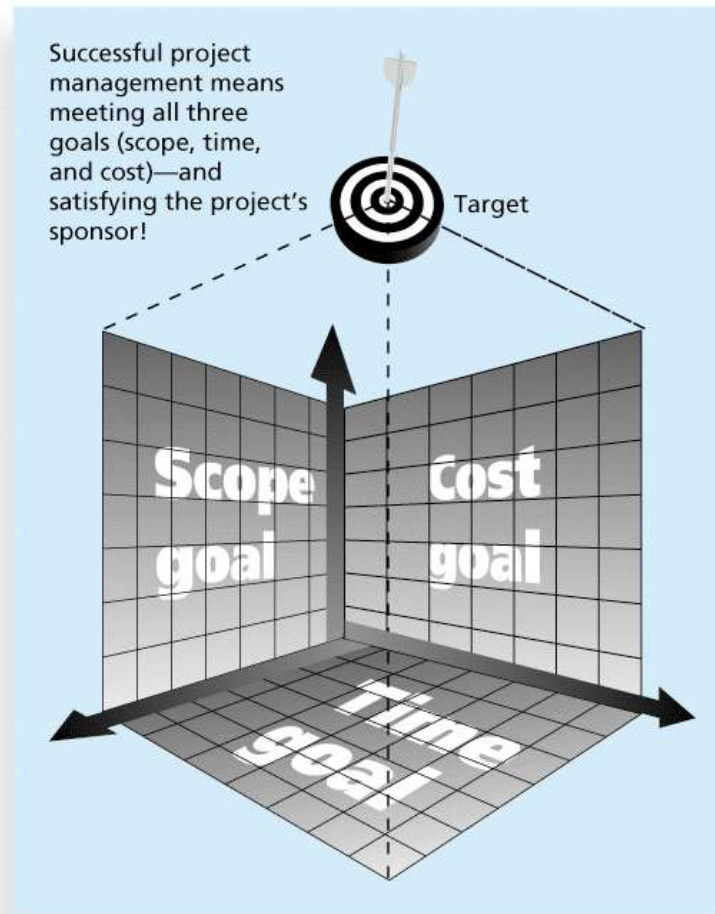
Project Attributes

- ▶ A project
 - has a unique purpose
 - is temporary
 - is developed using progressive elaboration
 - requires resources, often from various areas
 - should have a primary customer or sponsor
 - The **project sponsor** usually provides the direction and funding for the project
 - involves uncertainty

Project and Program Managers

- ▶ **Project managers** work with project sponsors, project team, and other people involved in a project to meet project goals
- ▶ **Program:** group of related projects managed in a coordinated way to obtain benefits and control not available from managing them individually (PMBOK® Guide, Fifth Edition, 2013)
- ▶ Program managers oversee programs; often act as bosses for project managers

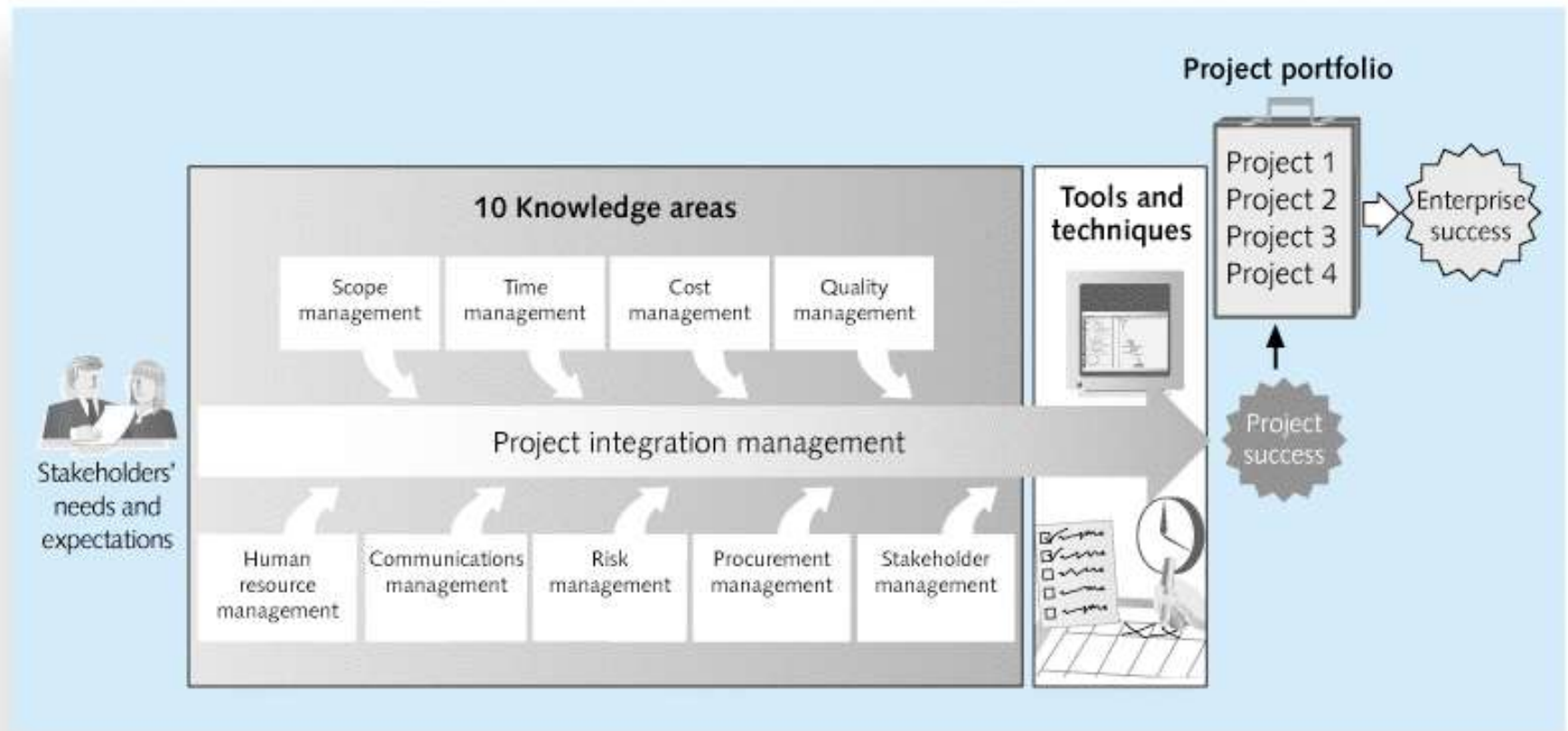
Figure 1-1 The Triple Constraint of Project Management



What is Project Management?

- ▶ **Project management** is “the application of knowledge, skills, tools and techniques to project activities to meet project requirements” (PMBOK® Guide, Fourth Edition, 2013)
- ▶ Project managers strive to meet the **triple constraint** (project scope, time, and cost goals) and also facilitate the entire process to meet the needs and expectations of project stakeholders

Figure 1-2 Project Management Framework



Project Stakeholders

- ▶ **Stakeholders** are the people involved in or affected by project activities
- ▶ Stakeholders include
 - the project sponsor
 - the project manager
 - the project team
 - support staff
 - customers
 - users
 - suppliers
 - opponents to the project

10 Project Management Knowledge Areas

- ▶ **Knowledge areas** describe the key competencies that project managers must develop
- ▶ Project managers must have knowledge and skills in all 10 knowledge areas (project integration, scope, time, cost, quality, human resource, communications, risk, procurement, and stakeholder management)
- ▶ This text includes an entire chapter on each knowledge area

Project Management Tools and Techniques

- ▶ **Project management tools and techniques** assist project managers and their teams in various aspects of project management
- ▶ Some specific ones include
 - Project charter, scope statement, and WBS (scope)
 - Gantt charts, network diagrams, critical path analysis, critical chain scheduling (time)
 - Cost estimates and earned value management (cost)
 - See Table 1-1 for many more

Super Tools

- ▶ **“Super tools”** are those tools that have high use and high potential for improving project success, such as:
 - Software for task scheduling (such as project management software)
 - Scope statements
 - Requirements analyses
 - Lessons-learned reports
- ▶ Tools already extensively used that have been found to improve project importance include:
 - Progress reports
 - Kick-off meetings
 - Gantt charts
 - Change requests

What Went Right? Improved Project Performance

The Standish Group's CHAOS studies show improvements in IT projects in the past decade:

- ▶ The number of successful IT projects has more than doubled, from 16 percent in 1994 to 39 percent in 2012
- ▶ The number of failed projects decreased from 31 percent in 1994 to 18 percent in 2012
- ▶ Success rates were much higher for small projects than large ones – 76 percent versus 10 percent

Project Success

- ▶ There are several ways to define project success:
 - The project met scope, time, and cost goals
 - The project satisfied the customer/sponsor
 - The results of the project met its main objective, such as making or saving a certain amount of money, providing a good return on investment, or simply making the sponsors happy

Table 1-2: What Helps Projects Succeed?*

1. Executive support
2. User involvement
3. Clear business objectives
4. Emotional maturity
5. Optimizing scope
6. Agile process
7. Project management expertise
8. Skilled resources
9. Execution
10. Tools and infrastructure

*The Standish Group, “CHAOS Manifesto 2013: Think Big, Act Small” (2013).

Top Three Reasons Why Federal Technology Project Succeed

- ▶ Adequate funding
- ▶ Staff expertise
- ▶ Engagement from all stakeholders

What the Winners Do...

- ▶ Recent research findings show that companies that excel in project delivery capability:
 - Use an integrated project management toolbox (use standard/advanced PM tools, lots of templates)
 - Grow project leaders, emphasizing business and soft skills
 - Develop a streamlined project delivery process
 - Measure project health using metrics, like customer satisfaction or return on investment

Program and Project Portfolio Management

- ▶ A **program** is “a group of related projects managed in a coordinated way to obtain benefits and control not available from managing them individually” (PMBOK® Guide, Fifth Edition, 2013)
- ▶ A **program manager** provides leadership and direction for the project managers heading the projects within the program
- ▶ Examples of common programs in the IT field include infrastructure, applications development, and user support

Project Portfolio Management

- ▶ As part of **project portfolio management**, organizations group and manage projects and programs as a portfolio of investments that contribute to the entire enterprise's success
- ▶ Portfolio managers help their organizations make wise investment decisions by helping to select and analyze projects from a strategic perspective

Figure 1-3. *Project Management Compared to Project Portfolio Management*



Best Practice

- ▶ A **best practice** is “an optimal way recognized by industry to achieve a stated goal or objective”*
- ▶ Robert Butrick *suggests that organizations* need to follow basic principles of project management, including these two mentioned earlier in this chapter:
 - Make sure your projects are driven by your strategy. Be able to demonstrate how each project you undertake fits your business strategy, and screen out unwanted projects as soon as possible
 - Engage your stakeholders. Ignoring stakeholders often leads to project failure. Be sure to engage stakeholders at all stages of a project, and encourage teamwork and commitment at all times

*Project Management Institute, *Organizational Project Management Maturity Model (OPM3) Knowledge Foundation* (2003), p. 13.

Figure 1-4. Sample Project Portfolio Approach

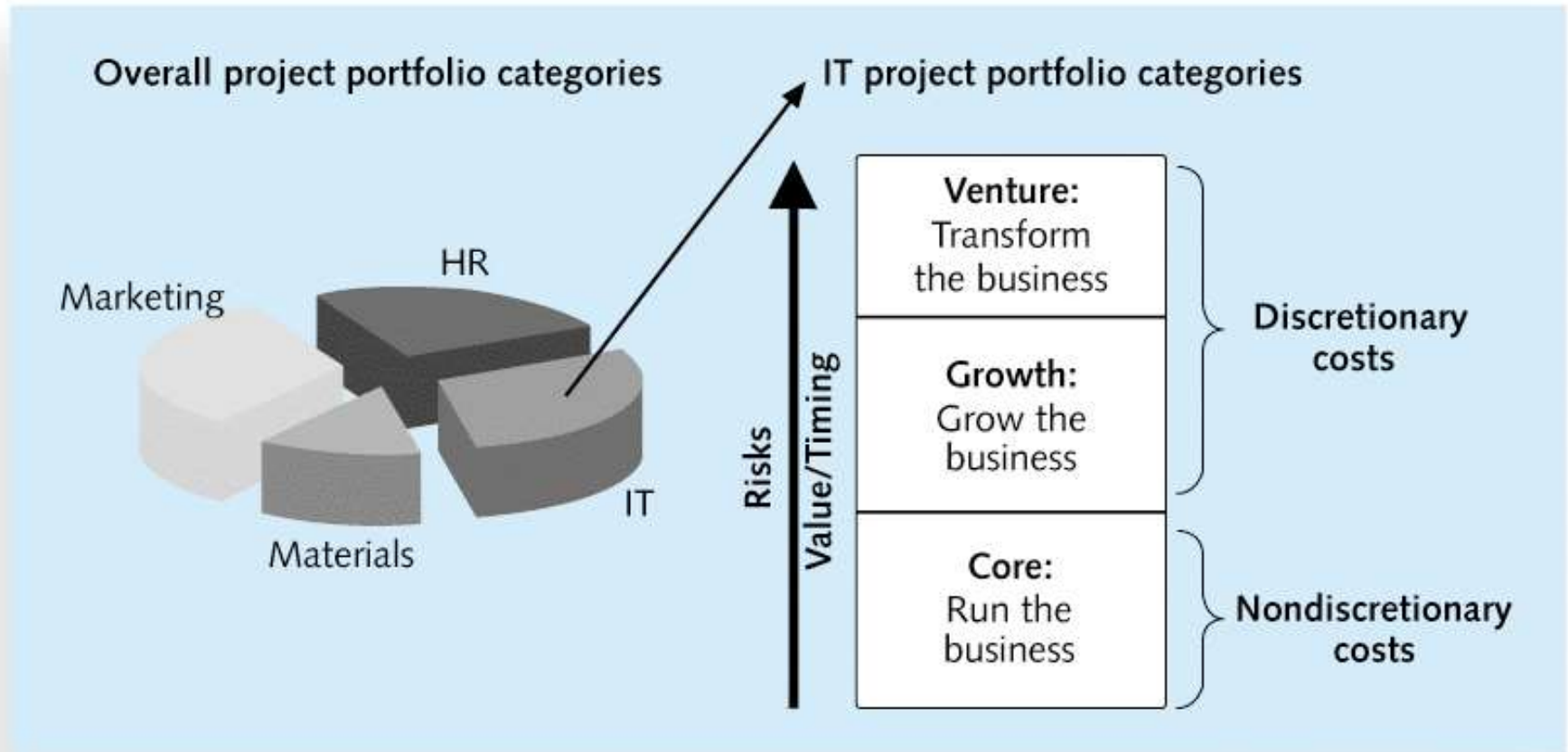


Figure 1-5. *Microsoft project portfolio management capabilities*



Source: Microsoft, "PPM Solution Guide" (October 2013).

The Role of the Project Manager

- ▶ Job descriptions vary, but most include responsibilities like planning, scheduling, coordinating, and working with people to achieve project goals
- ▶ Remember that 97% of successful projects were led by experienced project managers, who can often help influence success factors

Suggested Skills for Project Managers

- ▶ The Project Management Body of Knowledge
- ▶ Application area knowledge, standards, and regulations
- ▶ Project environment knowledge
- ▶ General management knowledge and skills
- ▶ Soft skills or human relations skills

Table 1-3 Ten Most Important Skills and Competencies for Project Managers

1. People skills
2. Leadership
3. Listening
4. Integrity, ethical behavior, consistent
5. Strong at building trust
6. Verbal communication
7. Strong at building teams
8. Conflict resolution, conflict management
9. Critical thinking, problem solving
10. Understands, balances priorities

Different Skills Needed in Different Situations

- ▶ Large projects: Leadership, relevant prior experience, planning, people skills, verbal communication, and team-building skills were most important
- ▶ High uncertainty projects: Risk management, expectation management, leadership, people skills, and planning skills were most important
- ▶ Very novel projects: Leadership, people skills, having vision and goals, self confidence, expectations management, and listening skills were most important

Importance of Leadership Skills

- ▶ Effective project managers provide leadership by example
- ▶ A **leader** focuses on long-term goals and big-picture objectives while inspiring people to reach those goals
- ▶ A **manager** deals with the day-to-day details of meeting specific goals
- ▶ Project managers often take on the role of both leader and manager

Careers for IT Project Managers

- ▶ In a 2014 survey, IT executives listed the “ten hottest skills” they planned to hire for in 2015
- ▶ Project management was second only to programming and application development
- ▶ Even if you choose to stay in a technical role, you still need project management knowledge and skills to help your team and organization

Table 1-4. Ten Hottest IT Skills

Skill	Percentage of Respondents
Programming and application development	48
Project management	35
Help desk/technical support	30
Security/compliance governance	28
Web development	28
Database administration	26
Business intelligence/analytics	24
Mobile application and device management	24
Networking	22
Big data	20

Source: Mary K. Pratt, “10 Hottest IT Skills for 2015,” Computerworld, November 18, 2014.

The Project Management Profession

- ▶ The profession of project management is growing at a very rapid pace
- ▶ It is helpful to understand the history of the field, the role of professional societies like the Project Management Institute, and the growth in project management software

History of Project Management

- ▶ Some people argue that building the Egyptian pyramids was a project, as was building the Great Wall of China
- ▶ Most people consider the ***Manhattan Project*** to be the first project to use “modern” project management
- ▶ This three-year, \$2 billion (in 1946 dollars) project had a separate project manager and a technical manager

Figure 1-6. Sample Gantt Chart Created with Project 2013

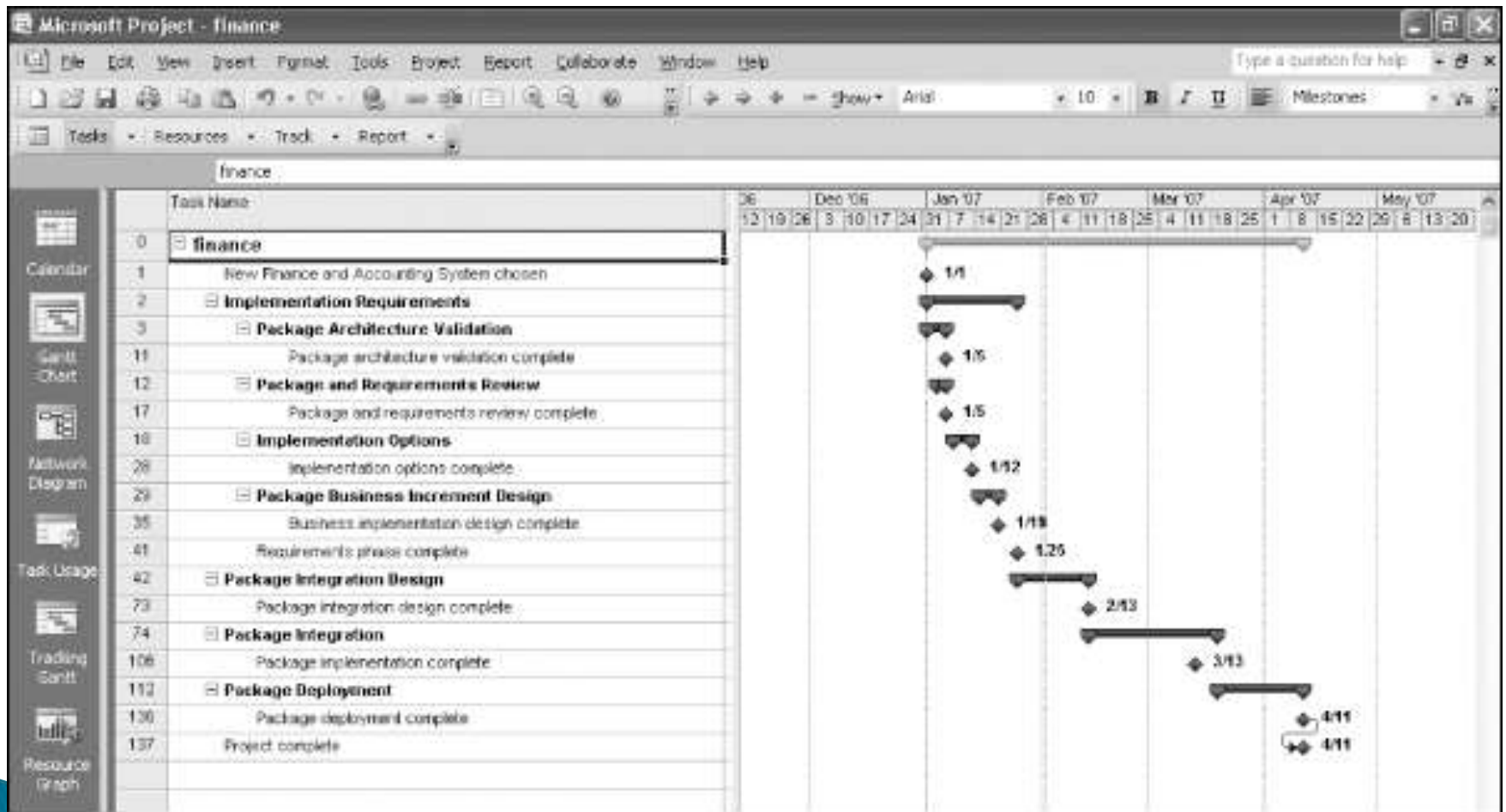
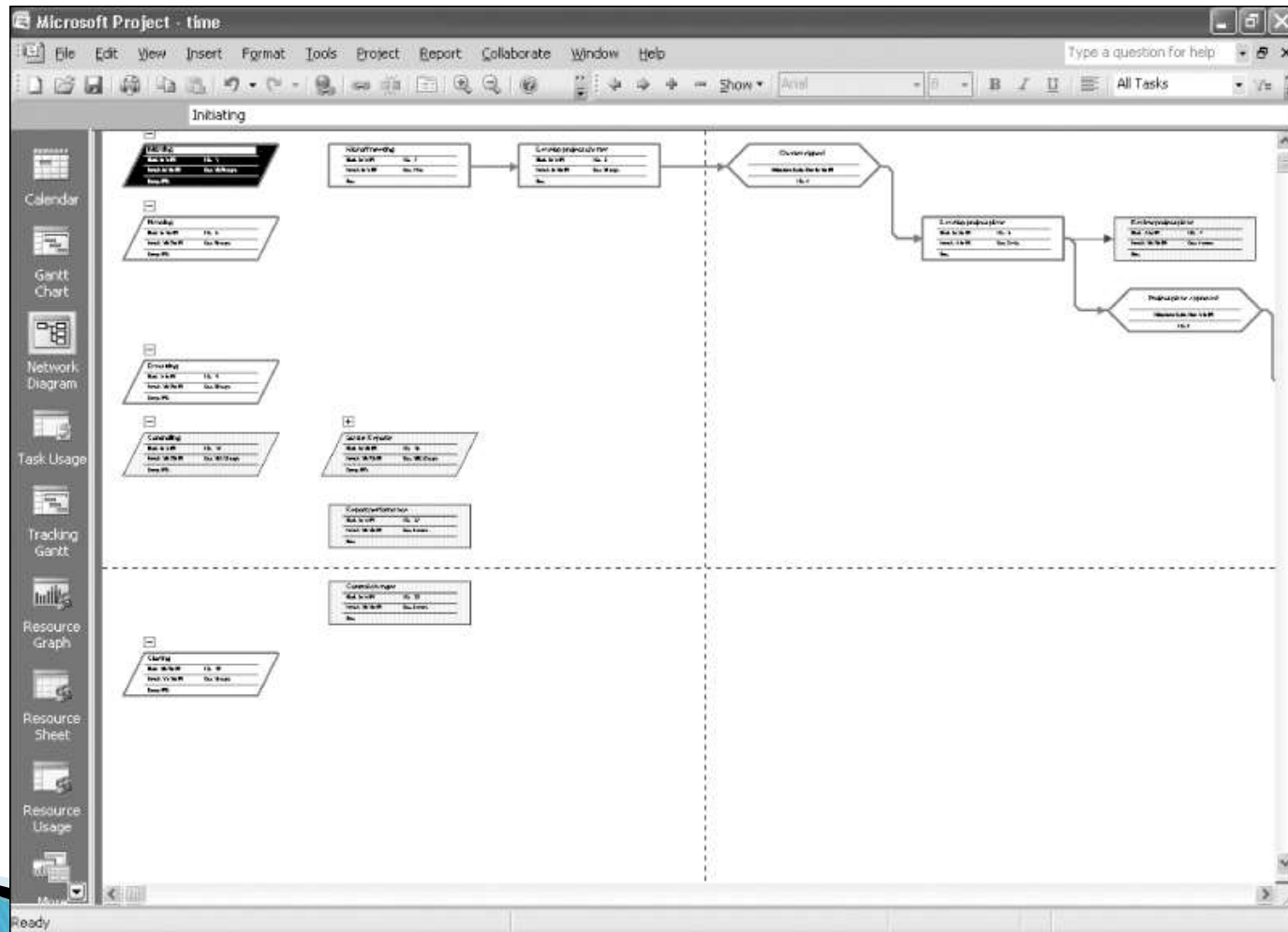


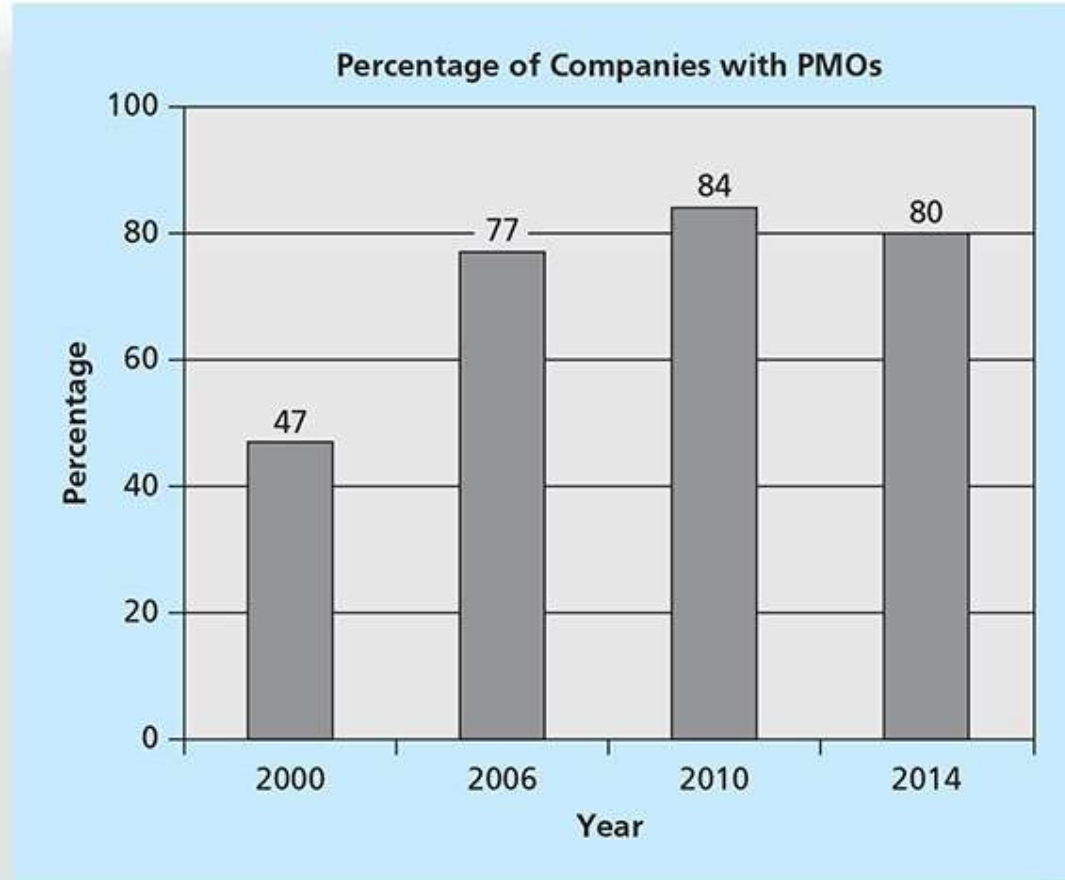
Figure 1-7. Sample Network Diagram Created with Project 2013



Project Management Offices

- ▶ In the 1900s, many companies began creating PMOs to help them handle the increasing number and complexity of projects
- ▶ A **Project Management Office (PMO)** is an organizational group responsible for coordinating the project management function throughout an organization

Figure 1-8. Growth in the Number of Project Management Offices



Source: PM Solutions, "The State of the PMO 2010" (2010).

Global Issues

- ▶ Several global dynamics are forcing organizations to rethink their practices:
 - Talent development for project and program managers is a top concern
 - Good project portfolio management is crucial in tight economic conditions
 - Basic project management techniques are core competencies
 - Organizations want to use more agile approaches to project management
 - Benefits realization of projects is a key metric

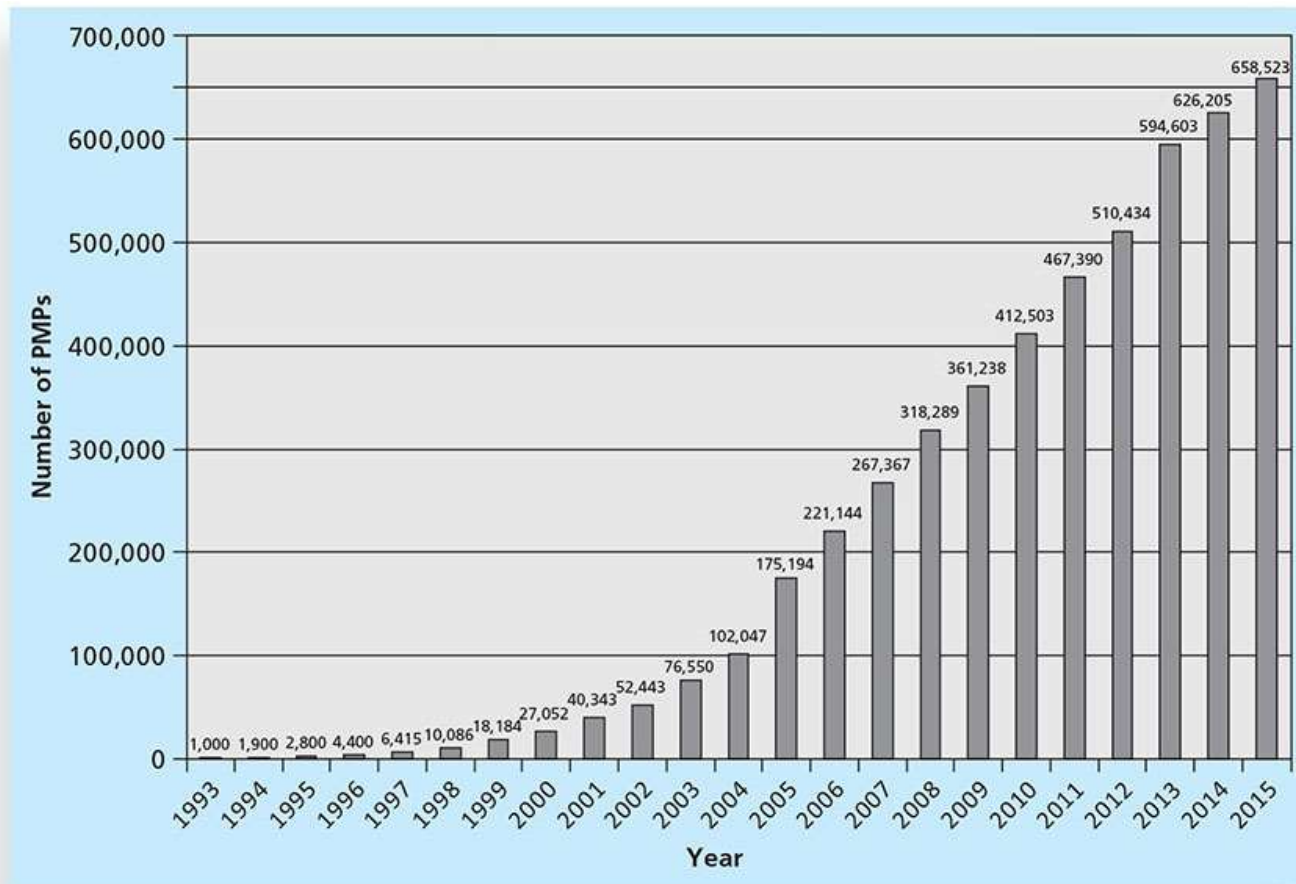
The Project Management Institute

- ▶ The Project Management Institute (PMI) is an international professional society for project managers founded in 1969
- ▶ PMI has continued to attract and retain members, reporting more than 449,000 members worldwide by late 2014
- ▶ There are communities of practices in many areas, like information systems, financial services, and health care
- ▶ Project management research and certification programs continue to grow
- ▶ Students can join PMI at a reduced fee and earn the Certified Associate in Project Management (CAPM) certification(see www.pmi.org for details)

Project Management Certification

- ▶ PMI provides certification as a **Project Management Professional (PMP)**
- ▶ A PMP has documented sufficient project experience, agreed to follow a code of ethics, and passed the PMP exam
- ▶ The number of people earning PMP certification is increasing quickly

Figure 1-9 Growth in PMP Certification, 1993-2014



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Ethics in Project Management

- ▶ **Ethics**, loosely defined, is a set of principles that guide our decision making based on personal values of what is “right” and “wrong”
- ▶ Project managers often face ethical dilemmas
- ▶ In order to earn PMP certification, applicants must agree to PMI’s Code of Ethics and Professional Conduct
- ▶ Several questions on the PMP exam are related to professional responsibility, including ethics

Project Management Software

- ▶ There are hundreds of different products to assist in performing project management
- ▶ Three main categories of tools:
 - Low-end tools: Handle single or smaller projects well, cost under \$200 per user
 - Midrange tools: Handle multiple projects and users, cost \$200-\$1,000 per user, Project 2013 most popular
 - High-end tools: Also called enterprise project management software, often licensed on a per-user basis
- ▶ Several free or open-source tools are also available

Chapter Summary

- ▶ A project is a temporary endeavor undertaken to create a unique product, service, or result
- ▶ Project management is the application of knowledge, skills, tools, and techniques to project activities to meet project requirements
- ▶ A program is a group of related projects managed in a coordinated way
- ▶ Project portfolio management involves organizing and managing projects and programs as a portfolio of investments
- ▶ Project managers play a key role in helping projects and organizations succeed
- ▶ The project management profession continues to grow and mature

Chapter 5: Project Scope Management

Information Technology Project Management, Eighth Edition

Note: See the text itself for full citations.



Learning Objectives, Part 1

- ▶ Understand the importance of good project scope management
- ▶ Describe the process of planning scope management
- ▶ Discuss methods for collecting and documenting requirements to meet stakeholder needs and expectations
- ▶ Explain the scope definition process and describe the contents of a project scope statement
- ▶ Discuss the process for creating a work breakdown structure using the analogy, top-down, bottom-up, and mind-mapping approaches

Learning Objectives, Part 2

- ▶ Explain the importance of validating scope and how it relates to defining and controlling scope
- ▶ Understand the importance of controlling scope and approaches for preventing scope-related problems on information technology (IT) projects
- ▶ Describe how software can assist in project scope management

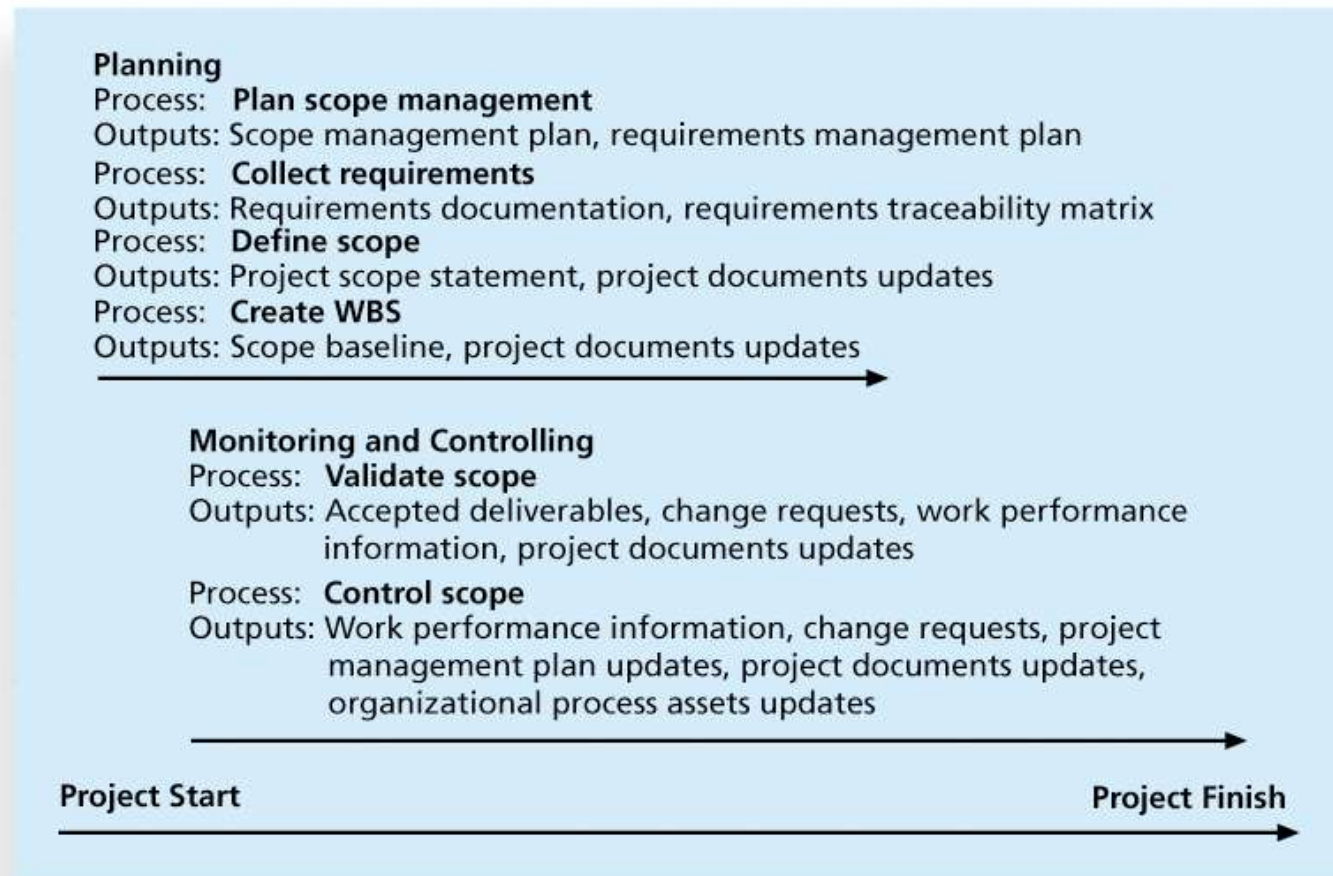
What is Project Scope Management?

- ▶ **Scope** refers to *all* the work involved in creating the products of the project and the processes used to create them
- ▶ A **deliverable** is a product produced as part of a project, such as hardware or software, planning documents, or meeting minutes
- ▶ Project scope management includes the processes involved in defining and controlling what is or is not included in a project

Project Scope Management Processes

- ▶ **Planning scope:** determining how the project's scope and requirements will be managed
- ▶ **Collecting requirements:** defining and documenting the features and functions of the products produced during the project as well as the processes used for creating them
- ▶ **Defining scope:** reviewing the project charter, requirements documents, and organizational process assets to create a scope statement
- ▶ **Creating the WBS:** subdividing the major project deliverables into smaller, more manageable components
- ▶ **Validating scope:** formalizing acceptance of the project deliverables
- ▶ **Controlling scope:** controlling changes to project scope throughout the life of the project

Figure 5-1. Project Scope Management Summary



Planning Scope Management

- ▶ The project team uses expert judgment and meetings to develop two important outputs: the scope management plan and the requirements management plan
- ▶ The scope management plan is a subsidiary part of the project management plan

Scope Management Plan Contents

- ▶ How to prepare a detailed project scope statement
- ▶ How to create a WBS
- ▶ How to maintain and approve the WBS
- ▶ How to obtain formal acceptance of the completed project deliverables
- ▶ How to control requests for changes to the project scope

Requirements Management Plan

- ▶ The PMBOK® Guide, Fifth Edition, describes requirements as “conditions or capabilities that must be met by the project or present in the product, service, or result to satisfy an agreement or other formally imposed specification”
- ▶ The **requirements management plan** documents how project requirements will be analyzed, documented, and managed

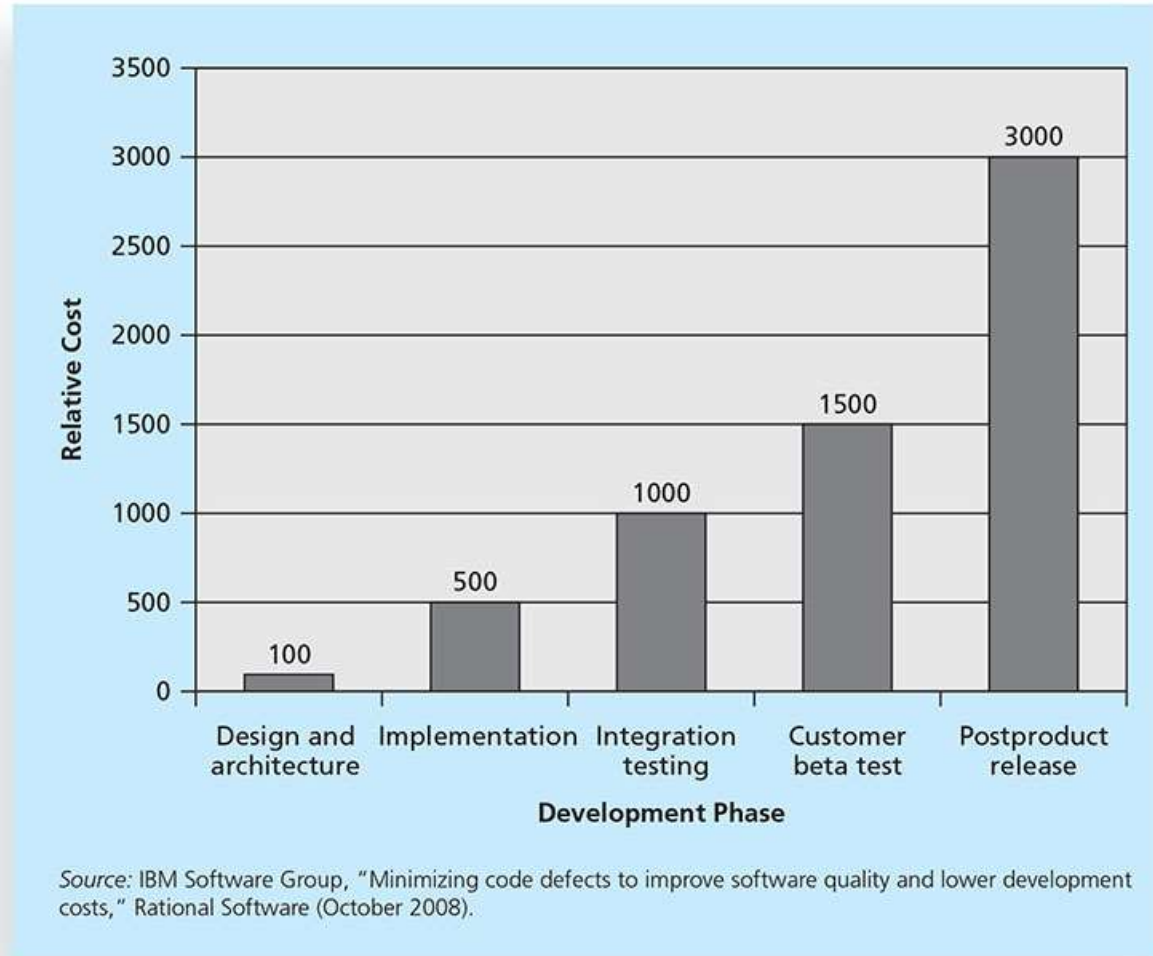
What Went Right?

- ▶ Projected number of jobs for business analysts expected to increase 19 percent by 2022
- ▶ Only 49 percent of survey respondents had the resources in place to do requirements management properly and 53 percent failed to use a formal process to validate requirements
- ▶ There are several certifications available for business analysis to help meet this need

Collecting Requirements

- ▶ For some IT projects, it is helpful to divide requirements development into categories called elicitation, analysis, specification, and validation
- ▶ It is important to use an iterative approach to defining requirements since they are often unclear early in a project

Figure 5-2. Relative Cost to Correct a Software Requirement Defect



Best Practice

- ▶ Book called “How Google Tests Software” describes how Google changed their culture as quality rests on the shoulders of those writing the code; they don’t rely on testers to ensure quality
- ▶ Google also does not believe in fads of buzzwords, including Agile

Methods for Collecting Requirements

- ▶ Interviewing
- ▶ Focus groups and facilitated workshops
- ▶ Using group creativity and decision-making techniques
- ▶ Questionnaires and surveys
- ▶ Observation
- ▶ Prototyping
- ▶ **Benchmarking**, or generating ideas by comparing specific project practices or product characteristics to those of other projects or products inside or outside the performing organization, can also be used to collect requirements

Statistics on Requirements for Software Projects (2011 Survey)*

- ▶ Eighty-eight percent of the software projects involved enhancing existing products instead of creating new ones
- ▶ Eighty-six percent of respondents said that customer satisfaction was the most important metric for measuring the success of development projects
- ▶ Eighty-three percent of software development teams still use Microsoft Office applications such as Word and Excel as their main tools to communicate requirements

*John Simpson, "2011: The State of Requirements Management" (2011).

Requirements Traceability Matrix

- ▶ A **requirements traceability matrix (RTM)** is a table that lists requirements, various attributes of each requirement, and the status of the requirements to ensure that all requirements are addressed
- ▶ Table 5-1. Sample entry in an RTM

Requirement No.	Name	Category	Source	Status
R32	Laptop memory	Hardware	Project charter and corporate laptop specifications	Complete. Laptops ordered meet requirement by having 4GB of memory.

Defining Scope

- ▶ **Project scope statements** should include at least a product scope description, product user acceptance criteria, and detailed information on all project deliverables. It is also helpful to document other scope-related information, such as the project boundaries, constraints, and assumptions. The project scope statement should also reference supporting documents, such as product specifications
- ▶ As time progresses, the scope of a project should become more clear and specific

Table 5-2. Sample Project Charter (partial)

Project Title: Information Technology (IT) Upgrade project

Project Start Date: March 4

Project Finish Date: December 4

Key Schedule Milestones:

- Inventory update completed April 15
- Hardware and software acquired August 1
- Installation complete October 1
- Testing complete November 15

Budget information: Budgeted \$1,000,000 for hardware and software costs and \$500,000 for labor costs.

Manager: Kim Nguyen, (310) 555- 2784, knguyen@course.com

Project objectives: Upgrade hardware and software for all employees (approximately 2,000) within nine months based on new corporate standards. See attached sheet describing the new standards. Upgrades may affect servers as well as associated network hardware and software.

Main Project Success Criteria: The hardware, software, and network upgrades must meet all written specifications, be thoroughly tested, and be completed in nine months. Employee work disruptions will be minimal.

Approach:

- Update the IT inventory database to determine upgrade needs.
- Develop detailed cost estimate for project and report to CIO
- Issue a request for quote to obtain hardware and software
- Use internal staff as much as possible for planning, analysis, and installation

Table 5-3: Further Defining Project Scope

Project Charter:

Upgrades may affect servers...(listed under Project Objective)

Project Scope Statement, Version 1:

Servers: If additional servers are required to support this project, they must be compatible with existing servers. If it is more economical to enhance existing servers, a detailed description of enhancements must be submitted to the CIO for approval. See current server specifications provided in Attachment 6. The CEO must approve a detailed plan describing the servers and their location at least 2 weeks before installation.

Project Scope Statement, Version 2

Servers: This project will required purchasing ten new servers to support Web, network, database, applications and printing functions. Virtualization will be used to maximize efficient. Detailed descriptions of the servers are provided in a product brochure in Appendix 8 along with a plan describing where they will be located.

Media Snapshot

- Inaccurate requirements gathering continues to be one of the main causes of project failure
- For every dollar spent on projects and programs, 5.1 percent is wasted due to poor requirements management
- Organizations need to develop people, processes, and culture to improve requirements management

Creating the Work Breakdown Structure (WBS)

- ▶ A **WBS** is a deliverable-oriented grouping of the work involved in a project that defines the total scope of the project
- ▶ WBS is a foundation document that provides the basis for planning and managing project schedules, costs, resources, and changes
- ▶ **Decomposition** is subdividing project deliverables into smaller pieces
- ▶ A **work package** is a task at the lowest level of the WBS
- ▶ The **scope baseline** includes the approved project scope statement and its associated WBS and WBS dictionary

Figure 5-3. Sample Intranet WBS Organized by Product

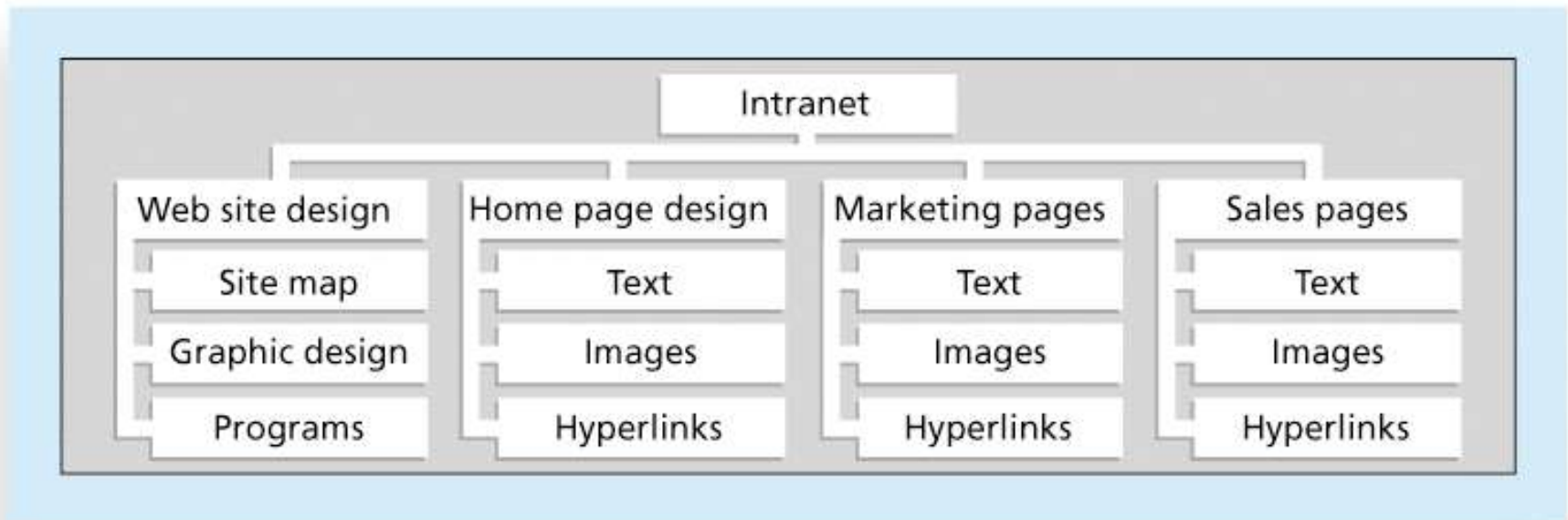


Figure 5-4. Sample Intranet WBS Organized by Phase

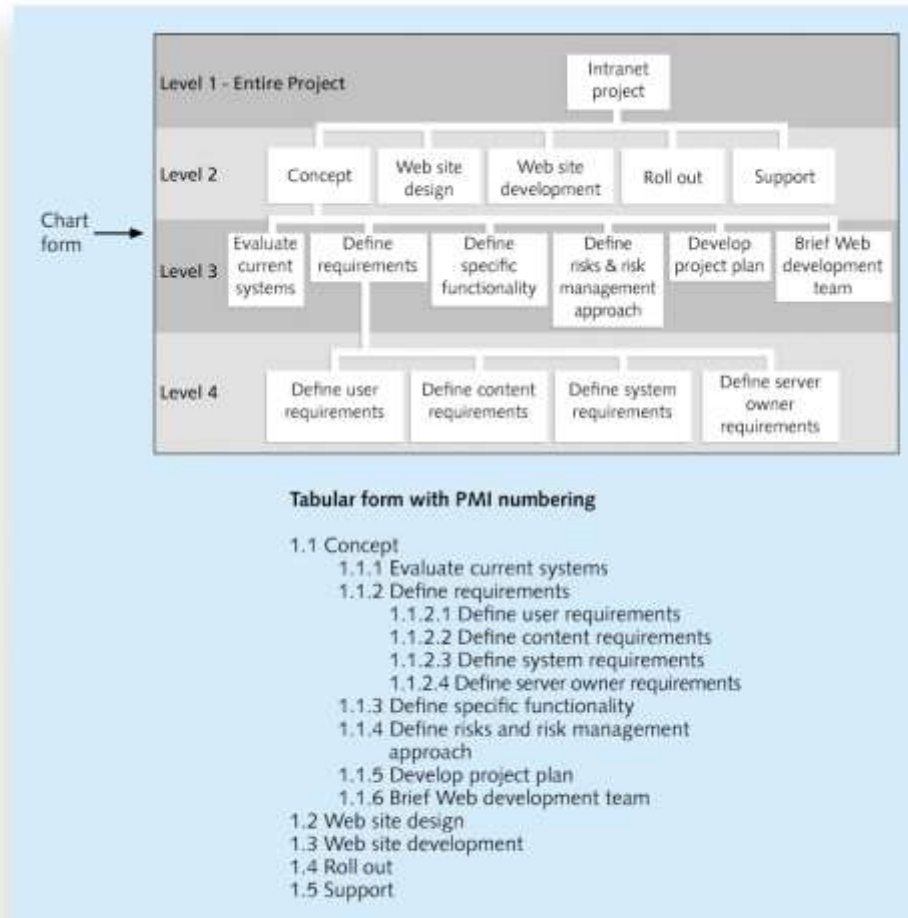


Figure 5-5. Intranet WBS and Gantt Chart in Microsoft Project

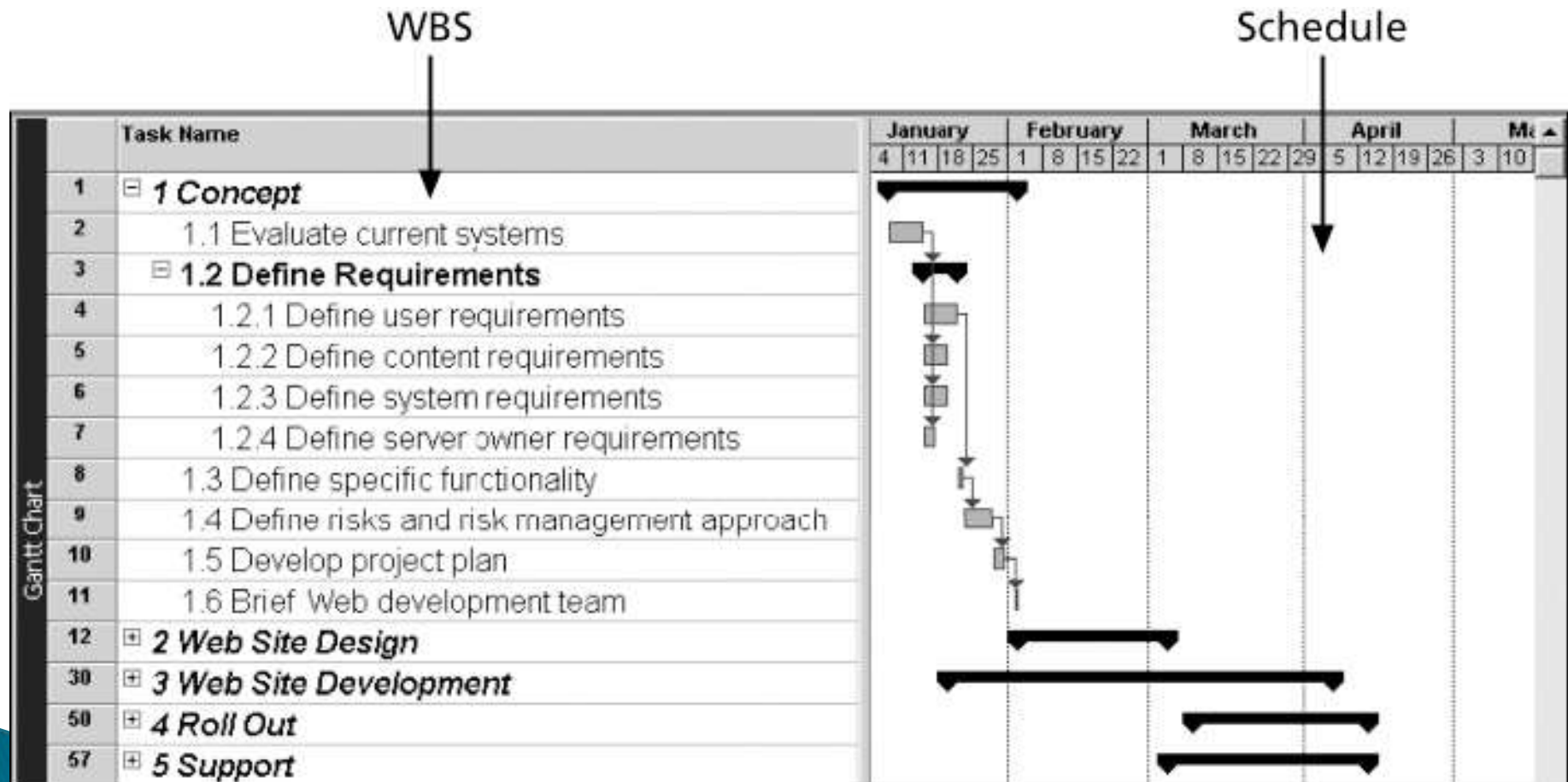


Figure 5-6. Intranet Gantt Chart Organized by Project Management Process Groups

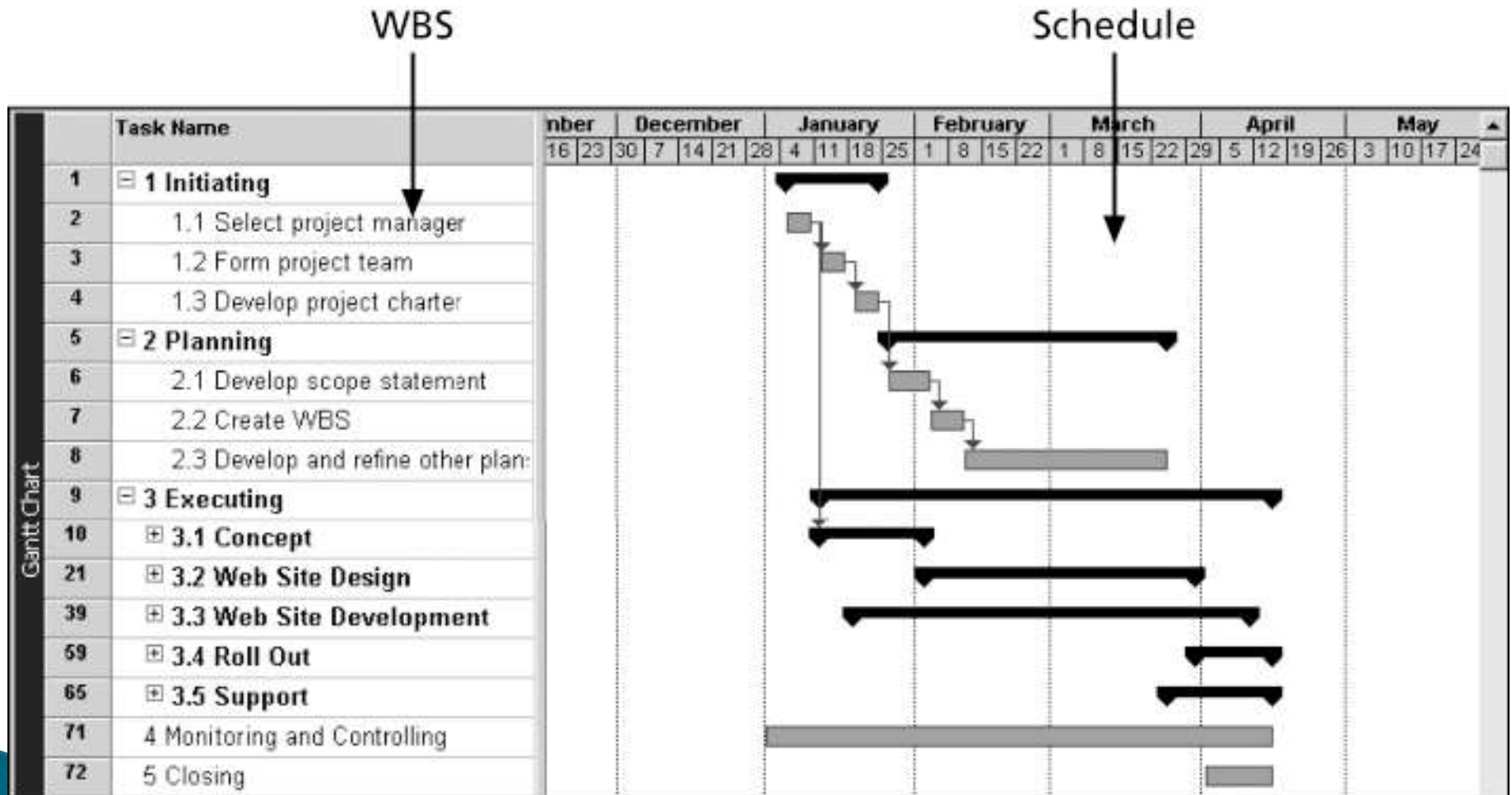


Table 5-4: Executing Tasks for JWD Consulting's WBS

3.0 Executing

3.1 Survey

3.2 User inputs

3.3 Intranet site content

3.3.1 Templates and Tools

3.3.2 Articles

3.3.3 Links

3.3.4 Ask the Expert

3.3.5 User requests feature

3.4 Intranet site design

3.5 Intranet site construction

3.6 Site testing

3.7 Site promotion

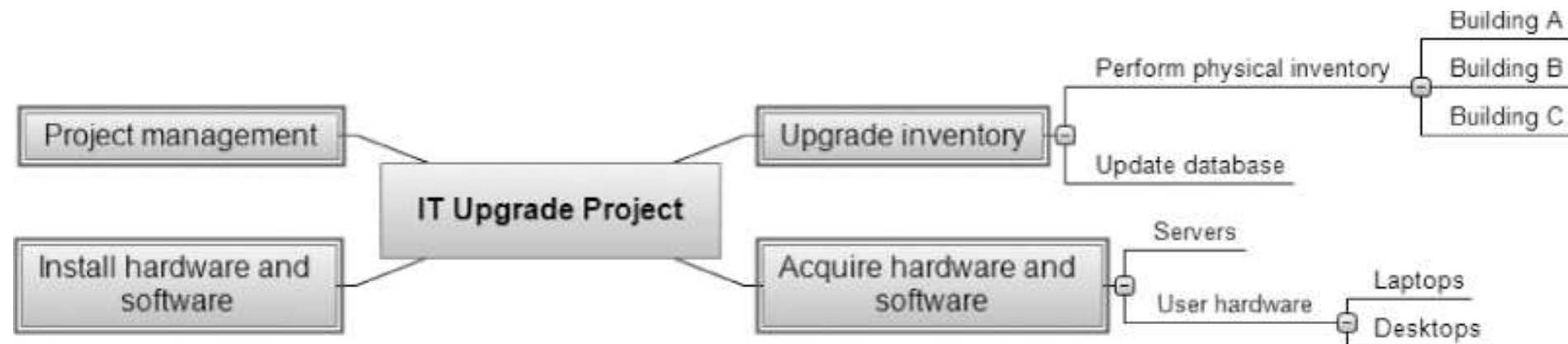
3.8 Site roll out

3.9 Project benefits measurement

Approaches to Developing WBSs

- ▶ Using guidelines: Some organizations, like the DOD, provide guidelines for preparing WBSs
- ▶ The **analogy approach**: Review WBSs of similar projects and tailor to your project
- ▶ The **top-down approach**: Start with the largest items of the project and break them down
- ▶ The **bottom-up approach**: Start with the specific tasks and roll them up
- ▶ Mind-mapping approach: **Mind mapping** is a technique that uses branches radiating out from a core idea to structure thoughts and ideas

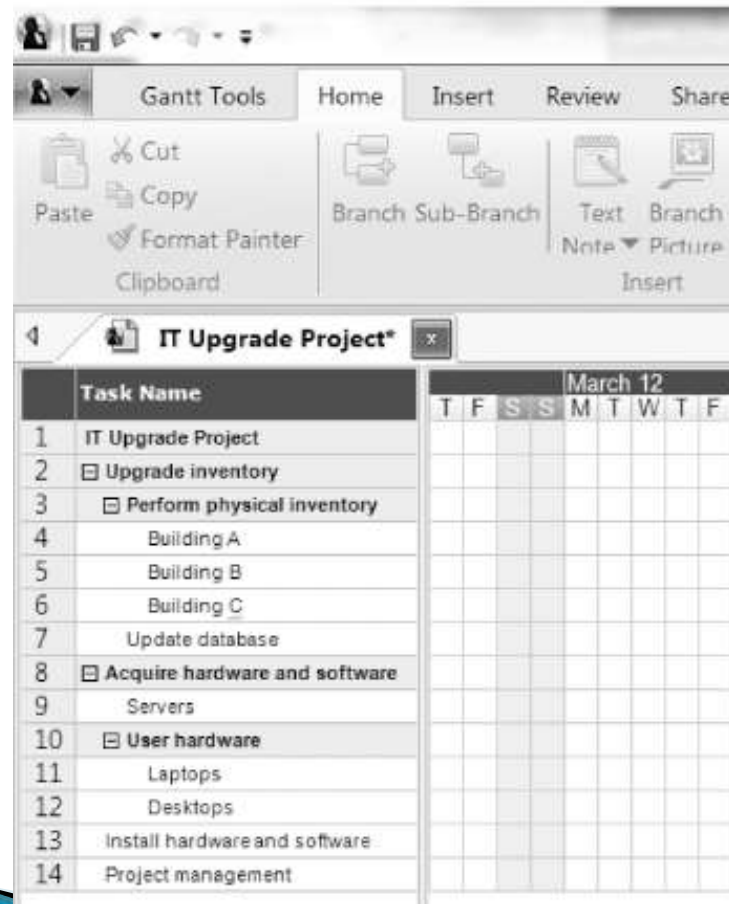
Figure 5-7. Sample Mind-Mapping Approach for Creating a WBS



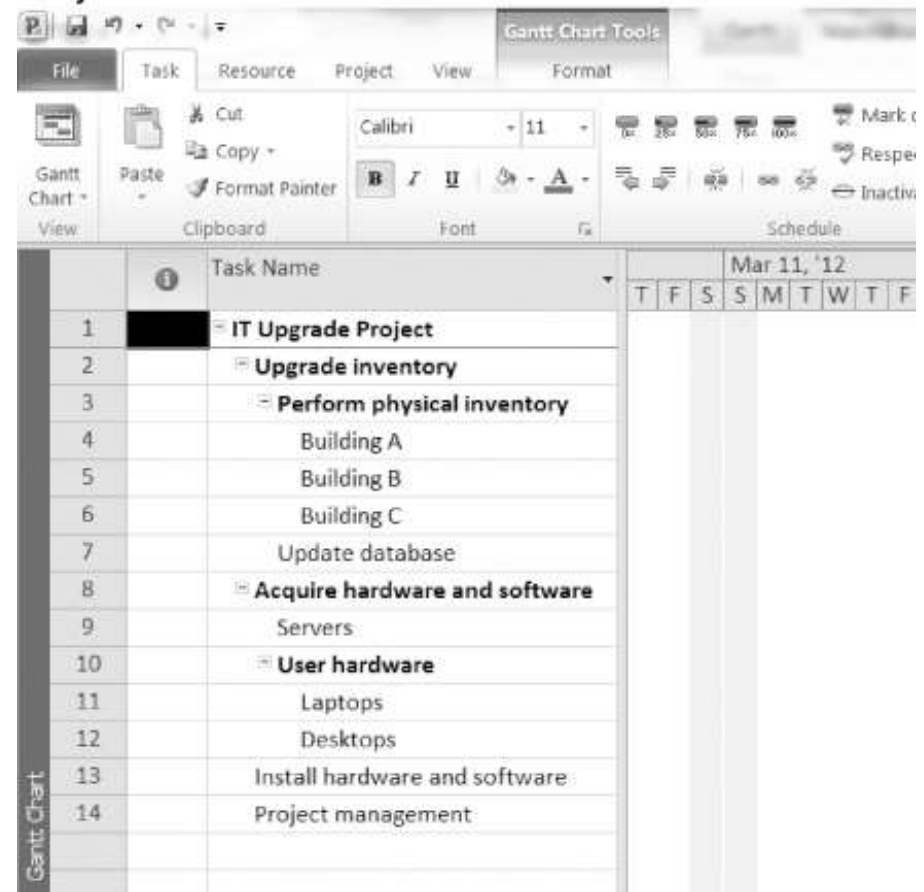
Source: MatchWare's MindView 4 Business Edition

Figure 5-8. Gantt Charts With WBS Generated From a Mind Map

MindView 4.0 Gantt Chart



Project 2010 Gantt Chart



The WBS Dictionary and Scope Baseline

- ▶ Many WBS tasks are vague and must be explained more so people know what to do and can estimate how long it will take and what it will cost to do the work
- ▶ A **WBS dictionary** is a document that describes detailed information about each WBS item

Table 5-5. Sample WBS Dictionary Entry

WBS Dictionary Entry March 20

Project Title: Information technology (IT) Upgrade Project

WBS Item Number: 2.2

WBS Item Name: Update Database

Description: The IT department maintains an online database of hardware and software on the corporate intranet. However, we need to make sure that we know exactly what hardware and software employees are currently using and if they have any unique needs before we decide what to order for the upgrade. This task will involve reviewing information from the current database, producing reports that list each department's employees and location, and updating the data after performing the physical inventory and receiving inputs from department managers. Our project sponsor will send a notice to all department managers to communicate the importance of this project and this particular task. In addition to general hardware and software upgrades, the project sponsors will ask the department managers to provide information for any unique requirements they might have that could affect the upgrades. This task also includes updating the inventory data for network hardware and software. After updating the inventory database, we will send an e-mail to each department manager to verify the information and make changes online as needed. Department managers will be responsible for ensuring that their people are available and cooperative during the physical inventory. Completing this task is dependent on WBS Item Number 2.1, Perform Physical Inventory, and must precede WBS Item Number 3.0, Acquire Hardware and Software.

Advice for Creating a WBS and WBS Dictionary, Part 1

- ▶ A unit of work should appear at only one place in the WBS.
- ▶ The work content of a WBS item is the sum of the WBS items below it
- ▶ A WBS item is the responsibility of only one individual, even though many people may be working on it
- ▶ The WBS must be consistent with the way in which work is actually going to be performed; it should serve the project team first, and other purposes only if practical

Advice for Creating a WBS and WBS Dictionary, Part 2

- ▶ Project team members should be involved in developing the WBS to ensure consistency and buy-in
- ▶ Each WBS item must be documented in a WBS dictionary to ensure accurate understanding of the scope of work included and not included in that item
- ▶ The WBS must be a flexible tool to accommodate inevitable changes while properly maintaining control of the work content in the project according to the scope statement

What Went Wrong?

- ▶ A project scope that is too broad and grandiose can cause severe problems
 - Scope creep and an overemphasis on technology for technology's sake resulted in the bankruptcy of a large pharmaceutical firm, Texas-based FoxMeyer Drug
 - In 2001, McDonald's fast-food chain initiated a project to create an intranet that would connect its headquarters with all of its restaurants to provide detailed operational information in real time. After spending \$170 million on consultants and initial implementation planning, McDonald's realized that the project was too much to handle and terminated it

Validating Scope

- ▶ It is very difficult to create a good scope statement and WBS for a project
- ▶ It is even more difficult to verify project scope and minimize scope changes
- ▶ **Scope validation** involves formal acceptance of the completed project deliverables
- ▶ Acceptance is often achieved by a customer inspection and then sign-off on key deliverables

Global Issues

- ▶ Many countries have had difficulties controlling the scope of large projects, especially those that involve advanced technologies and many different users
- ▶ For example, the state government of Victoria, Australia, has a Web site for its public transportation smart card at www.myki.com.au.
- ▶ There were many problems in developing and implementing the smart card

Controlling Scope

- ▶ Scope control involves controlling changes to the project scope
- ▶ Goals of scope control are to
 - influence the factors that cause scope changes
 - assure changes are processed according to procedures developed as part of integrated change control, and
 - manage changes when they occur
- ▶ **Variance** is the difference between planned and actual performance

Suggestions for Improving User Input

- ▶ Develop a good project selection process and insist that sponsors are from the user organization
- ▶ Have users on the project team in important roles
- ▶ Have regular meetings with defined agendas, and have users sign off on key deliverables presented at meetings
- ▶ Deliver something to users and sponsors on a regular basis
- ▶ Don't promise to deliver when you know you can't
- ▶ Co-locate users with developers

Suggestions for Reducing Incomplete and Changing Requirements, Part 1

- ▶ Develop and follow a requirements management process
- ▶ Use techniques such as prototyping, use case modeling, and JAD to get more user involvement
- ▶ Put requirements in writing and keep them current
- ▶ Create a requirements management database for documenting and controlling requirements

Suggestions for Reducing Incomplete and Changing Requirements, Part 2

- ▶ Provide adequate testing and conduct testing throughout the project life cycle
- ▶ Review changes from a systems perspective
- ▶ Emphasize completion dates to help focus on what's most important
- ▶ Allocate resources specifically for handling change requests/enhancements like NWA did with ResNet

Using Software to Assist in Project Scope Management

- ▶ Word-processing software helps create several scope-related documents
- ▶ Spreadsheets help to perform financial calculations, weighed scoring models, and develop charts and graphs
- ▶ Communication software like e-mail and the Web help clarify and communicate scope information
- ▶ Project management software helps in creating a WBS, the basis for tasks on a Gantt chart
- ▶ Specialized software is available to assist in project scope management

Chapter Summary

- ▶ Project scope management includes the processes required to ensure that the project addresses all the work required, and only the work required, to complete the project successfully
- ▶ Main processes include
 - Define scope management
 - Collect requirements
 - Define scope
 - Create WBS
 - Validate scope
 - Control scope

Chapter 6: Project Time Management

Information Technology Project Management, Eighth Edition

Note: See the text itself for full citations.



Learning Objectives, Part 1

- ▶ Understand the importance of project schedules and good project time management
- ▶ Discuss the process of planning schedule management
- ▶ Define activities as the basis for developing project schedules
- ▶ Describe how project managers use network diagrams and dependencies to assist in activity sequencing
- ▶ Understand the relationship between estimating resources and project schedules
- ▶ Explain how various tools and techniques help project managers perform activity duration estimates

Learning Objectives, Part 2

- ▶ Use a Gantt chart for planning and tracking schedule information, find the critical path for a project, and describe how critical chain scheduling and the Program Evaluation and Review Technique (PERT) affect schedule development
- ▶ Understand how time management is addressed using Agile
- ▶ Discuss how reality checks and discipline are involved in controlling and managing changes to the project schedule
- ▶ Describe how project management software can assist in project time management and review words of caution before using this software

Importance of Project Schedules

- ▶ Managers often cite delivering projects on time as one of their biggest challenges
- ▶ Time has the least amount of flexibility; it passes no matter what happens on a project
- ▶ Schedule issues are the main reason for conflicts on projects, especially during the second half of projects

Individual Work Styles and Cultural Differences Cause Schedule Conflicts

- ▶ One dimension of the Meyers-Briggs Type Indicator focuses on peoples' attitudes toward structure and deadline
- ▶ Some people prefer to follow schedules and meet deadlines while others do not (J vs. P)
- ▶ Different cultures and even entire countries have different attitudes about schedules

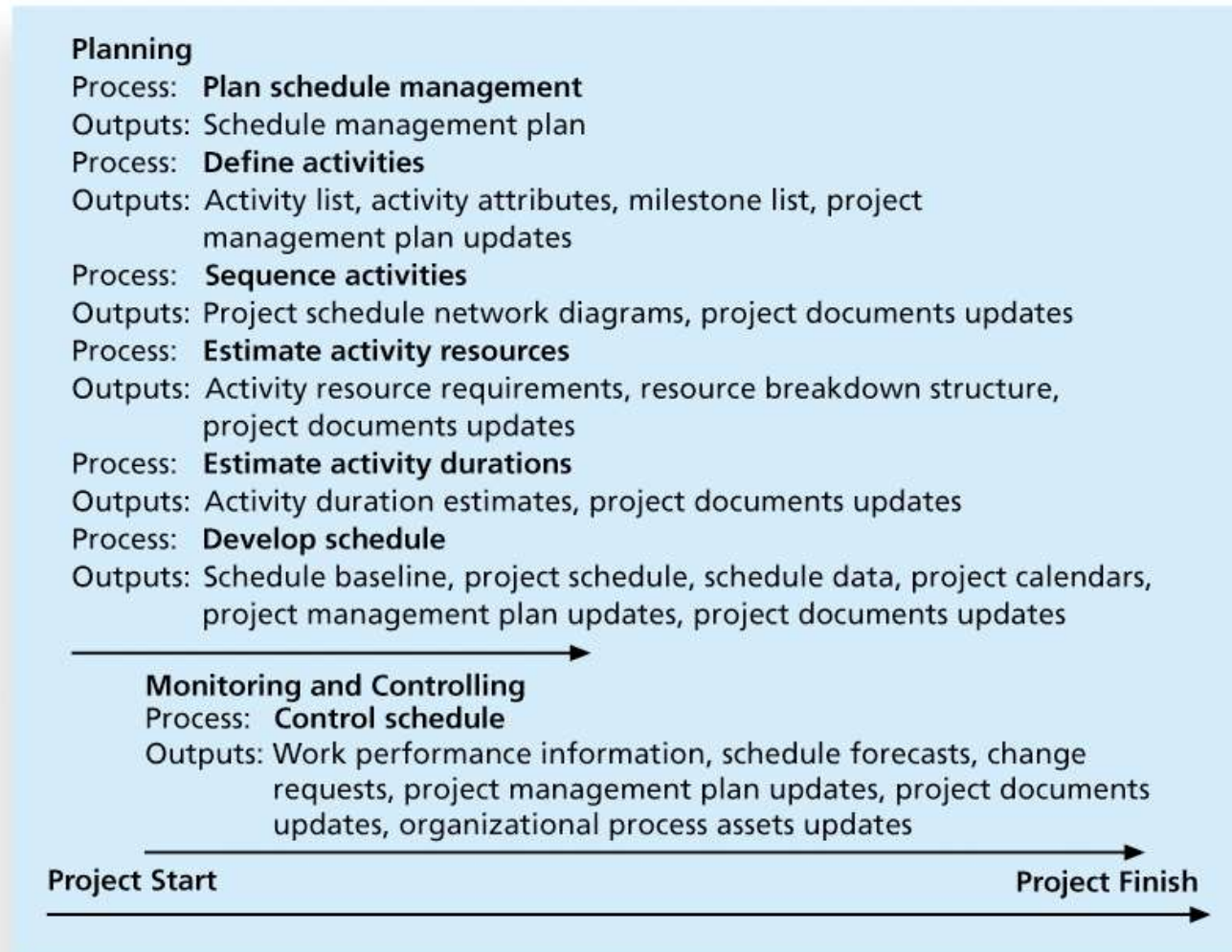
Media Snapshot

- In contrast to the 2002 Salt Lake City Winter Olympic Games (see Chapter 4's Media Snapshot), planning and scheduling was very different for the 2004 Athens Summer Olympic Games and the 2014 Sochi Winter Olympic Games
- Many articles were written before the opening ceremonies in Athens predicting that the facilities would not be ready in time. ...the Greeks even made fun of critics by having construction workers pretend to still be working as the ceremonies began...but the games cost more than twice the planned budget
- The 2014 Winter Olympic Games in Sochi, Russia, suffered even greater financial loss and were the most expensive games in history

Project Time Management Processes

- ▶ **Planning schedule management:** determining the policies, procedures, and documentation that will be used for planning, executing, and controlling the project schedule
- ▶ **Defining activities:** identifying the specific activities that the project team members and stakeholders must perform to produce the project deliverables
- ▶ **Sequencing activities:** identifying and documenting the relationships between project activities
- ▶ **Estimating activity resources:** estimating how many **resources** a project team should use to perform project activities
- ▶ **Estimating activity durations:** estimating the number of work periods that are needed to complete individual activities
- ▶ **Developing the schedule:** analyzing activity sequences, activity resource estimates, and activity duration estimates to create the project schedule
- ▶ **Controlling the schedule:** controlling and managing changes to the project schedule

Figure 6-1. Project Time Management Summary



Planning Schedule Management

- ▶ The project team uses expert judgment, analytical techniques, and meetings to develop the schedule management plan
- ▶ A schedule management plan includes:
 - Project schedule model development
 - The scheduling methodology
 - Level of accuracy and units of measure
 - Control thresholds
 - Rules of performance measurement
 - Reporting formats
 - Process descriptions

Defining Activities

- ▶ An **activity** or **task** is an element of work normally found on the work breakdown structure (WBS) that has an expected duration, a cost, and resource requirements
- ▶ Activity definition involves developing a more detailed WBS and supporting explanations to understand all the work to be done so you can develop realistic cost and duration estimates

Activity Lists and Attributes

- ▶ An **activity list** is a tabulation of activities to be included on a project schedule that includes
 - the activity name
 - an activity identifier or number
 - a brief description of the activity
- ▶ **Activity attributes** provide more information such as predecessors, successors, logical relationships, leads and lags, resource requirements, constraints, imposed dates, and assumptions related to the activity

Milestones

- ▶ A **milestone** is a significant event that normally has no duration
- ▶ It often takes several activities and a lot of work to complete a milestone
- ▶ They're useful tools for setting schedule goals and monitoring progress
- ▶ Examples include obtaining customer sign-off on key documents or completion of specific products

What Went Wrong?

- ▶ At the U.S. Federal Bureau of Investigation (FBI), poor time management was one of the reasons behind the failure of Trilogy, a “disastrous, unbelievably expensive piece of vaporware, which was more than four years in the (un)making. The system was supposed to enable FBI agents to integrate intelligence from isolated information silos within the Bureau.”*
- ▶ In May 2006, the Government Accounting Agency said that the Trilogy project failed at its core mission of improving the FBI’s investigative abilities and was plagued with missed milestones and escalating costs. Sentinel replaced Trilogy in 2007.
- ▶ During a test exercise in 2011, Sentinel experienced two outages, and the FBI determined that the current hardware structure was inadequate. In 2014, the system still wasn’t working well.

*Roberts, Paul, “Frustrated contractor sentenced for hacking FBI to speed deployment,” *InfoWorld Tech Watch*, (July 6, 2006).

Sequencing Activities

- ▶ Involves reviewing activities and determining dependencies
- ▶ A **dependency** or **relationship** is the sequencing of project activities or tasks
- ▶ You *must* determine dependencies in order to use critical path analysis

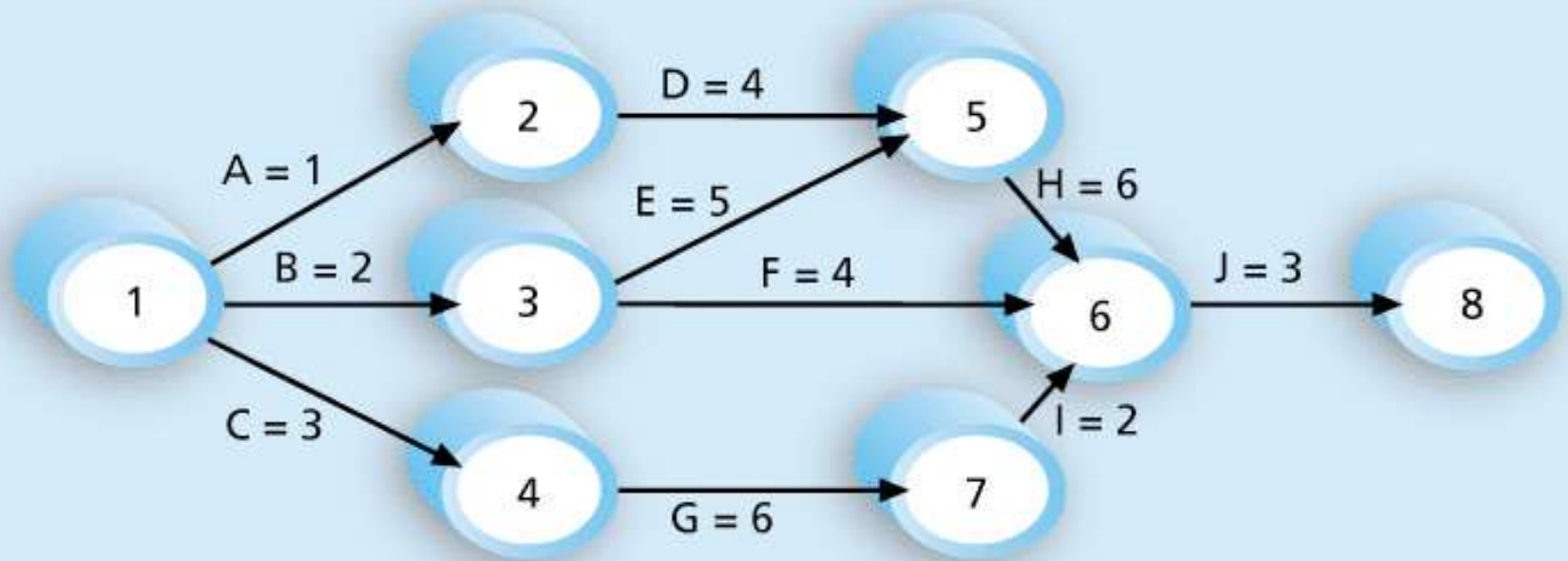
Three types of Dependencies

- ▶ **Mandatory dependencies:** inherent in the nature of the work being performed on a project, sometimes referred to as hard logic
- ▶ **Discretionary dependencies:** defined by the project team., sometimes referred to as soft logic and should be used with care since they may limit later scheduling options
- ▶ **External dependencies:** involve relationships between project and non-project activities

Network Diagrams

- ▶ Network diagrams are the preferred technique for showing activity sequencing
- ▶ A **network diagram** is a schematic display of the logical relationships among, or sequencing of, project activities
- ▶ Two main formats are the arrow and precedence diagramming methods

Figure 6-2. Network Diagram for Project X



Note: Assume all durations are in days; A=1 means Activity A has a duration of 1 day.

Arrow Diagramming Method (ADM)

- ▶ Also called activity-on-arrow (AOA) network diagrams
- ▶ Activities are represented by arrows
- ▶ Nodes or circles are the starting and ending points of activities
- ▶ Can only show finish-to-start dependencies

Process for Creating AOA Diagrams

1. Find all of the activities that start at node 1. Draw their finish nodes and draw arrows between node 1 and those finish nodes. Put the activity letter or name and duration estimate on the associated arrow
2. Continuing drawing the network diagram, working from left to right. Look for bursts and merges. **Bursts** occur when a single node is followed by two or more activities. A **merge** occurs when two or more nodes precede a single node
3. Continue drawing the project network diagram until all activities are included on the diagram that have dependencies
4. As a rule of thumb, all arrowheads should face toward the right, and no arrows should cross on an AOA network diagram

Precedence Diagramming Method (PDM)

- ▶ Activities are represented by boxes
- ▶ Arrows show relationships between activities
- ▶ More popular than ADM method and used by project management software
- ▶ Better at showing different types of dependencies

Figure 6-3. Task Dependency Types

Task dependencies

The nature of the relationship between two linked tasks. You link tasks by defining a dependency between their finish and start dates. For example, the “Contact caterers” task must finish before the start of the “Determine menus” task. There are four kinds of task dependencies in Microsoft Project.

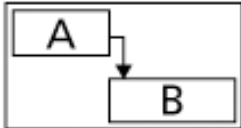
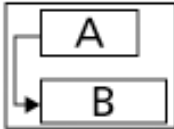
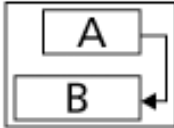
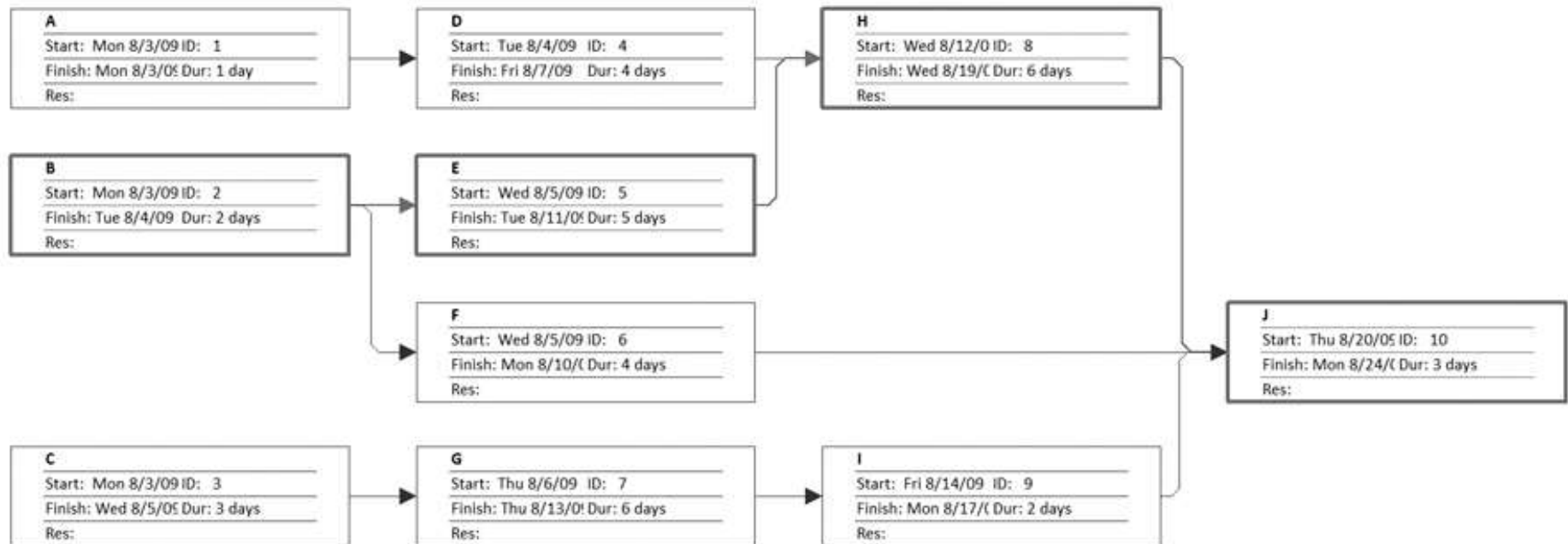
Task dependency	Example	Description
Finish-to-start (FS)		Task (B) cannot start until task (A) finishes.
Start-to-start (SS)		Task (B) cannot start until task (A) starts.
Finish-to-finish (FF)		Task (B) cannot finish until task (A) finishes.
Start-to-finish (SF)		Task (B) cannot finish until task (A) starts.

Figure 6-4. Sample PDM Network Diagram



Estimating Activity Resources

- ▶ Before estimating activity durations, you must have a good idea of the quantity and type of resources that will be assigned to each activity; **resources** are people, equipment, and materials
- ▶ Consider important issues in estimating resources
 - How difficult will it be to do specific activities on this project?
 - What is the organization's history in doing similar activities?
 - Are the required resources available?
- ▶ A **resource breakdown structure** is a hierarchical structure that identifies the project's resources by category and type

Activity Duration Estimating

- ▶ **Duration** includes the actual amount of time worked on an activity *plus* elapsed time
- ▶ **Effort** is the number of workdays or work hours required to complete a task
- ▶ Effort does not normally equal duration
- ▶ People doing the work should help create estimates, and an expert should review them

Three-Point Estimates

- ▶ Instead of providing activity estimates as a discrete number, such as four weeks, it's often helpful to create a **three-point estimate**
 - an estimate that includes an optimistic, most likely, and pessimistic estimate, such as three weeks for the optimistic, four weeks for the most likely, and five weeks for the pessimistic estimate
- ▶ Three-point estimates are needed for PERT and Monte Carlo simulations

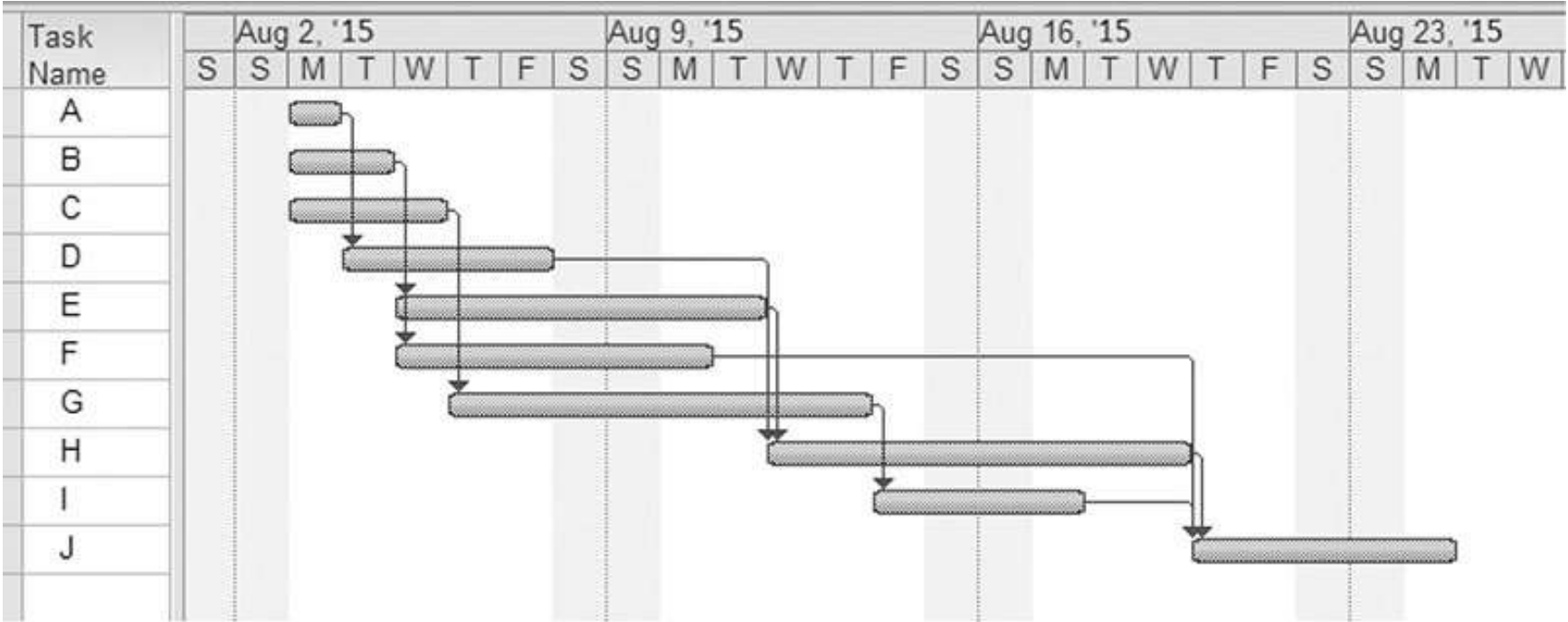
Developing the Schedule

- ▶ Uses results of the other time management processes to determine the start and end date of the project
- ▶ Ultimate goal is to create a realistic project schedule that provides a basis for monitoring project progress for the time dimension of the project
- ▶ Important tools and techniques include Gantt charts, critical path analysis, and critical chain scheduling, and PERT analysis

Gantt Charts

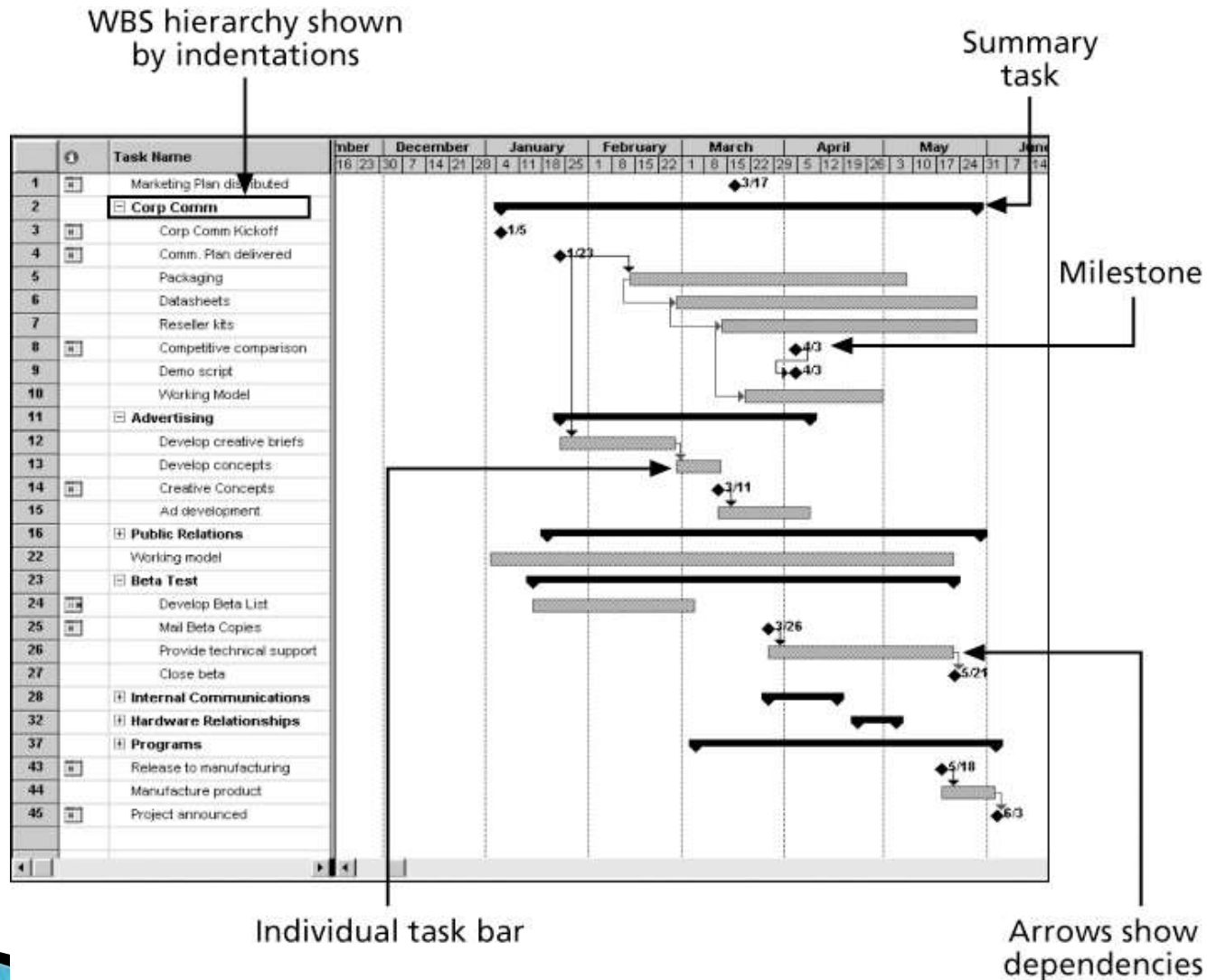
- ▶ **Gantt charts** provide a standard format for displaying project schedule information by listing project activities and their corresponding start and finish dates in a calendar format
- ▶ Symbols include:
 - A black diamond: a milestones
 - Thick black bars: summary tasks
 - Lighter horizontal bars: durations of tasks
 - Arrows: dependencies between tasks

Figure 6-5. Gantt Chart for Project X



Used with permission from Microsoft Corporation

Figure 6-6. Gantt Chart for Software Launch Project



Adding Milestones to Gantt Charts

- ▶ Many people like to focus on meeting milestones, especially for large projects
- ▶ Milestones emphasize important events or accomplishments on projects
- ▶ Normally create milestone by entering tasks with a zero duration, or you can mark any task as a milestone

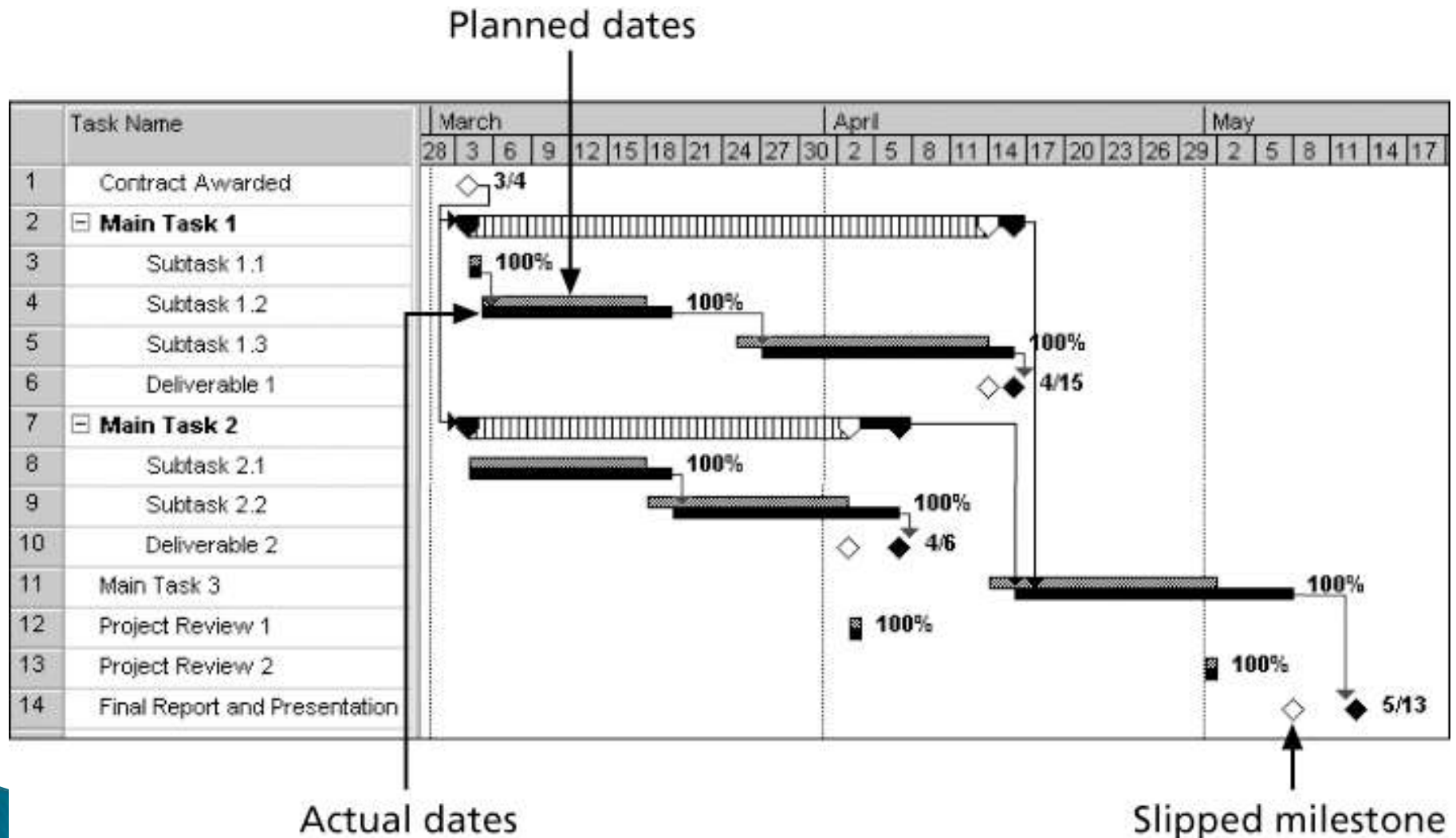
SMART Criteria

- ▶ Milestones should be
 - **S**pecific
 - **M**easurable
 - **A**ssignable
 - **R**ealistic
 - **T**ime-framed

Best Practice

- ▶ Shawn Achor suggests the 20-second rule in his book, *The Happiness Advantage*
- ▶ People prefer the path of least resistance. For example, if you have to wait in line 20 seconds to get a second scoop of ice cream, you might resist it
- ▶ Achor recommends making it more difficult for yourself to be distracted at work by keeping email or websites closed while you are working
- ▶ Save time by adding time to the distracting behaviors at work

Figure 6-7. Sample Tracking Gantt Chart



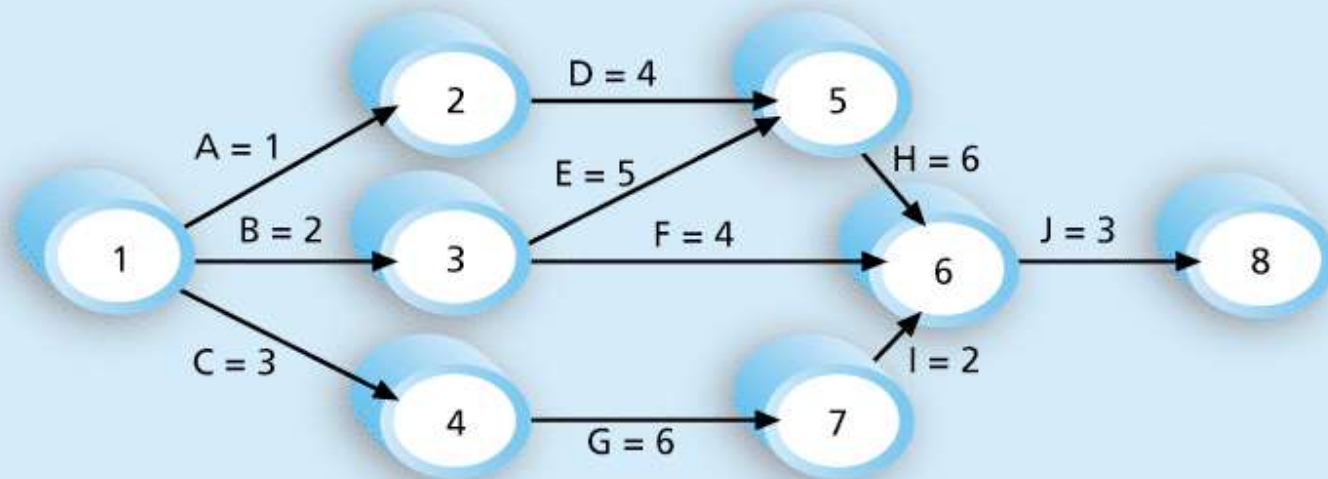
Critical Path Method (CPM)

- ▶ **CPM** is a network diagramming technique used to predict total project duration
- ▶ A **critical path** for a project is the series of activities that determines the *earliest time* by which the project can be completed
- ▶ The critical path is the *longest path* through the network diagram and has the least amount of slack or float
- ▶ **Slack** or **float** is the amount of time an activity may be delayed without delaying a succeeding activity or the project finish date

Calculating the Critical Path

- ▶ First develop a good network diagram
- ▶ Add the duration estimates for all activities on each path through the network diagram
- ▶ The longest path is the critical path
- ▶ If one or more of the activities on the critical path takes longer than planned, the whole project schedule will slip *unless* the project manager takes corrective action

Figure 6-8. Determining the Critical Path for Project X



Note: Assume all durations are in days.

Path 1: A-D-H-J Length = $1+4+6+3 = 14$ days

Path 2: B-E-H-J Length = $2+5+6+3 = 16$ days

Path 3: B-F-J Length = $2+4+3 = 9$ days

Path 4: C-G-I-J Length = $3+6+2+3 = 14$ days

Since the critical path is the longest path through the network diagram, Path 2, B-E-H-J, is the critical path for Project X.

More on the Critical Path

- ▶ A project team at Apple computer put a stuffed gorilla on the top of the cubicle of the person currently managing critical task
- ▶ The critical path is *not* the one with all the critical activities; it only accounts for time
 - Remember the example of **growing grass** being on the critical path for Disney's Animal Kingdom
- ▶ There can be more than one critical path if the lengths of two or more paths are the same
- ▶ The critical path can change as the project progresses

Using Critical Path Analysis to Make Schedule Trade-offs

- ▶ **Free slack** or **free float** is the amount of time an activity can be delayed without delaying the early start of any immediately following activities
- ▶ **Total slack** or **total float** is the amount of time an activity may be delayed from its early start without delaying the planned project finish date
- ▶ A **forward pass** through the network diagram determines the early start and finish dates
- ▶ A **backward pass** determines the late start and finish dates

Figure 6-9. Calculating Early and Late Start and Finish Dates

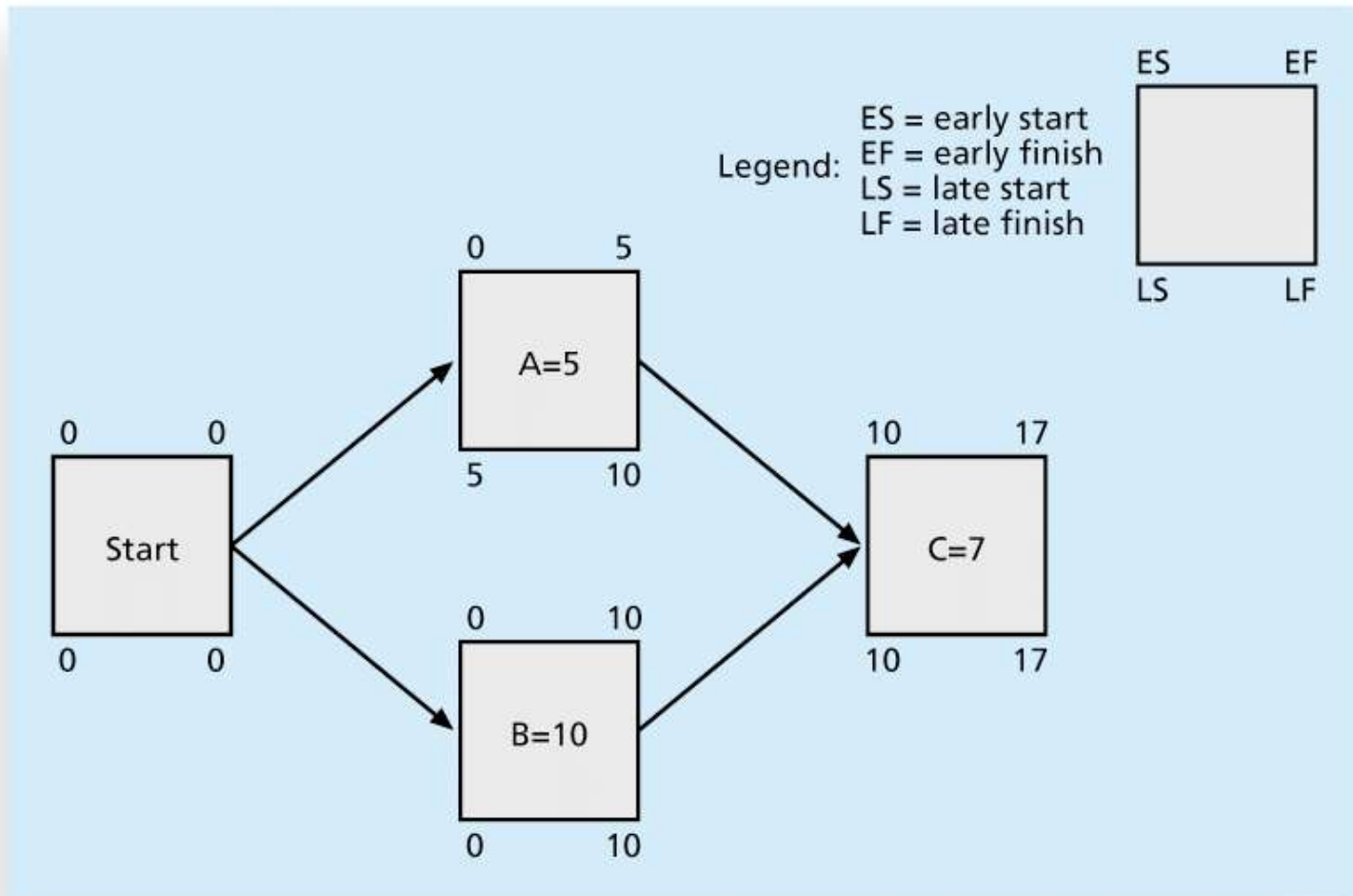


Table 6-1. Free and Total Float or Slack for Project X

Task Name	Start	Finish	Late Start	Late Finish	Free Slack	Total Slack
A	8/3/15	8/3/15	8/5/15	8/5/15	0d	2d
B	8/3/15	8/4/15	8/3/15	8/4/15	0d	0d
C	8/3/15	8/5/15	8/5/15	8/7/15	0d	2d
D	8/4/15	8/7/15	8/6/15	8/11/15	2d	2d
E	8/5/15	8/11/15	8/5/15	8/11/15	0d	0d
F	8/5/15	8/10/15	8/14/15	8/17/15	7d	7d
G	8/6/15	8/13/15	8/10/15	8/17/15	0d	2d
H	8/12/15	8/19/15	8/12/15	8/19/15	0d	0d
I	8/14/15	8/17/15	8/18/15	8/19/15	2d	2d
J	8/20/15	8/24/15	8/20/15	8/24/15	0d	0d

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Using the Critical Path to Shorten a Project Schedule

- ▶ Three main techniques for shortening schedules
 - Shortening durations of critical activities/tasks by adding more resources or changing their scope
 - **Crashing** activities by obtaining the greatest amount of schedule compression for the least incremental cost
 - **Fast tracking** activities by doing them in parallel or overlapping them

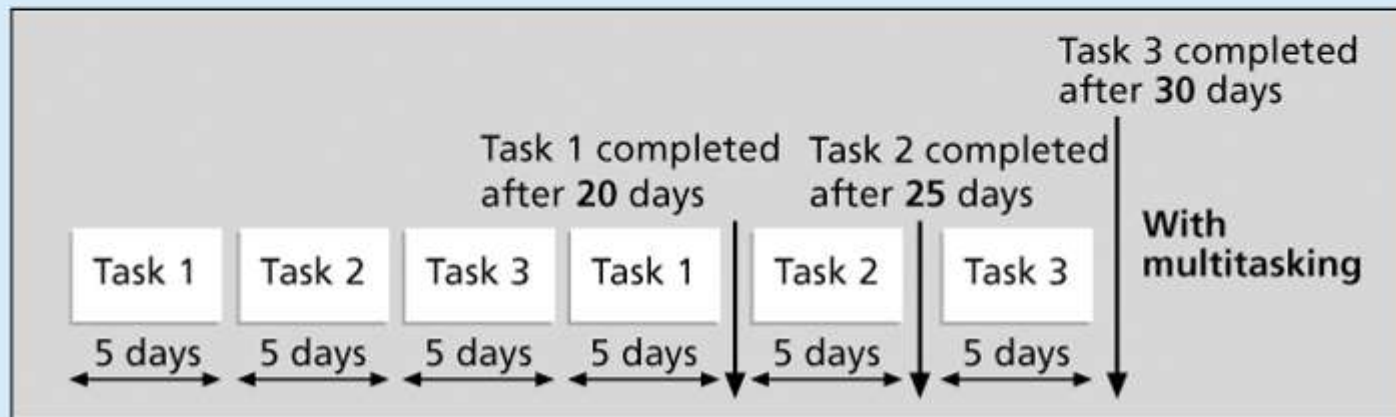
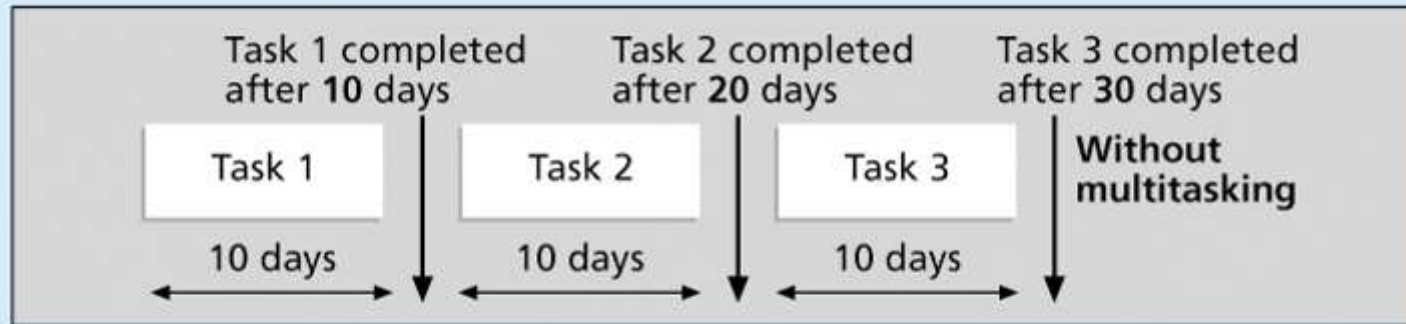
Importance of Updating Critical Path Data

- ▶ It is important to update project schedule information to meet time goals for a project
- ▶ The critical path may change as you enter actual start and finish dates
- ▶ If you know the project completion date will slip, negotiate with the project sponsor

Critical Chain Scheduling

- ▶ **Critical chain scheduling**
 - a method of scheduling that considers limited resources when creating a project schedule and includes buffers to protect the project completion date
- ▶ Uses the **Theory of Constraints (TOC)**
 - a management philosophy developed by Eliyahu M. Goldratt and introduced in his book *The Goal*.
- ▶ Attempts to minimize **multitasking**
 - when a resource works on more than one task at a time

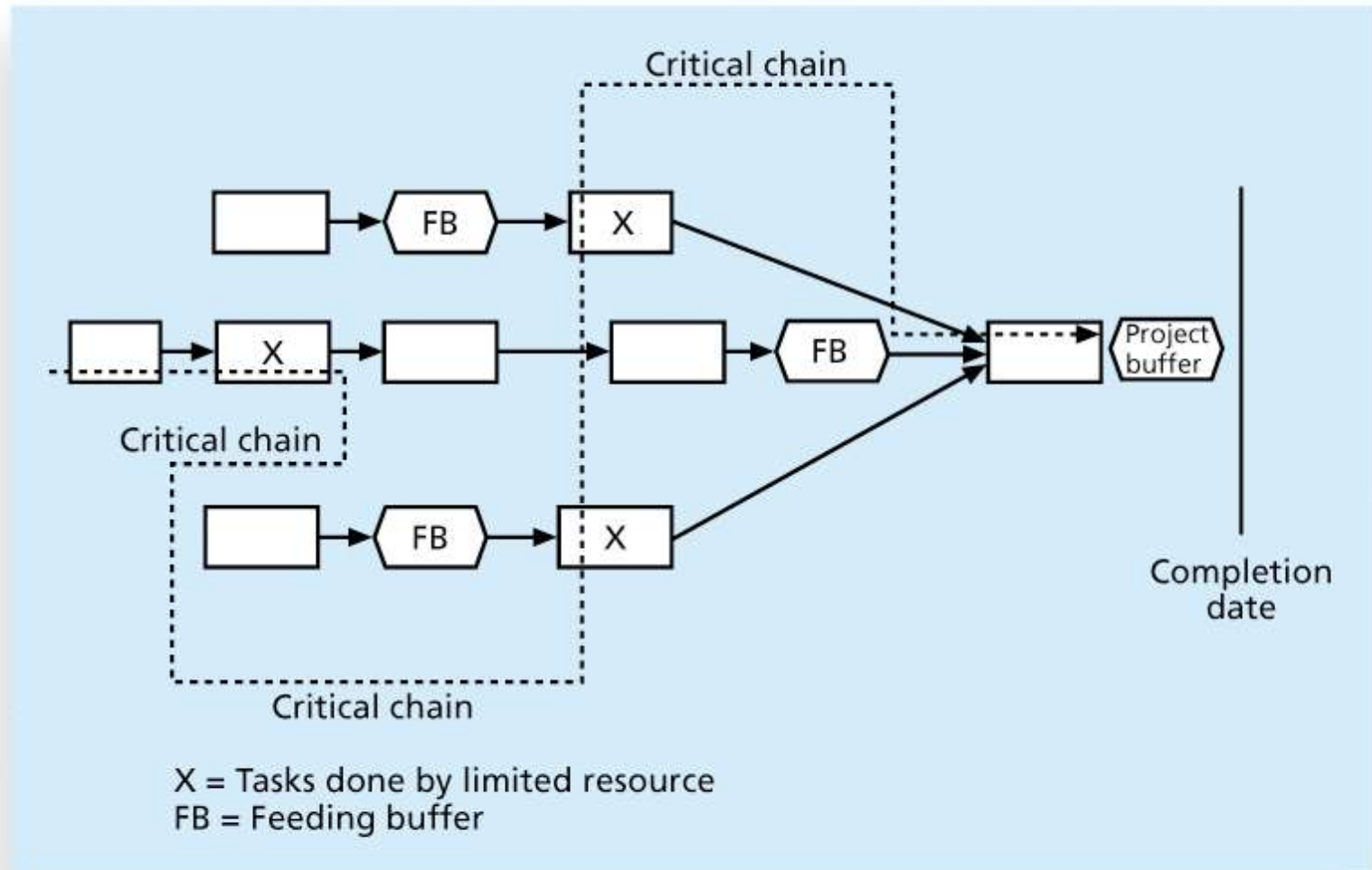
Figures 6-10.a and b. Multitasking Example



Buffers and Critical Chain

- ▶ A **buffer** is additional time to complete a task
- ▶ **Murphy's Law** states that if something can go wrong, it will
- ▶ **Parkinson's Law** states that work expands to fill the time allowed
- ▶ In traditional estimates, people often add a buffer to each task and use it if it's needed or not
- ▶ Critical chain scheduling removes buffers from individual tasks and instead creates
 - a **project buffer** or additional time added before the project's due date
 - **feeding buffers** or additional time added before tasks on the critical path

Figure 6-11. Example of Critical Chain Scheduling



What Went Right?

- ▶ Scheduling at healthcare clinic's can be more efficient by using critical chain scheduling
- ▶ The National University Hospital in Singapore decreased patient admission times by more than 50 percent
- ▶ See www.goldratt.com for more information

Program Evaluation and Review Technique (PERT)

- ▶ **PERT** is a network analysis technique used to estimate project duration when there is a high degree of uncertainty about the individual activity duration estimates
- ▶ PERT uses **probabilistic time estimates**
 - duration estimates based on using optimistic, most likely, and pessimistic estimates of activity durations, or a three-point estimate

PERT Formula and Example

▶ PERT weighted average =
$$\frac{\text{optimistic time} + 4 \times \text{most likely time} + \text{pessimistic time}}{6}$$

▶ Example:

PERT weighted average =
$$\frac{8 \text{ workdays} + 4 \times 10 \text{ workdays} + 24 \text{ workdays}}{6} = \mathbf{12 \text{ days}}$$

where optimistic time = 8 days
most likely time = **10 days**, and
pessimistic time = 24 days

Therefore, you'd use **12 days** on the network diagram instead of 10 when using PERT for the above example

Agile and Time Management

- ▶ Core values of the Manifesto for Agile Software Development are
 - Customer collaboration over contract negotiation
 - Responding to change over following a plan
- ▶ The product owner defines and prioritizes the work to be done within a spring, so collaboration and time management are designed into the process
- ▶ Teams focus on producing a useful product in a specified timeframe with strong customer input
- ▶ Don't emphasize defining all the work before scheduling it

Schedule Control Suggestions

- ▶ Perform reality checks on schedules
- ▶ Allow for contingencies
- ▶ Don't plan for everyone to work at 100% capacity all the time
- ▶ Hold progress meetings with stakeholders and be clear and honest in communicating schedule issues

Controlling the Schedule

- ▶ Goals are to know the status of the schedule, influence factors that cause schedule changes, determine that the schedule has changed, and manage changes when they occur
- ▶ Tools and techniques include
 - Progress reports
 - A schedule change control system
 - Project management software, including schedule comparison charts like the tracking Gantt chart
 - Variance analysis, such as analyzing float or slack
 - Performance management, such as earned value (chapter 7)

Reality Checks on Scheduling

- ▶ First review the draft schedule or estimated completion date in the project charter
- ▶ Prepare a more detailed schedule with the project team
- ▶ Make sure the schedule is realistic and followed
- ▶ Alert top management well in advance if there are schedule problems

Working with People Issues

- ▶ Strong leadership helps projects succeed more than good PERT charts
- ▶ Project managers should use
 - empowerment
 - incentives
 - discipline
 - negotiation

Using Software to Assist in Time Management

- ▶ Software for facilitating communications helps people exchange schedule-related information
- ▶ Decision support models help analyze trade-offs that can be made
- ▶ Project management software can help in various time management areas

Global Issues

- ▶ Microsoft tell the customer story of Mexico's Secretary of Economy, who wanted to ensure that IT initiatives aligned with business goals and improved project management efficiency
- ▶ After implementing new software, their IT team could handle four times the number of concurrent projects without adding more staff

Words of Caution on Using Project Management Software

- ▶ Many people misuse project management software because they don't understand important concepts and have not had training
- ▶ You must enter dependencies to have dates adjust automatically and to determine the critical path
- ▶ You must enter actual schedule information to compare planned and actual progress

Chapter Summary

- ▶ Project time management is often cited as the main source of conflict on projects, and most IT projects exceed time estimates
- ▶ Main processes include
 - Plan schedule management
 - Define activities
 - Sequence activities
 - Estimate activity resources
 - Estimate activity durations
 - Develop schedule
 - Control schedule

Chapter 7

Project Cost Management

Information Technology Project Management, Eighth Edition

Note: See the text itself for full citations.



Learning Objectives, Part 1

- ▶ Understand the importance of project cost management
- ▶ Explain basic project cost management principles, concepts, and terms
- ▶ Describe the process of planning cost management
- ▶ Discuss different types of cost estimates and methods for preparing them

Learning Objectives, Part 2

- ▶ Understand the processes of determining a budget and preparing a cost estimate for an information technology (IT) project
- ▶ Understand the benefits of earned value management and project portfolio management to assist in cost control
- ▶ Describe how project management software can assist in project cost management

The Importance of Project Cost Management

- ▶ IT projects have a poor track record for meeting budget goals
- ▶ A cost **overrun** is the additional percentage or dollar amount by which actual costs exceed estimates
- ▶ A 2011 Harvard Business Review study reported an average cost overrun of 27 percent. The most important finding was the discovery of a large number of gigantic overages or “black swans”

What Went Wrong?

- ▶ The United Kingdom's National Health Service IT modernization program was called the greatest IT disaster in history with an estimated **\$26 billion overrun**
- ▶ The program had problems due to incompatible systems, resistance from physicians, and arguments among contractors about who's responsible for what
- ▶ It was finally scrapped in 2011

What is Cost and Project Cost Management?

- ▶ **Cost** is a resource sacrificed or foregone to achieve a specific objective or something given up in exchange
- ▶ Costs are usually measured in monetary units like dollars
- ▶ **Project cost management** includes the processes required to ensure that the project is completed within an approved budget

Project Cost Management Processes

- ▶ **Planning cost management** :determining the policies, procedures, and documentation that will be used for planning, executing, and controlling project cost.
- ▶ **Estimating costs**: developing an approximation or estimate of the costs of the resources needed to complete a project
- ▶ **Determining the budget**: allocating the overall cost estimate to individual work items to establish a baseline for measuring performance
- ▶ **Controlling costs**: controlling changes to the project budget

Figure 7-1. Project Cost Management Summary

Planning

Process: **Plan cost management**

Outputs: Cost management plan

Process: **Estimate costs**

Outputs: Activity cost estimates, basis of estimates, project documents updates

Process: **Determine budget**

Outputs: Cost baseline, project funding requirements, project documents updates

Monitoring and Controlling

Process: **Control costs**

Outputs: Work performance information, cost forecasts, change requests, project management plan updates, project documents updates, organizational process assets updates

Project Start

Project Finish

Basic Principles of Cost Management

- ▶ Most members of an executive board better understand and are more interested in financial terms than IT terms , so IT project managers must speak their language
 - **Profits** are revenues minus expenditures
 - **Profit margin** is the ratio of revenues to profits
 - **Life cycle costing** considers the total cost of ownership, or development plus support costs, for a project
 - **Cash flow analysis** determines the estimated annual costs and benefits for a project and the resulting annual cash flow

Media Snapshot

- ▶ You cannot measure ROI unless you have a benefits measurement process in place
- ▶ A 2015 report by PMI found:
 - Only 20 percent of organizations report having a high level of benefits realization maturity
 - 39 percent of high-performing organizations report high benefits realization maturity compared to 9 percent of low performers

What Went Right?

- ▶ Investing in green IT and other initiatives has helped both the environment and companies' bottom lines
- ▶ Michael Dell, CEO of Dell, reached his goal to make his company "carbon neutral" in 2008
- ▶ As of March 2012, Dell had helped its customers save almost \$7 billion in energy costs
- ▶ In 2014 Dell reported being on track toward reaching their goal of recovering 2 billion pounds of used electronics by 2020

Types of Costs and Benefits

- ▶ **Tangible costs or benefits** are those costs or benefits that an organization can easily measure in dollars
- ▶ **Intangible costs or benefits** are costs or benefits that are difficult to measure in monetary terms
- ▶ **Direct costs** are costs that can be directly related to producing the products and services of the project
- ▶ **Indirect costs** are costs that are not directly related to the products or services of the project, but are indirectly related to performing the project
- ▶ **Sunk cost** is money that has been spent in the past; when deciding what projects to invest in or continue, you should *not* include sunk costs

More Basic Principles of Cost Management

- ▶ **Learning curve theory** states that when many items are produced repetitively, the unit cost of those items decreases in a regular pattern as more units are produced
- ▶ **Reserves** are dollars included in a cost estimate to mitigate cost risk by allowing for future situations that are difficult to predict
 - **Contingency reserves** allow for future situations that may be partially planned for (sometimes called **known unknowns**) and are included in the project cost baseline
 - **Management reserves** allow for future situations that are unpredictable (sometimes called **unknown unknowns**)

Planning Cost Management

- ▶ The project team uses expert judgment, analytical techniques, and meetings to develop the cost management plan
- ▶ A cost management plan includes:
 - Level of accuracy and units of measure
 - Organizational procedure links
 - Control thresholds
 - Rules of performance measurement
 - Reporting formats
 - Process descriptions

Estimating Costs

- ▶ Project managers must take cost estimates seriously if they want to complete projects within budget constraints
- ▶ It's important to know the types of cost estimates, how to prepare cost estimates, and typical problems associated with IT cost estimates

Table 7-1. Types of Cost Estimates

TYPE OF ESTIMATE	WHEN DONE	WHY DON'T	HOW ACCURATE
Rough order of magnitude (ROM)	Very early in the project life cycle, often 3-5 years before project completion	Provides estimate of cost for selection decisions	-50% to +100%
Budgetary	Early, 1-2 years out	Puts dollars in the budget plans	-10% to +25%
Definitive	Later in the project, less than 1 year out	Provides details for purchases, estimates actual costs	-5% to +10%

More on Cost Estimates

- ▶ The number and type of cost estimates vary by application area. The Association for the Advancement of Cost Engineering International identifies five types of cost estimates for construction projects: order of magnitude, conceptual, preliminary, definitive, and control
- ▶ Estimates are usually done at various stages of a project and should become more accurate as time progresses
- ▶ A large percentage of total project costs are often labor costs

Table 7-2. Maximum FTE by Department by Year

Department	Year 1	Year 2	Year 3	Year 4	Year 5	Totals
Information system	24	31	35	13	13	116
Marketing systems	3	3	3	3	3	15
Reservations	12	29	33	9	7	90
Contractors	2	3	1	0	0	6
Totals	41	66	72	25	23	227

Cost Estimation Tools and Techniques

- ▶ Basic tools and techniques for cost estimates:
 - **Analogous or top-down estimates:** use the actual cost of a previous, similar project as the basis for estimating the cost of the current project
 - **Bottom-up estimates:** involve estimating individual work items or activities and summing them to get a project total
 - **Parametric modeling** uses project characteristics (parameters) in a mathematical model to estimate project costs

Typical Problems with IT Cost Estimates

- ▶ Estimates are done too quickly
- ▶ People lack estimating experience
- ▶ Human beings are biased toward underestimation
- ▶ Management desires accuracy

Sample Cost Estimate

- ▶ See the text for a detailed example of creating a cost estimate for the Surveyor Pro project described in the opening case
- ▶ Before creating an estimate, know what it will be used for, gather as much information as possible, and clarify the ground rules and assumptions for the estimate
- ▶ If possible, estimate costs by major WBS categories
- ▶ Create a cost model to make it easy to make changes to and document the estimate

Figure 7-2. Surveyor Pro Project Cost Estimate

Surveyor Pro Project Cost Estimate October 5

	# Units/Hrs.	Cost/Unit/Hr.	Subtotals	WBS Level 2 Totals	% of Total
WBS Items					
1. Project Management				\$306,300	20%
Project manager	960	\$100	\$96,000		
Project team members	1920	\$75	\$144,000		
Contractors (10% of software development and testing)			\$66,300		
2. Hardware				\$76,000	5%
2.1 Handheld devices	100	\$600	\$60,000		
2.2 Servers	4	\$4,000	\$16,000		
3. Software				\$614,000	40%
3.1 Licensed software	100	\$200	\$20,000		
3.2 Software development*			\$594,000		
4. Testing (10% of total hardware and software costs)			\$69,000	\$69,000	5%
5. Training and Support			Empty cell	\$202,400	13%
Trainee cost	100	\$500	\$50,000		
Travel cost	12	\$700	\$8,400		
Project team members	1920	\$75	\$144,000		
6. Reserves (20% of total estimate)			\$253,540	\$253,540	17%
Total project cost estimate				\$1,521,240	

*See software development estimate

Figure 7-3. Surveyor Pro Software Development Estimate

1. Labor Estimate	# Units/Hrs.	Cost/Unit/Hr.	Subtotals	Calculations
Contractor labor estimate	3000	\$150	\$450,000	3000*150
Project team member estimate	1920	\$75	\$144,000	1920*75
Total labor estimate			\$594,000	Sum above two values
Empty cell				
2. Function point estimate	Quantity	Conversion Factor	Function Points	Calculations
External inputs	10	4	40	10*4
External interface files	3	7	21	3*7
External outputs	4	5	20	4*5
External queries	6	4	24	6*4
Logical internal tables	7	10	70	7*10
Total function points			175	Sum above function point values
Java 2 language equivalency value			46	Assumed value from reference
Source lines of code (SLOC) estimate			8,050	175*46
ProductivityxKSLOC*Penalty (in months)			29.28	3.13*8.05^1.072 (see reference)
Total labor hours (27 hours/function point)*			4,725	27*175
Cost/labor hour (\$120/hour)			\$120	Assumed value from budget expert
Total function point estimate			\$567.000	4,725*120

*Based on historical data

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Best Practice

- ▶ Alvin Alexander wrote a book called *Cost Estimating in an Agile Development Environment* in 2015
- ▶ Function points are a means of measuring software size in terms that are meaningful to end users
- ▶ User stories are a common way to describe requirements in a simple, concise way
- ▶ Developers can analyze user stories to estimate function points and person-hours

Determining the Budget

- ▶ Cost budgeting involves allocating the project cost estimate to individual work items over time
- ▶ The WBS is a required input to the cost budgeting process since it defines the work items
- ▶ Important goal is to produce a **cost baseline**
 - a time-phased budget that project managers use to measure and monitor cost performance

Figure 7-4. Surveyor Pro Project Cost Baseline

Surveyor Pro Project Cost Baseline Created October 10*

WBS Items	1	2	3	4	5	6	7	8	9	10	11	12	Totals
1. Project Management													
1.1 Project manager	8,000	8,000	8,000	8,000	8,000	8,000	8,000	8,000	8,000	8,000	8,000	8,000	96,000
1.2 Project team members	12,000	12,000	12,000	12,000	12,000	12,000	12,000	12,000	12,000	12,000	12,000	12,000	144,000
1.3 Contractors		6,027	6,027	6,027	6,027	6,027	6,027	6,027	6,027	6,027	6,027	6,027	66,300
2. Hardware													
2.1 Handheld devices				30,000	30,000								60,000
2.2 Servers				8,000	8,000								16,000
3. Software													
3.1 Licensed software				10,000	10,000								20,000
3.2 Software development		60,000	60,000	80,000	127,000	127,000	90,000	50,000					594,000
4. Testing			6,000	8,000	12,000	15,000	15,000	13,000					69,000
5. Training and Support													
5.1 Trainee cost									50,000				50,000
5.2 Travel cost									8,400				8,400
5.3 Project team members							24,000	24,000	24,000	24,000	24,000	24,000	144,000
6. Reserves				10,000	10,000	30,000	30,000	60,000	40,000	40,000	30,000	3,540	253,540
	20,000	86,027	92,027	172,027	223,027	198,027	173,027	148,427	90,027	80,027	53,567	53,567	1,521,240

*See the lecture slides for this chapter on the companion Web site for a larger view of this and other figures in this chapter.
Numbers are rounded, so some totals appear to be off.

Controlling Costs

- ▶ Project cost control includes
 - Monitoring cost performance
 - Ensuring that only appropriate project changes are included in a revised cost baseline
 - Informing project stakeholders of authorized changes to the project that will affect costs
- ▶ Many organizations around the globe have problems with cost control

Earned Value Management (EVM)

- ▶ **EVM** is a project performance measurement technique that integrates scope, time, and cost data
- ▶ Given a **baseline** (original plan plus approved changes), you can determine how well the project is meeting its goals
- ▶ You must enter actual information periodically to use EVM
- ▶ More and more organizations around the world are using EVM to help control project costs

Earned Value Management Terms

- ▶ The **planned value (PV)**, formerly called the budgeted cost of work scheduled (BCWS), also called the budget, is that portion of the approved total cost estimate planned to be spent on an activity during a given period
- ▶ **Actual cost (AC)**, formerly called actual cost of work performed (ACWP), is the total of direct and indirect costs incurred in accomplishing work on an activity during a given period
- ▶ The **earned value (EV)**, formerly called the budgeted cost of work performed (BCWP), is an estimate of the value of the physical work actually completed
- ▶ EV is based on the original planned costs for the project or activity and the rate at which the team is completing work on the project or activity to date

Rate of Performance

- ▶ **Rate of performance (RP)** is the ratio of actual work completed to the percentage of work planned to have been completed at any given time during the life of the project or activity
- ▶ Brenda Taylor, Senior Project Manager in South Africa, suggests this term and approach for estimating earned value
- ▶ For example, suppose the server installation was halfway completed by the end of week 1. The rate of performance would be 50% because by the end of week 1, the planned schedule reflects that the task should be 100 percent complete and only 50 percent of that work has been completed

Table 7-3. Earned Value Calculations for One Activity After Week One

ACTIVITY	WEEK 1
Earned Value (EV)	5,000
Planned Value (PV)	10,000
Actual Cost (AC)	15,000
Cost Variance (CV)	-10,000
Schedule Variance (SV)	-5,000
Cost Performance Index (CPI)	33%
Schedule Performance Index (SPI)	50%

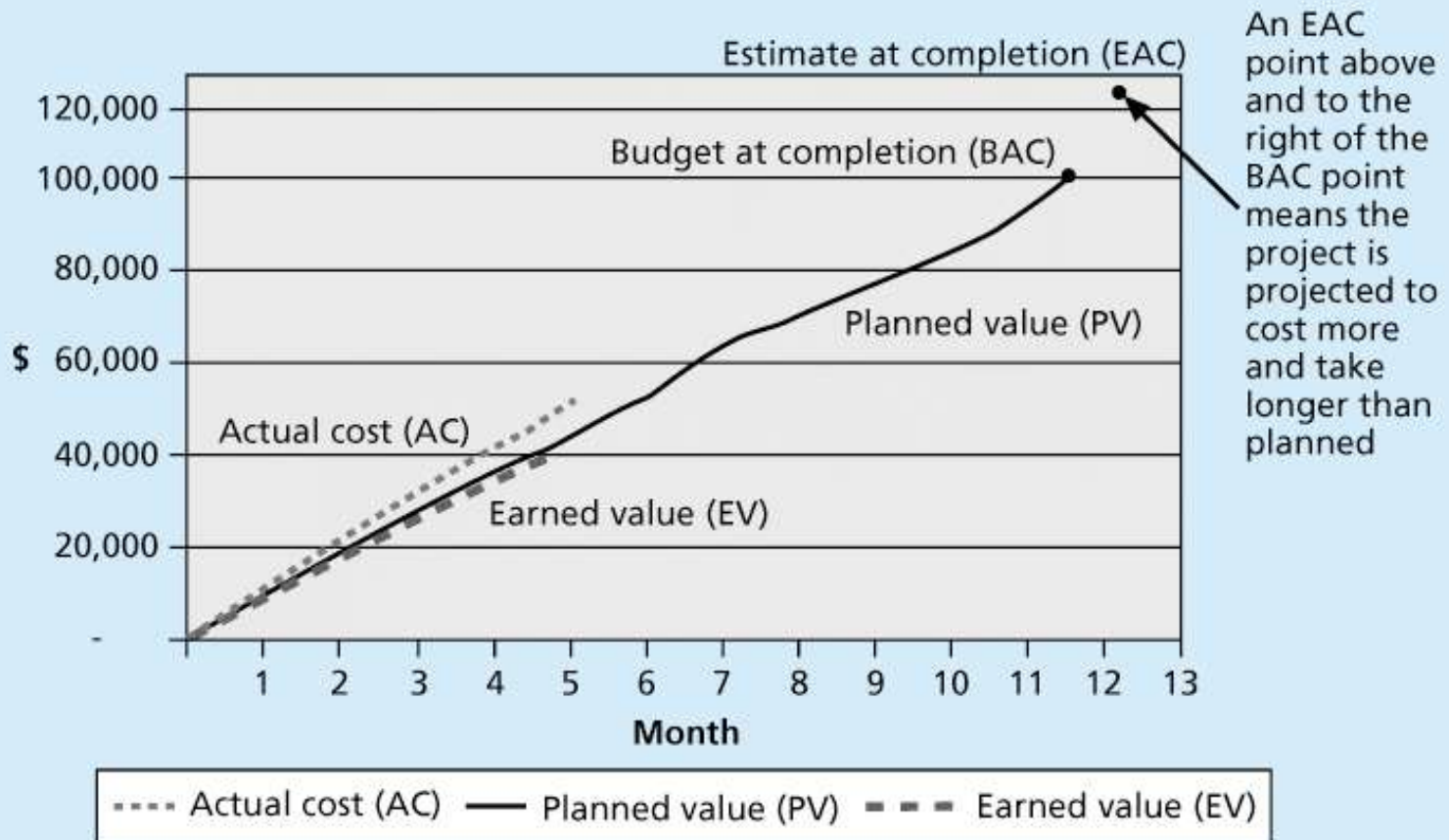
Table 7-4. Earned Value Formulas

Term	Formula
Earned value (EV)	$EV = PV \text{ to date} * RP$
Cost variance (CV)	$CV = EV - AC$
Schedule variance (SV)	$SV = EV - PV$
Cost performance index (CPI)	$CPI = EV/AC$
Schedule performance index (SPI)	$SPI = EV/PV$
Estimate at completion (EAC)	$EAC = BAC/CPI$
Estimated time to complete	Original time estimate/SPI

Rules of Thumb for Earned Value Numbers

- ▶ Negative numbers for cost and schedule variance indicate problems in those areas
- ▶ CPI and SPI less than 100% indicate problems
- ▶ Problems mean the project is costing more than planned (over budget) or taking longer than planned (behind schedule)
- ▶ The CPI can be used to calculate the **estimate at completion** (EAC)—an estimate of what it will cost to complete the project based on performance to date. The **budget at completion** (BAC) is the original total budget for the project

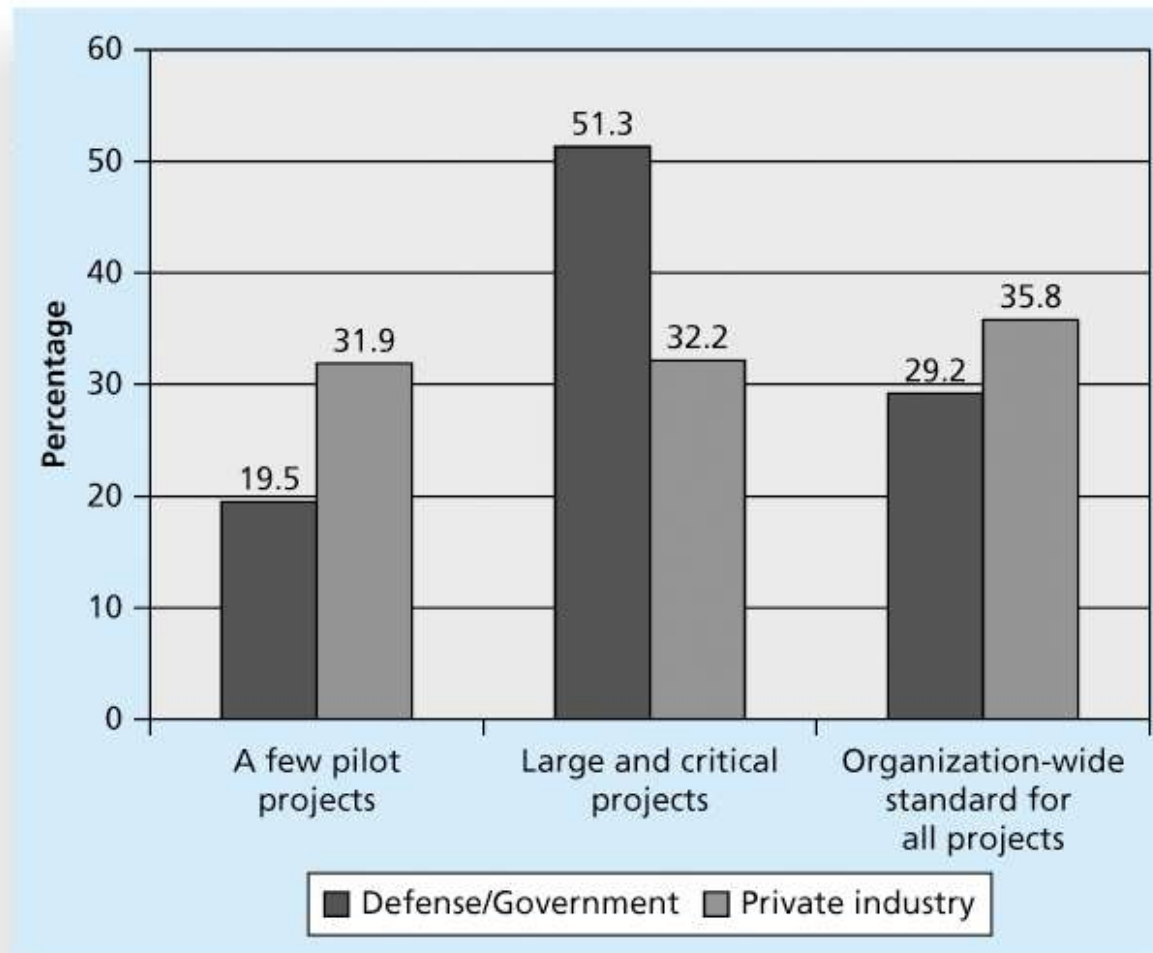
Figure 7-5. Earned Value Chart for Project after Five Months



Global Issues

- ▶ EVM is used worldwide, and it is particularly popular in the Middle East, South Asia, Canada, and Europe
- ▶ Most countries require EVM for large defense or government projects, as shown in Figure 7-6
- ▶ EVM is also used in such private-industry sectors as IT, construction, energy, and manufacturing.
- ▶ However, most private companies have not yet applied EVM to their projects because management does not require it, feeling it is too complex and not cost effective

Figure 7-6. Earned Value Usage



Project Portfolio Management

- ▶ Many organizations collect and control an entire suite of projects or investments as one set of interrelated activities in a portfolio
- ▶ Five levels for project portfolio management
 1. Put all your projects in one database
 2. Prioritize the projects in your database
 3. Divide your projects into two or three budgets based on type of investment
 4. Automate the repository
 5. Apply modern portfolio theory, including risk-return tools that map project risk on a curve

Benefits of Portfolio Management

- ▶ Schlumberger saved \$3 million in one year by organizing 120 information technology projects into a portfolio
- ▶ Reduced redundant projects and coordinated those with overlap
- ▶ IT projects can be huge investments, so it makes sense to view them as portfolios and track their progress as a whole

Using Software to Assist in Cost Management

- ▶ Spreadsheets are a common tool for resource planning, cost estimating, cost budgeting, and cost control
- ▶ Many companies use more sophisticated and centralized financial applications software for cost information
- ▶ Project management software has many cost-related features, especially enterprise PM software
- ▶ Portfolio management software can help reduce costs

Recent Studies on PPM Software

- ▶ 2014 Gartner report says the market continues to grow, with annual sales over \$1.65 billion in 2014 report. The pace of change is driving the demand for enterprise software to help manage projects
- ▶ Forrester estimates ROIs of 250 percent from PPM tools
- ▶ Pfizer and Ford use PPM software to improve transparency of the many projects they manage

Chapter Summary

- ▶ Project cost management is a traditionally weak area of IT projects, and project managers must work to improve their ability to deliver projects within approved budgets
- ▶ Main processes include
 - Plan cost management
 - Estimate costs
 - Determine the budget
 - Control costs

Chapter 8:

Project Quality Management

Information Technology Project
Management



Learning Objectives, Part 1

- Understand the importance of project quality management for information technology (IT) products and services
- Define project quality management and understand how quality relates to various aspects of IT projects
- Describe quality management planning and how quality and scope management are related
- Discuss the importance of quality assurance
- Explain the main outputs of the quality control process

Learning Objectives, Part 2

- Understand the tools and techniques for quality control, such as the Seven Basic Tools of Quality, statistical sampling, Six Sigma, and testing
- Summarize the contributions of noteworthy quality experts to modern quality management
- Describe how leadership, the cost of quality, organizational influences, expectations, cultural differences, and maturity models relate to improving quality in IT projects
- Discuss how software can assist in project quality management

The Importance of Project Quality Management

- Many people joke about the poor quality of IT products (see cars and computers joke)
- People seem to accept systems being down occasionally or needing to reboot their PCs
- But quality is very important in many IT projects

What Went Wrong?

- In 1986, two hospital patients died after receiving fatal doses of radiation from a Therac 25 machine after a software problem caused the machine to ignore calibration data
- In one of the biggest software errors in banking history, Chemical Bank mistakenly deducted about \$15 million from more than 100,000 customer accounts
- In 2015, the United States Department of Justice unsealed indictments in what was described as “the largest data break of names and e-mail addresses in the history of the internet”

Continuous Improvement is **CRUCIAL!**



Commitment to quality pays off !

**NO MORE
ERRORS**



**Happier
Customers!**

What Is Project Quality?

- The International Organization for Standardization (ISO) defines **quality** as “the degree to which a set of inherent characteristics fulfils requirements” (ISO9000:2000)
- Other experts define quality based on:
 - **Conformance to requirements:** The project’s processes and products meet written specifications
 - **Fitness for use:** A product can be used as it was intended

<https://www.youtube.com/watch?v=eQ0GvN2DsFo>

What Is Project Quality Management?

- **Project quality management** ensures that the project will satisfy the needs for which it was undertaken
- Processes include:
 - **Planning quality management**: Identifying which quality standards are relevant to the project and how to satisfy them; a **metric** is a standard of measurement
 - **Performing quality assurance**: Periodically evaluating overall project performance to ensure the project will satisfy the relevant quality standards
 - **Performing quality control**: Monitoring specific project results to ensure that they comply with the relevant quality standards

Figure 8-1. Project Quality Management Summary



Planning Quality

- Implies the ability to anticipate situations and prepare actions to bring about the desired outcome
- Important to prevent defects by:
 - Selecting proper materials
 - Training and inculcating people in quality
 - Planning a process that ensures the appropriate outcome

Scope Aspects of IT Projects

- **Functionality** is the degree to which a system performs its intended function
- **Features** are the system's special characteristics that appeal to users
- **System outputs** are the screens and reports the system generates
- **Performance** addresses how well a product or service performs the customer's intended use
- **Reliability** is the ability of a product or service to perform as expected under normal conditions
- **Maintainability** addresses the ease of performing maintenance on a product

Function vs Feature

	Function	Feature
Definition	A goal or objective accomplished by a product or service.	A tool that helps to achieve functions.
Examples	Tasks Activities Calculations Outputs	Doors Buttons Menus Robotic Arms

Who's Responsible for the Quality of Projects?

- Project managers are ultimately responsible for quality management on their projects
- Several organizations and references can help project managers and their teams understand quality
 - International Organization for Standardization (www.iso.org)
 - IEEE (www.ieee.org)

Performing Quality Assurance

- **Quality assurance** includes all the activities related to satisfying the relevant quality standards for a project
- Another goal of quality assurance is continuous quality improvement. (**Kaizen** is the Japanese word for improvement or change for the better)
- **Lean** involves evaluating processes to maximize customer value while minimizing waste.

(Lean methodology, which originated in manufacturing but have since been applied to various industries, including service sectors like healthcare and software development. In Lean thinking, the primary goal is to optimize processes to deliver maximum value to the customer while eliminating or minimizing any activities that do not contribute to that value)

- **Benchmarking** generates ideas for quality improvements by comparing specific project practices or product characteristics to those of other projects or products within or outside the performing organization
- A **quality audit** is a structured review of specific quality management activities that help identify lessons learned that could improve performance on current or future projects

What Went Right?

- Kanban uses five core properties:
 - Visual workflow
 - Limit work-in-process
 - Measure and manage flow
 - Make process policies explicit
 - Use models to recognize improvement opportunities
- The application of Kanban is different for every team
- *Kanban is a methodology used in project management, particularly in software development and manufacturing, to visualize work, limit work-in-progress (WIP), and maximize efficiency.*

Kanban:

1.Visualize the workflow: This involves creating a visual representation of the work process, typically using a Kanban board with columns representing different stages of the workflow and cards representing individual tasks or work items.

2.Limit work-in-progress (WIP): Setting limits on the number of tasks that can be in progress at any given time helps prevent overloading the team and ensures that work flows smoothly through the system.

3.Manage flow: Kanban focuses on optimizing the flow of work through the system, aiming to minimize delays and bottlenecks and maximize throughput.

4.Make process policies explicit: Clearly defining the rules and policies that govern how work is done and how tasks move through the workflow helps ensure consistency and alignment within the team.

5.Improve collaboratively, evolve experimentally: Kanban encourages continuous improvement through collaboration and experimentation, with teams regularly reviewing and refining their processes to identify opportunities for optimization and innovation.

Controlling Quality

- The main outputs of quality control are:
 - Acceptance decisions
 - Rework
 - Process adjustments
- There are Seven Basic Tools of Quality that help in performing quality control

Cause-and-Effect Diagrams

- **Cause-and-effect diagrams** trace complaints about quality problems back to the responsible production operations
- They help you find the root cause of a problem
- Also known as **fishbone** or **Ishikawa diagrams**
- Can also use the **5 whys** technique where you repeated ask the question “Why” (five is a good rule of thumb) to peel away the layers of symptoms that can lead to the root cause

Figure 8-2. Sample Cause-and-Effect Diagram

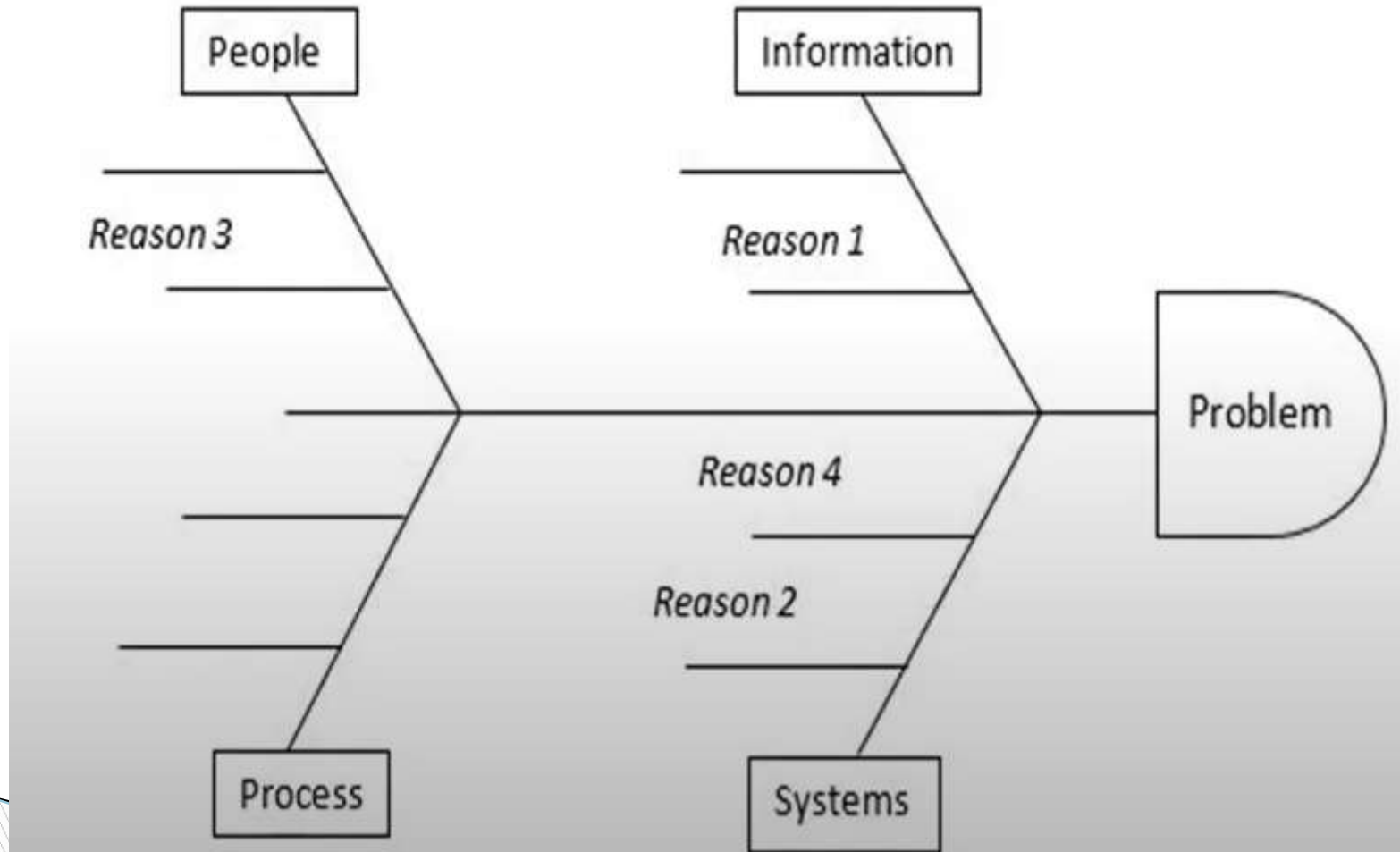
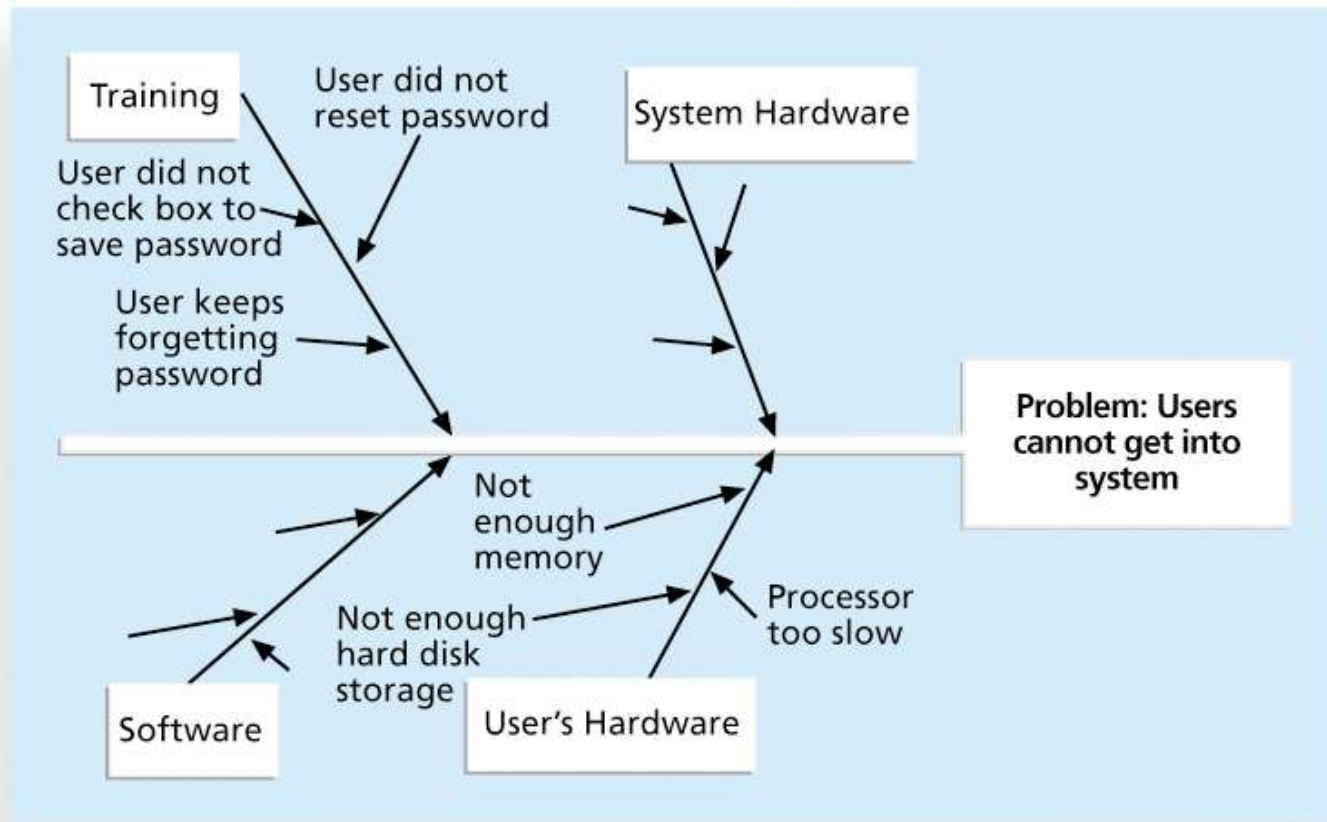


Figure 8-2. Sample Cause-and-Effect Diagram



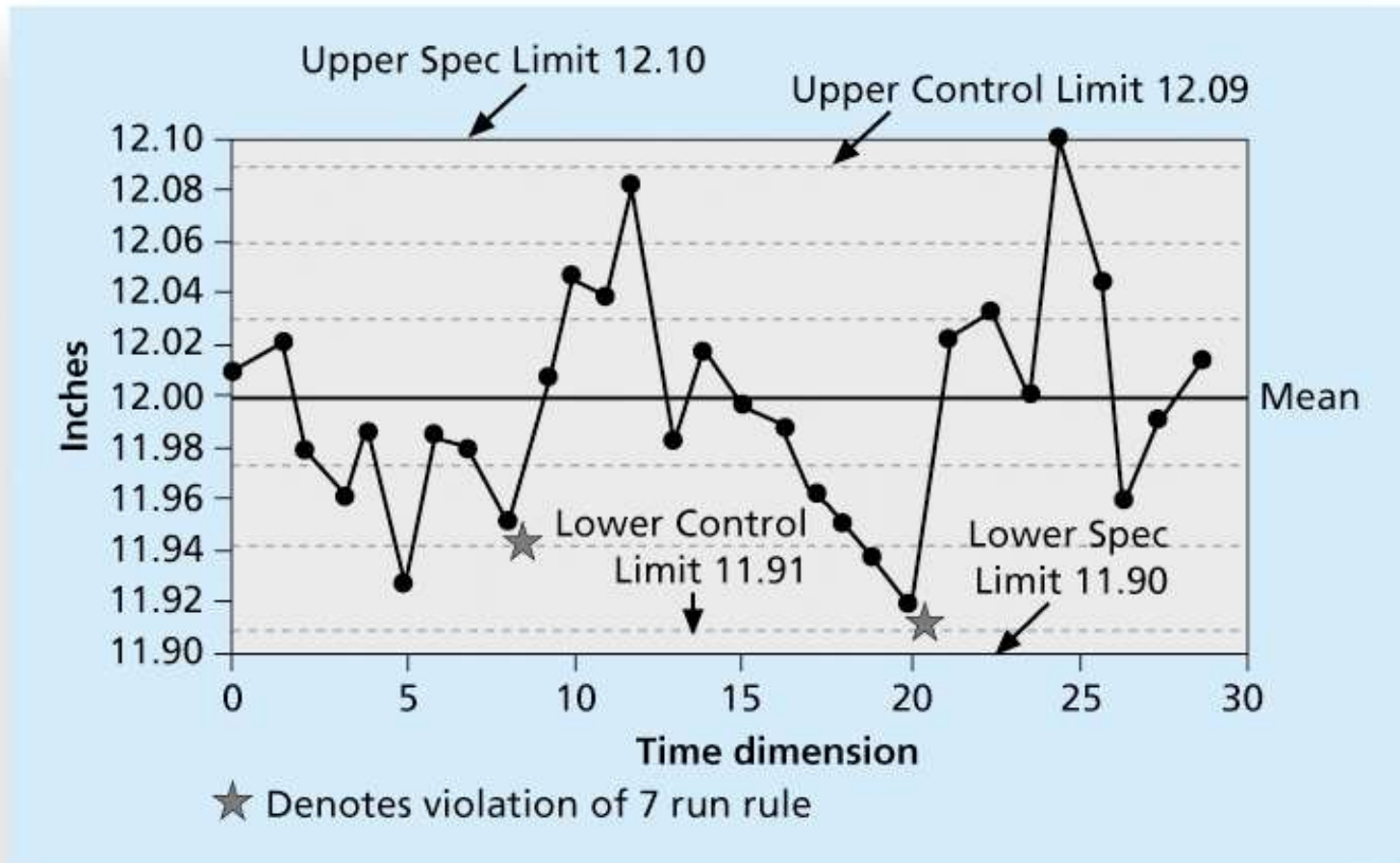
Quality Control Charts

- A **control chart** is a graphic display of data that illustrates the results of a process over time
- The main use of control charts is to **prevent defects**, rather than to detect or reject them
- Quality control charts allow you to determine whether a process is in control or out of control
 - When a process is **in control**, any variations in the results of the process are created by random events; processes that are in control **do not need to be adjusted**
 - When a process is **out of control**, variations in the results of the process are caused by non-random events; you need to **identify the causes** of those non-random events **and adjust the process to correct or eliminate them.**

The Seven Run Rule

- You can use **quality control charts** and the **seven-run rule** to look for patterns in data
- The **seven-run rule** states that if seven data points in a row are all **below the mean**, **above the mean**, or are all increasing or decreasing, **then the process needs to be examined for non-random problems**

Figure 8-3. Sample Quality Control Chart



Checksheet

- A **checksheet** is used to collect and analyze data
- It is sometimes called a **tally sheet** or **checklist**, depending on its format

Figure 8-4. Sample Checksheet

System Complaints								
Source	Day							
	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday	Total
Email								12
Text								29
Phone call								8
Total	11	10	8	6	7	3	4	49

- ✓ In the example in Figure 8-4, most complaints arrive via text message, and there are more complaints on Monday and Wednesday than on other days of the week
- ✓ This information might be useful in improving the process for handling complaints

Scatter diagram

- A **scatter diagram** helps to show if there is a relationship between two variables
- The **closer data points are to a diagonal line**, the more closely the two variables are related. (*high user satisfaction for customers aged 20 – 40 yo*)

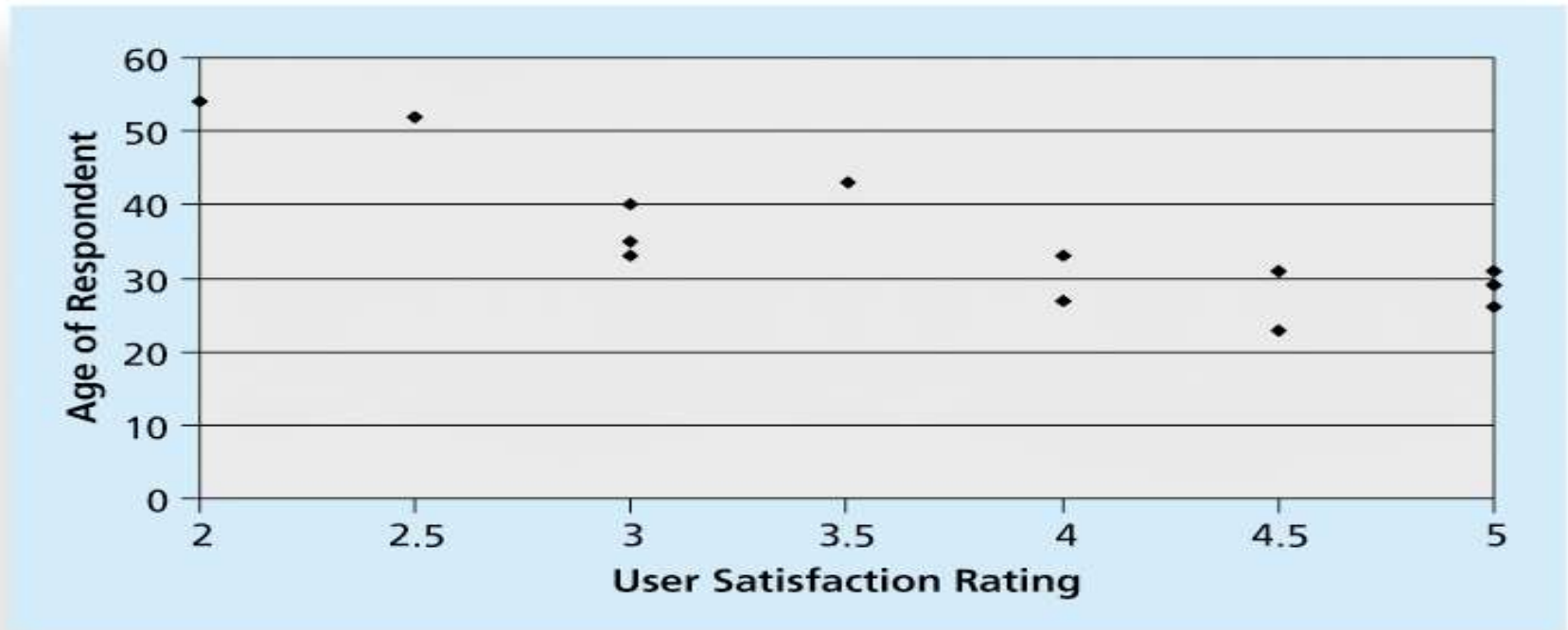
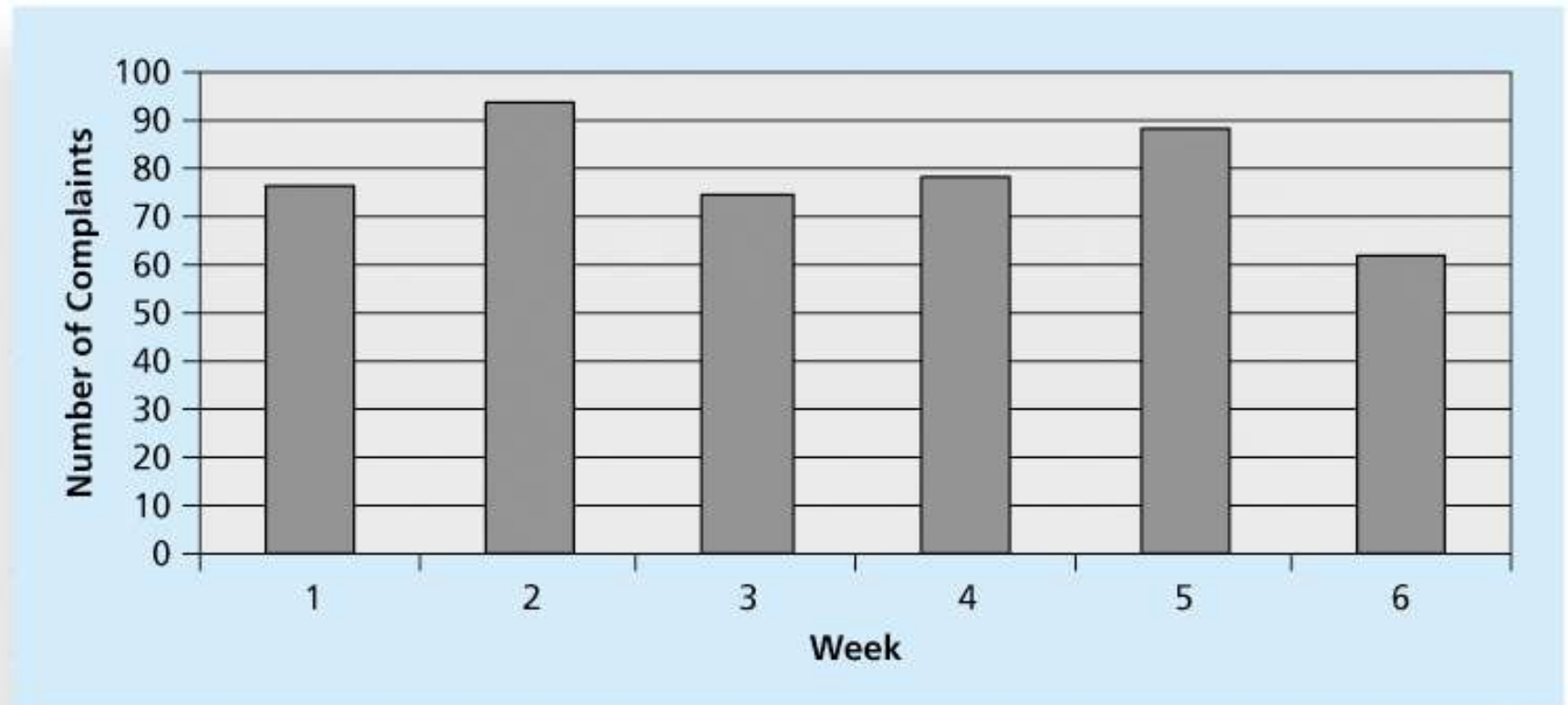


Figure 8-5. Sample Scatter Diagram

Histograms

- A **histogram** is a bar graph of a distribution of variables
- Each bar **represents** an attribute or characteristic of a **problem or situation**, and the **height** of the bar **represents its frequency**.

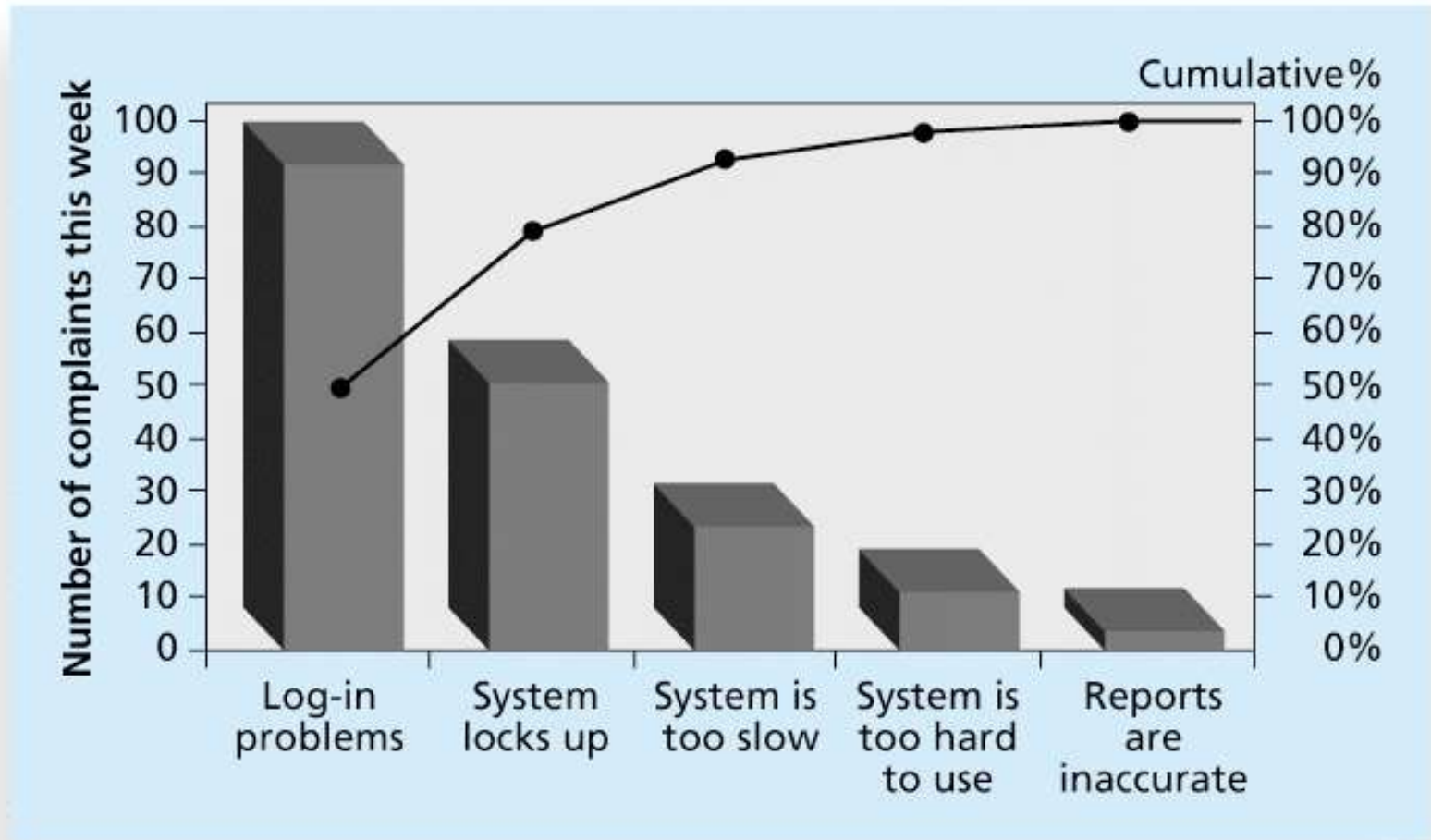
Figure 8-6. Sample Histogram



Pareto Charts

- A **Pareto chart** is a histogram that can help you identify and [prioritize problem areas](#)
- **Pareto analysis** is also called the 80-20 rule, meaning that 80 percent of problems are often due to 20 percent of the causes
- **Example:**
 - *ICT project management context: a team is experiencing frequent delays in project delivery. Upon conducting a Pareto analysis, they found that 80% of the project delays are caused by issues related to software bugs and system integration, while the remaining 20% of delays are due to factors like communication issues and unexpected hardware failures.*
 - *This example illustrates the 80-20 rule in action within an ICT project. By identifying and addressing the critical 20% of causes (such as software bugs and system integration issues), which contribute to 80% of the project delays, the team can prioritize their efforts and resources more effectively to improve project performance and delivery timelines.*

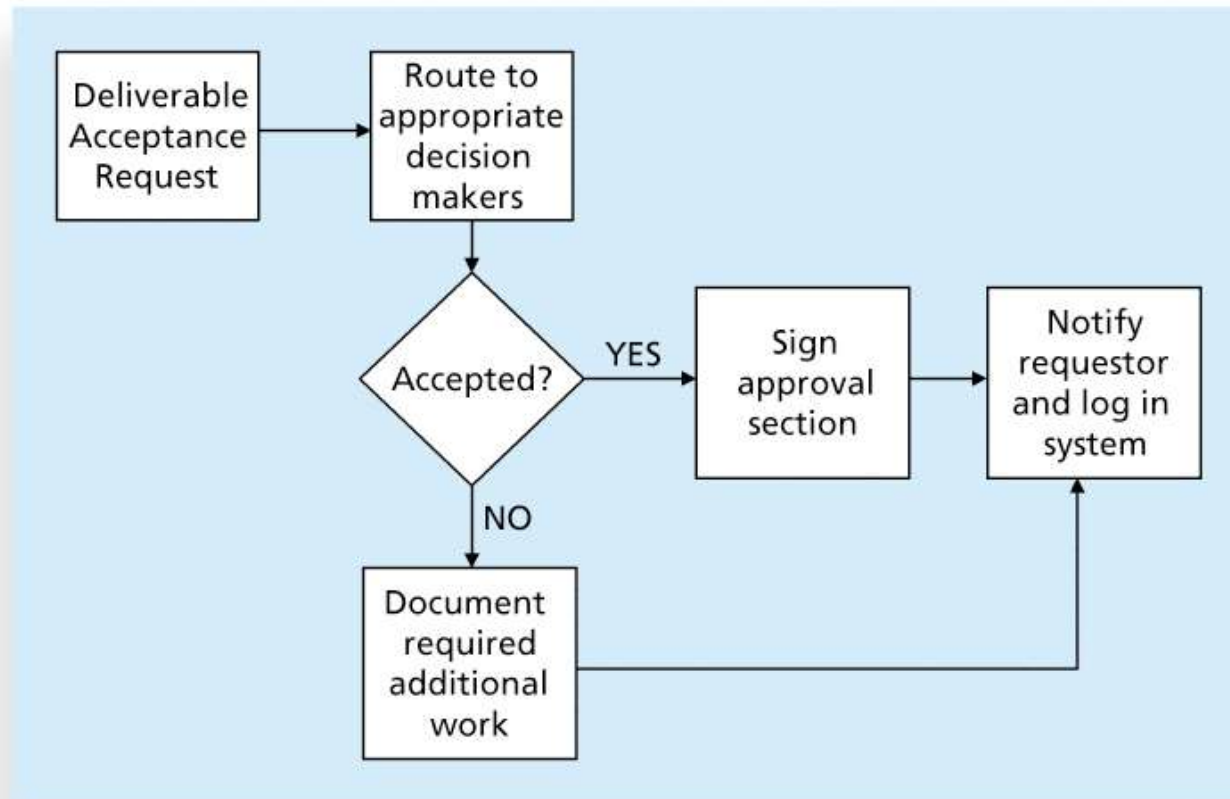
Figure 8-7. Sample Pareto Chart



Flowcharts

- Flowcharts are **graphic displays of the logic** and flow of processes that help you analyze how problems occur and how processes can be improved
- They show activities, decision points, and the order of which information is processed

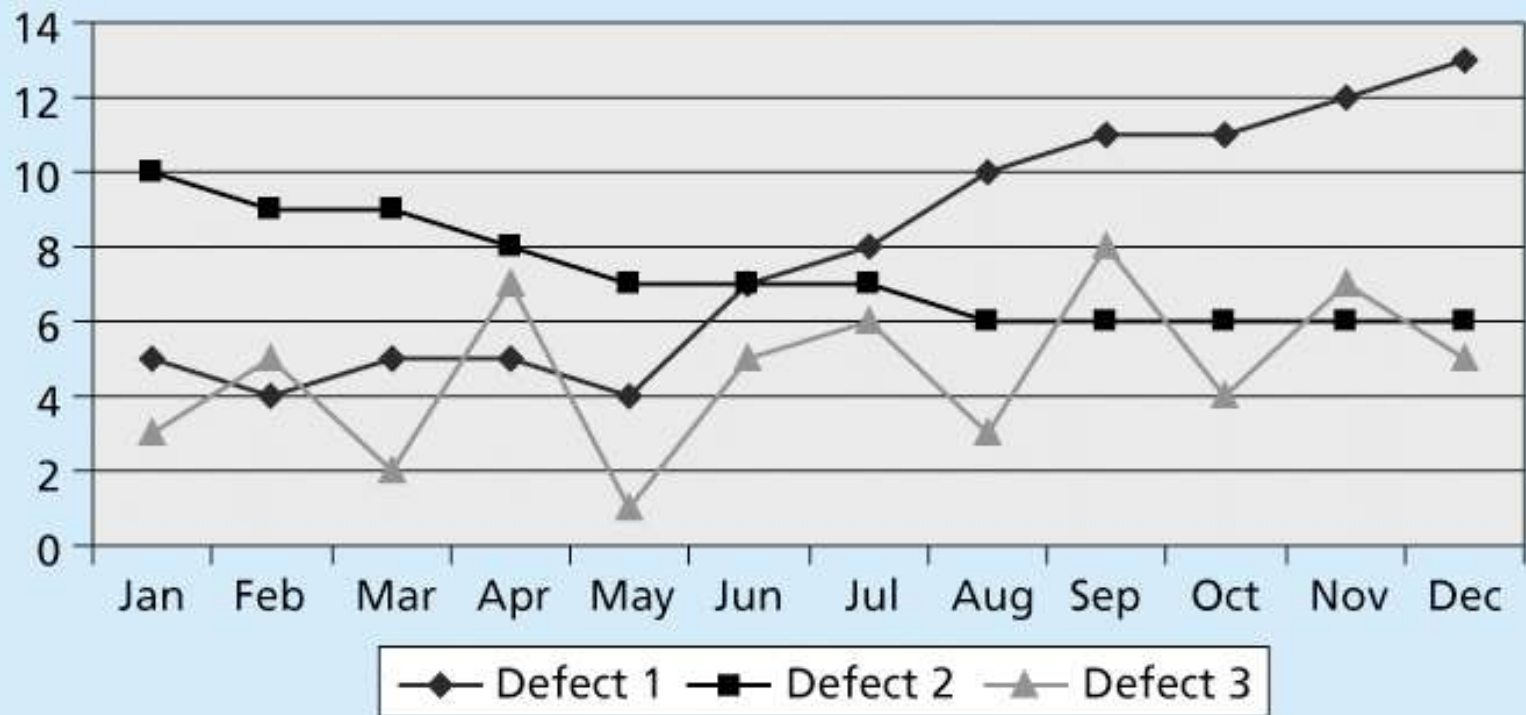
Figure 8-8. Sample Flowchart



Run Charts

- In addition to flowcharts, run charts are also used for **stratification** (*layer*), a technique that shows data from a variety of sources to see if a pattern emerges
- A **run chart** displays the history and pattern of variation of a process over time.
- You can use run charts **to perform trend analysis** and **forecast future outcomes** based on historical results.

Figure 8-9. Sample Run Chart



Statistical Sampling

- **Statistical sampling** involves choosing part of a population of interest for inspection
- The size of a sample depends on how representative you want the sample to be
- Sample size formula:

Sample size = $.25 \times (\text{certainty factor/acceptable error})^2$

$$k = \text{Skip interval} = \frac{\text{Population size}}{\text{Sample size}}$$

$$\text{Probability of selection} = \frac{\text{Sample size}}{\text{Population size}}$$

- Be sure to consult with an expert when using statistical analysis

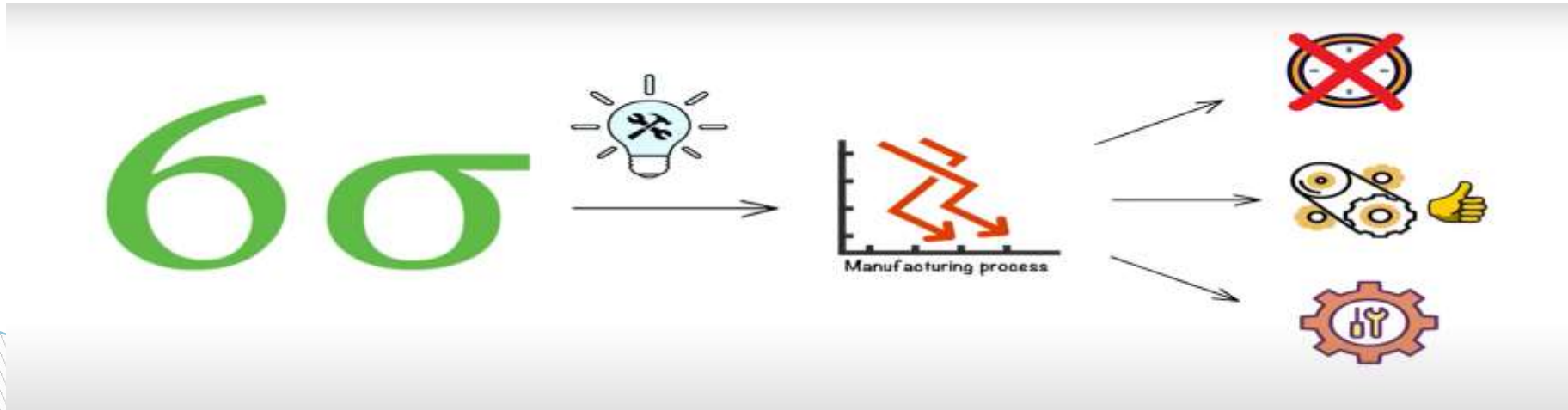
Table 8-1. Commonly Used Certainty Factors

Desired Certainty	Certainty Factor
95%	1.960
90%	1.645
80%	1.281

Six Sigma

- **Six Sigma** is “a comprehensive and flexible system for achieving, sustaining, and maximizing business success. Six Sigma is uniquely driven by a close understanding of customer needs, disciplined use of facts, data, and statistical analysis, and diligent attention to managing, improving, and reinventing business processes”*

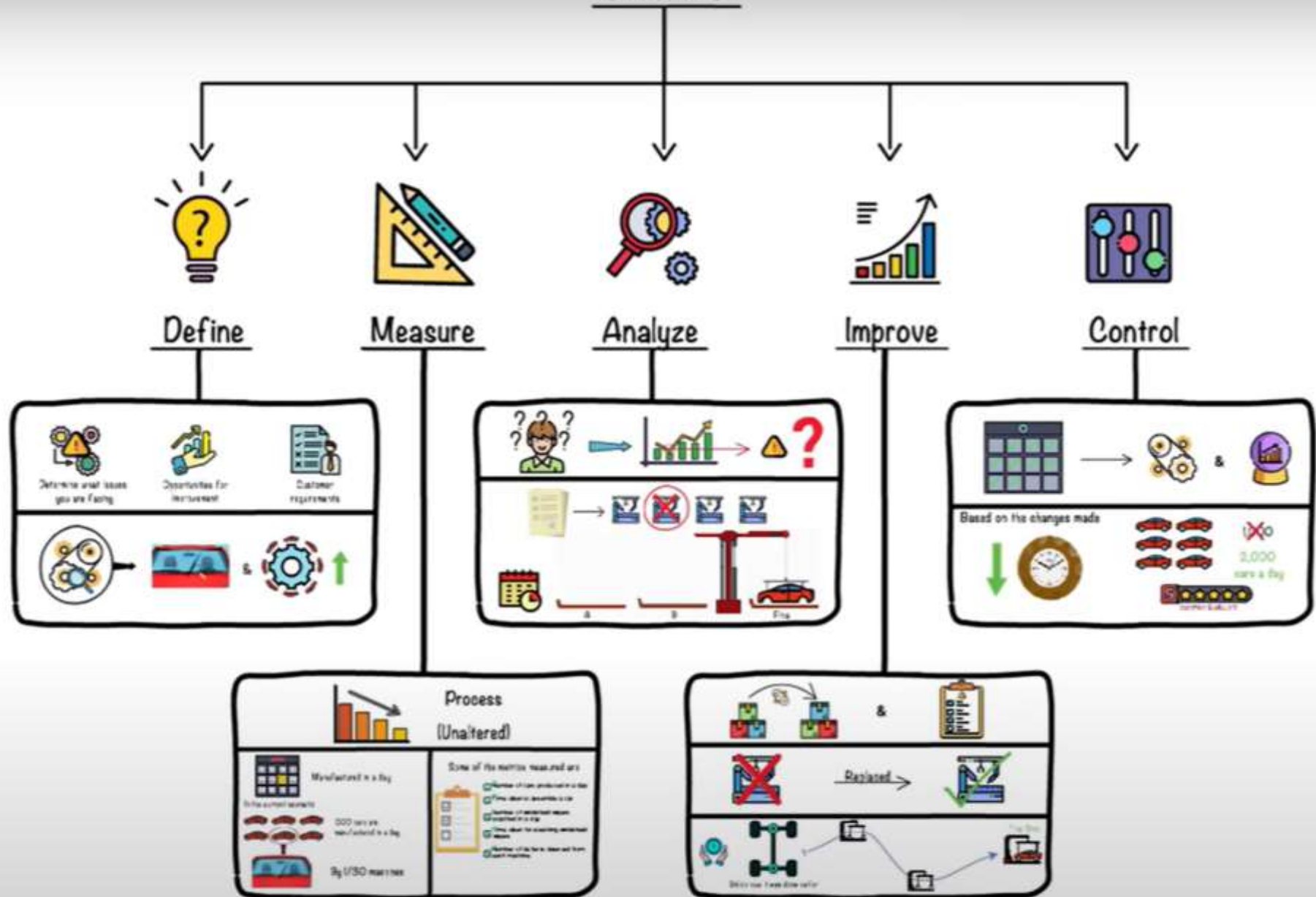
*Pande, Peter S., Robert P. Neuman, and Roland R. Cavanagh, The Six Sigma Way, New York: McGraw-Hill, 2000, p. xi.



Basic Information on Six Sigma

- The target for perfection is the achievement of no more than **3.4 defects per million opportunities**
- The principles can apply to a wide variety of processes
- Six Sigma projects normally follow a **five-phase improvement process called DMAIC** (*Define, Measure, Analyze, Improve, Change*)

DMAIC

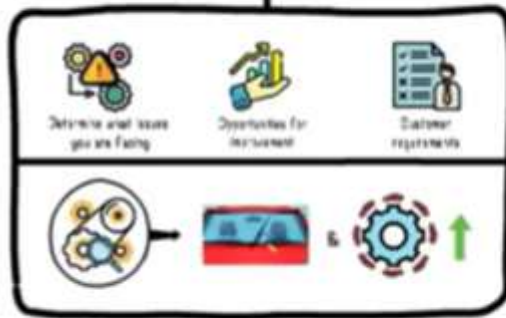


DMAIC is one of the most commonly used methodologies in the world

DMAIC

- **DMAIC** is a systematic, closed-loop process for continued improvement that is scientific, and fact based
- DMAIC stands for:
 - **Define:** Define the problem/opportunity, process, and customer requirements
 - **Measure:** Define measures, then collect, compile, and display data
 - **Analyze:** Scrutinize process details to find improvement opportunities
 - **Improve:** Generate solutions and ideas for improving the problem
 - **Control:** Track and verify the stability of the improvements and the predictability of the solution

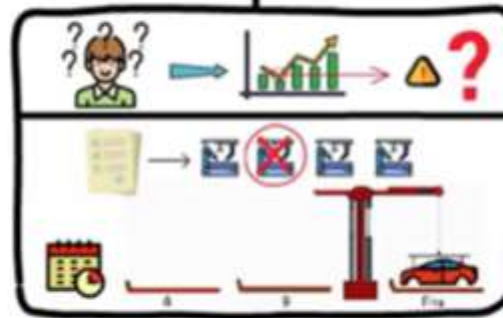
Define



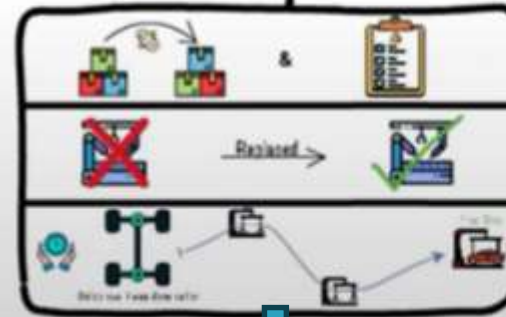
Measure



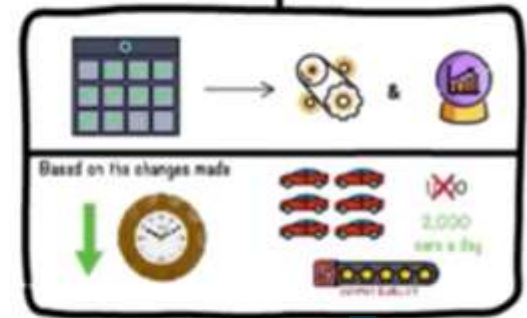
Analyze



Improve



Control



Define Issue: Car Manufacturing reduce by time.

Measure: How many cars are manufactured in a day & other metrics.

Analyze: What causes the manufacturing to slow down?
Example: one of the line machines/robots malfunctioned/broken.

Improve: Change or remove the broken machine/robot. And the result show more car manufactured due to faster car part fixing.

Control: Make regular checking & adjustment according to new process or needs

How is Six Sigma Quality Control Unique?

- It requires an organization-wide commitment.
- Training follows the “Belt” system

*(The "Belt" system refers to a structured training and certification program within Six Sigma. Similar to martial arts, **Six Sigma uses different colored belts to signify different levels of expertise and responsibility.** The main belts include Yellow Belt, Green Belt, Black Belt, and Master Black Belt, each representing a different level of proficiency in Six Sigma methodologies and tools.)*

- Six Sigma organizations have the ability and willingness to adopt contrary objectives, such as **reducing errors and getting things done faster**
- It is an operating philosophy that is customer-focused and strives to drive out waste, raise levels of quality, and improve financial performance at *breakthrough* levels

Six Sigma and Project Management

- Joseph M. Juran stated, “*All improvement takes place project by project, and in no other way*”*
- It’s important to select projects carefully and apply higher quality where it makes sense; companies that use Six Sigma do not always boost their stock values
- As Mikel Harry puts it, “I could genetically engineer a Six Sigma goat, but if a rodeo is the marketplace, people are still going to buy a Four Sigma horse.”**
- Six Sigma projects must focus on a quality problem or gap between the current and desired performance and not have a clearly understood problem or a predetermined solution

*“What You Need to Know About Six Sigma,” *Productivity Digest* (December 2001), p. 38.

**Clifford, Lee, “Why You Can Safely Ignore Six Sigma,” *Fortune* (January 22, 2001), p. 140.

Six Sigma Projects Use Project Management

- The training for Six Sigma includes many project management concepts, tools, and techniques
- For example, Six Sigma projects often use business cases, project charters, schedules, budgets, and so on
- Six Sigma projects are done in teams; the **project manager** is often called the **team leader**, and the **sponsor** is called the **champion**

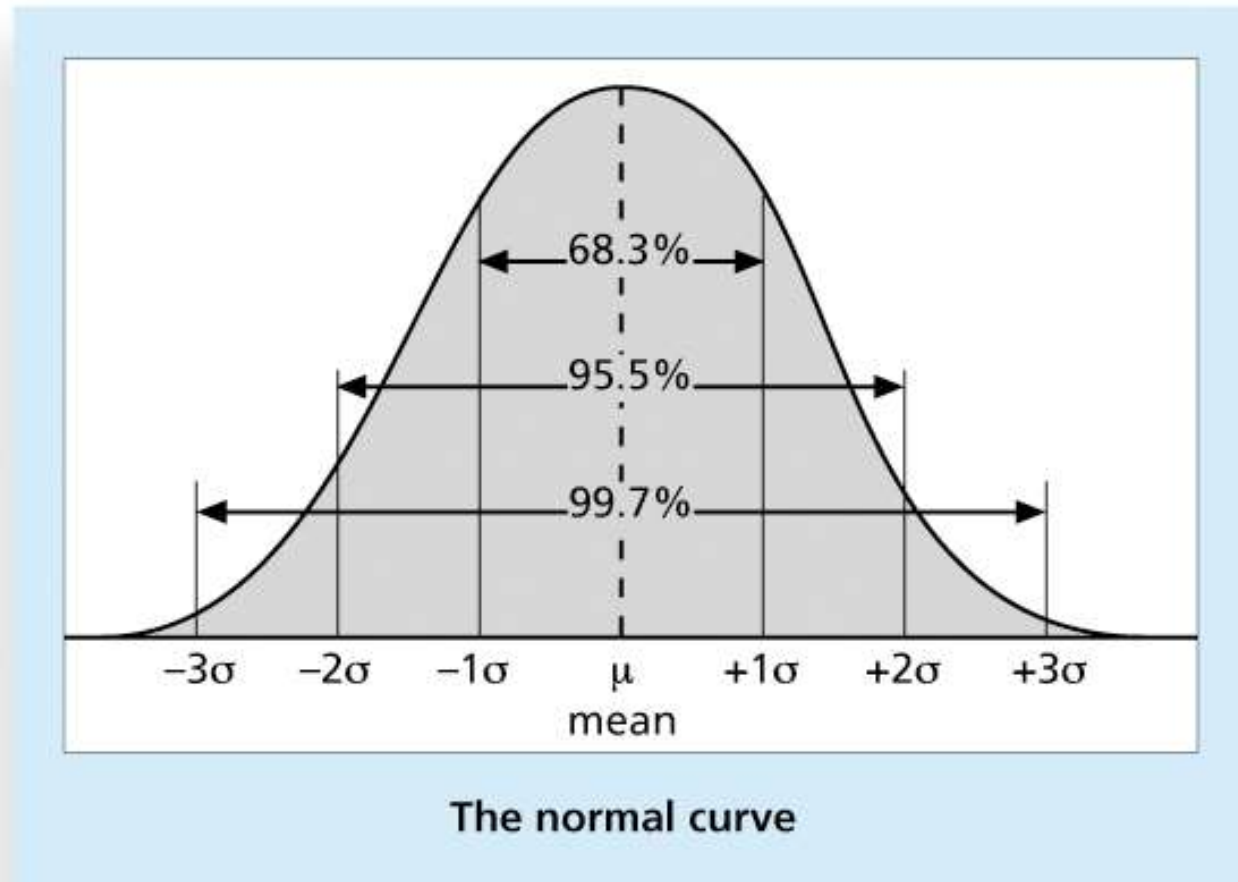
Six Sigma and Statistics

- The term *sigma* means **standard deviation**
- **Standard deviation** measures how much variation exists in a distribution of data
- Standard deviation is a key factor in determining the acceptable number of defective units found in a population
- Six Sigma projects strive for no more than 3.4 defects per million opportunities, yet this number is confusing to many statisticians

Six Sigma Uses a Conversion Table

- Using a normal curve, if a process is at six sigma, there would be no more than two defective units per billion produced
- Six Sigma uses a scoring system that accounts for time, an important factor in determining process variations
- **Yield** represents the number of units handled correctly through the process steps
- A **defect** is any instance where the product or service fails to meet customer requirements
- There can be several opportunities to have a defect

Figure 8-10. Normal Distribution and Standard Deviation



Calculating Sigma Level

$$\frac{\text{Number of Opportunities} - \text{Number of Defects}}{\text{Number of Opportunities}} \times 100 = \text{Yield (\%)}$$

Examples

- Assume that the process inserts 30,000 letters in preaddressed envelopes each day. In a given business week, the process outputs 150,000 letters.
- Marketing department randomly selects 1,000 letters from the next Weeks batch and finds that 5 has errors.
- Applying to the total amount, they estimate that as many as 750 letters could have errors.

$$\frac{150,000 - 750}{150,000} \times 100 = 99.5 (\%)$$

Calculating Sigma Level

$$\frac{\text{Number of Opportunities} - \text{Number of Defects}}{\text{Number of Opportunities}} \times 100 = \text{Yield (\%)}$$

Examples

$$\frac{150,000 - 750}{150,000} \times 100 = 99.5 (\%)$$

Yield %	Defects per Million Opportunities (DPMO)	Sigma Level
99.7450	2,550	4.3
99.6540	3,460	4.2
99.5340	4,550	4.1
99.3790	6,210	4.0
99.1810	8,190	3.9

Table 8-2. Sigma and Defective Units

Specification Range (in plus or minus Sigmas)	Percent of population within Range	Defective Units per Billion
1	68.27	317,300,000
2	95.45	45,400,000
3	99.73	2,700,000
4	99.9937	63,000
5	99.999943	57
6	99.9999998	2

Table 8-3: Sigma Conversion Table

Sigma	Yield	Defects per Million Opportunities (DPMO)
1	31.0%	690,000
2	69.2%	308,000
3	93.3%	66,800
4	99.4%	6,210
5	99.97%	230
6	99.99966%	3.4

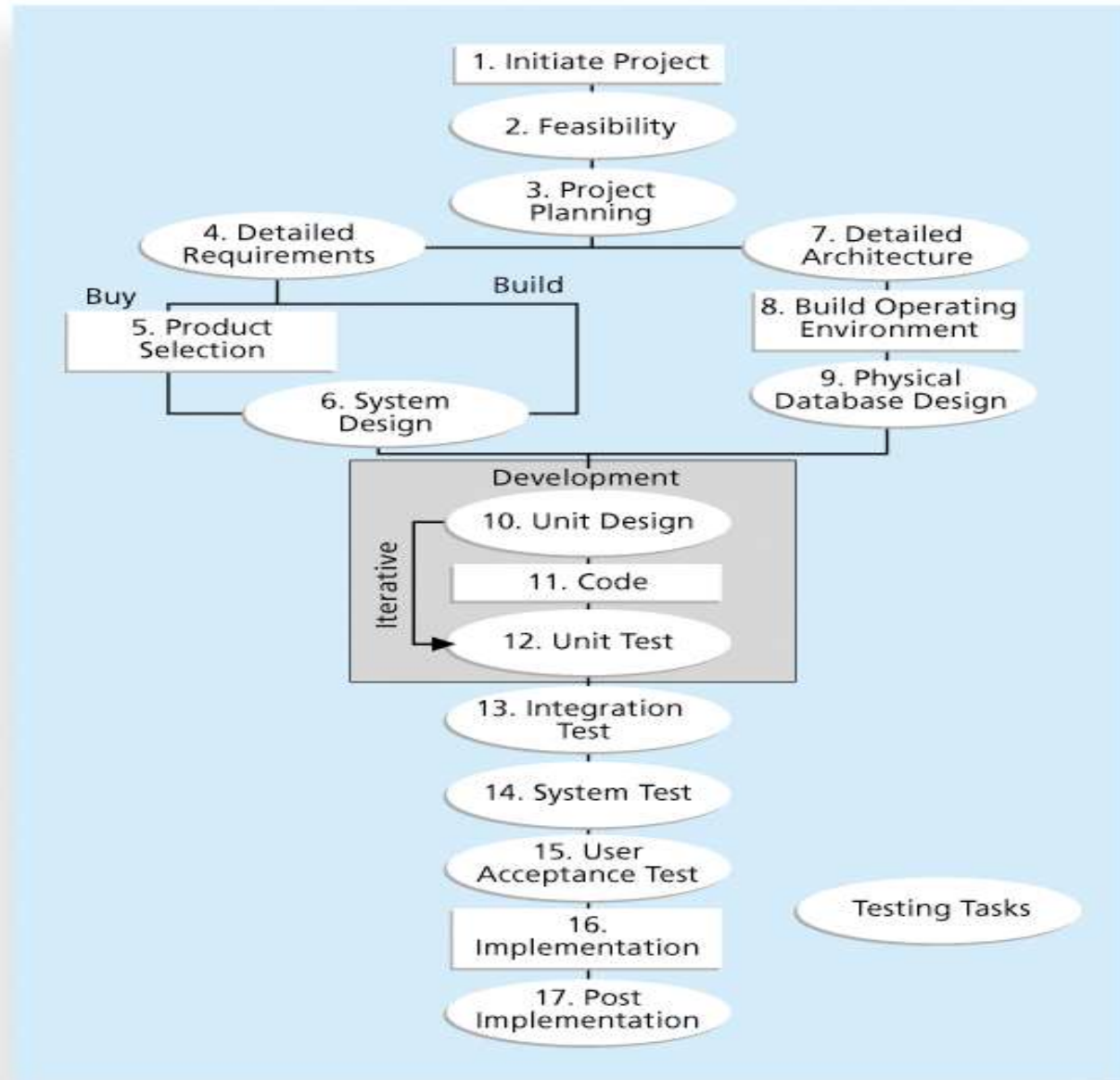
Six 9s of Quality

- **Six 9s of quality** is a measure of quality control equal to **1 fault in 1 million opportunities**
- In the telecommunications industry, it means 99.9999 percent service availability or *30 seconds of down time a year*
- This level of quality has also been stated as the target goal for the number of errors in a communications circuit, system failures, or errors in lines of code

Testing

- Many IT professionals think of testing as a stage that comes near the end of IT product development
- **Testing** should be done **during almost every phase** of the IT product development **life cycle**

Figure 8-11. Testing Tasks in the Software Development Life Cycle



Types of Tests

- **Unit testing** tests each individual component (often a program) to ensure it is as defect-free as possible
- **Integration testing** occurs between unit and system testing to test functionally grouped components
- **System testing** tests the entire system as one entity
- **User acceptance testing** is an independent test performed by end users prior to accepting the delivered system
- **Final Acceptance Test (FAT)**

Testing Alone Is Not Enough

- Watts S. Humphrey, a renowned expert on software quality, defines a **software defect** as anything that must be changed before delivery of the program
- Testing **does not** sufficiently prevent **software defects** because:
 - The number of ways to test a complex system is huge
 - Users will continue to invent new ways to use a system that its developers never considered
- Humphrey suggests that people rethink the software development process to provide *no* potential defects when you enter system testing; developers must be responsible for providing error-free code at each stage of testing

Modern Quality Management

- Modern quality management:
 - Requires customer satisfaction
 - Prefers prevention to inspection
 - Recognizes management responsibility for quality
- Noteworthy quality experts include Deming, Juran, Crosby, Ishikawa, Taguchi, and Feigenbaum (*the most influential and renowned experts in the field of quality management and quality improvement methodologies*)

Quality Experts

- Deming was famous for his work in rebuilding Japan and his 14 Points for Management
- Juran wrote the *Quality Control Handbook* and ten steps to quality improvement
- Crosby wrote *Quality is Free* and suggested that organizations strive for zero defects
- Ishikawa developed the concepts of quality circles and fishbone diagrams
- Taguchi developed methods for optimizing the process of engineering experimentation
- Feigenbaum developed the concept of total quality control

Malcolm Baldrige Award

- The **Malcolm Baldrige National Quality Award** originated in 1987 to recognize companies that have achieved a level of world-class competition through quality management
- Given by the President of the United States to U.S. businesses
- Three awards each year in different categories:
 - Manufacturing
 - Service
 - Small business
 - Education and health care

ISO Standards for Quality

- **ISO 9000** is a quality system standard that:
 - Is a three-part, continuous cycle of planning, controlling, and documenting quality in an organization
 - Provides minimum requirements needed for an organization to meet its quality certification standards
 - Helps organizations around the world reduce costs and improve customer satisfaction
- See www.iso.org for more information

Global Issues

- In 2015, 15 electric cars were introduced throughout the world
- Driverless cars are also being tested
- Google's director of self-driving cars is striving to improve their quality to reduce accident rates



Improving Information Technology Project Quality

- Several suggestions for improving quality for IT projects include:
 - Establish leadership that promotes quality
 - Understand the cost of quality
 - Focus on organizational influences and workplace factors that affect quality
 - Follow maturity models

Leadership

- As Joseph M. Juran said in 1945, “It is most important that top management be quality-minded. In the absence of sincere manifestation of interest at the top, little will happen below”*
- A large percentage of quality problems are associated with management, not technical issues.

*American Society for Quality (ASQ), (www.asqc.org/about/history/juran.html).

The Cost of Quality

- The **cost of quality** is the cost of conformance plus the cost of nonconformance
 - **Conformance** means delivering products that meet requirements and fitness for use
 - **Cost of nonconformance** means taking responsibility for failures or not meeting quality expectations (*eg: buying the electrical product with a warranty*)

- A study reported that software bugs cost the U.S. economy \$59.6 billion each year and that one third of the bugs could be eliminated by an improved testing infrastructure

Five Cost Categories Related to Quality

- **Prevention cost:** Cost of planning and executing a project so it is error-free or within an acceptable error range
- **Appraisal cost:** Cost of evaluating processes and their outputs to ensure quality
- **Internal failure cost:** Cost incurred to correct an identified defect before the customer receives the product
- **External failure cost:** Cost that relates to all errors not detected and corrected before delivery to the customer
- **Measurement and test equipment costs:** Capital cost of equipment used to perform prevention and appraisal activities

Media Snapshot

- ❑ Computer viruses and malware software have been a **quality concern** for years
- ❑ In a new twist, consumers are now being warned that e-cigarettes can be bad for computers
- ❑ Anything can infect your computer if it can be inserted into a USB port
- ❑ Other consumer products like smart TVs can invade on privacy

Organizational Influences, Workplace Factors, and Quality

- ❑ Study by DeMarco and Lister showed that organizational issues had a much **greater influence on programmer productivity than the technical environment** or programming languages
- ❑ Programmer productivity varied by a factor of one to ten across organizations, but only by 21 percent within the same organization
- ❑ Study found no correlation between productivity and programming language, years of experience, or salary.
- ❑ **A dedicated workspace and a quiet work environment were key factors in improving programmer productivity**

Expectations and Cultural Differences in Quality

- Project managers must understand and manage stakeholder expectations.
- Expectations also vary by:
 - Organization's culture
 - Geographic regions

Maturity Models

- **Maturity models** are frameworks for helping organizations improve their processes and systems
 - The **Software Quality Function Deployment Model** focuses on defining user requirements and planning software projects
 - The Software Engineering Institute's **Capability Maturity Model Integration (CMMI)** is a process improvement approach that provides organizations with the essential elements of effective processes

CMMI Levels

- CMMI levels, from **lowest to highest**, are:
 - 0. Incomplete
 - 1. Performed
 - 2. Managed
 - 3. Defined
 - 4. Quantitatively Managed
 - 5. Optimizing
- Companies may not get to bid on government projects unless they have a CMMI Level 3

PMI's Maturity Model

- PMI released the **Organizational Project Management Maturity Model (OPM3)** in December 2003
- Model is based on market research surveys sent to more than 30,000 project management professionals and incorporates 180 best practices and more than 2,400 capabilities, outcomes, and key performance indicators
- **Addresses standards for excellence in project, program, and portfolio management best practices and explains the capabilities necessary to achieve those best practices**

Best Practice

- **OPM3** provides the following example to illustrate a best practice, capability, outcome, and key performance indicator:
 - **Best practice:** Establish internal project management communities
 - **Capability:** Facilitate project management activities
 - **Outcome:** Local initiatives, meaning the organization develops pockets of consensus around areas of special interest
 - **Key performance indicator:** Community addresses local issues

Using Software to Assist in Project Quality Management

- **Spreadsheet** and charting software helps create Pareto diagrams, fishbone diagrams, and so on
- **Statistical software** packages help perform statistical analysis
- **Specialized software** products help manage Six Sigma projects or create quality control charts
- **Project management software** helps create Gantt charts and other tools to help plan and track work related to quality management

Chapter Summary

- **Project quality management ensures that the project will satisfy the needs for which it was undertaken**
- **Main processes include:**
 - **Plan quality**
 - **Perform quality assurance**
 - **Perform quality control**